



Department of Electronics and Communications Engineering
Faculty of Engineering
Alexandria University

Design of a 400 Gb/s Silicon Photonics Transceiver for IEEE 802.3 DR4 Intra-DC Interconnects

Submitted by:

Omar Mahfouz Farah Elsayed Mark Zakria
Mahmoud Magdy Islam Ibrahim Alyedeen Hassan
Khalid Khalil

Supervisor:

Dr. Eslam El-fiky

Co-Supervisors:

Dr. Eman A. Elzahaby Dr. Mai Banawan

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List of Abbreviations

Abbreviation	Meaning
APD	Avalanche photodiode
BOX	Buried oxide
CMOS	Complementary metal-oxide-semiconductor
CSPR	Carrier-to-signal power ratio
CW	Continuous wave
DR4	400GBASE-DR4 parallel single-mode fiber interface
DSP	Digital signal processing
EQE	External quantum efficiency
FCA	Free-carrier absorption
FDTD	Finite-difference time-domain
IQE	Internal quantum efficiency
NEP	Noise-equivalent power
OOK	On-off keying
PAM-4	Four-level pulse-amplitude modulation
PDL	Polarization-dependent loss
PIN	p-intrinsic-n photodiode
PML	Perfectly matched layer
RC	Resistance-capacitance
RF	Radio frequency
SOI	Silicon-on-insulator
SRH	Shockley-Read-Hall
SNR	Signal-to-noise ratio
SNI	Surface-normal illumination
SSBI	Signal-signal beating interference
TDD	Threading dislocation density
UTC	Uni-travelling-carrier
WG	Waveguide

List of Symbols

Symbol	Meaning	Unit
α	Absorption coefficient	cm^{-1}
$\tilde{n}$	Complex refractive index	—
n	Real part of refractive index	—
κ	Extinction coefficient	—
ω	Angular frequency	rad s^{-1}
μ_0	Permeability of free space	H m^{-1}
ε_0	Permittivity of free space	F m^{-1}
$\tilde{\varepsilon}$	Complex permittivity	F m^{-1}
$\tilde{k}$	Complex wave number	m^{-1}
c	Speed of light in vacuum	m s^{-1}
E	Electric field	V m^{-1}
H	Magnetic field	A m^{-1}
S	Poynting vector	W m^{-2}
G_{opt}	Optical generation rate	$\text{cm}^{-3} \text{s}^{-1}$
$\hbar$	Reduced Planck constant	J s
i_{ph}	Photocurrent	A
$\mathcal{R}$	Responsivity	A W^{-1}
P	Optical power	W
η_0	Impedance of free space	Ω
$\tilde{s}$	Complex envelope	—
ω_0	Carrier frequency	rad s^{-1}
c	DC carrier amplitude	—
$\tilde{u}$	Modulated signal amplitude	—
$CSPR$	Carrier-to-signal power ratio	dB
$E_{g,\text{ind}}$	Indirect bandgap energy	eV
$E_{g,\text{dir}}$	Direct bandgap energy	eV

Chapter 1

Design and Simulation of a Ge-on-Si Waveguide Photodetector for O-Band 400G Reception

1.1 Introduction and Chapter Scope

The relentless growth of data-centre interconnect capacity has driven the O-band silicon-photonics ecosystem toward 400G and beyond. IEEE 802.3 Clause 140 specifies 400GBASE-DR4 as a four-lane PAM-4 interface at 1310 nm, placing each lane at $106.25 \text{ Gbit s}^{-1}$ raw line rate and demanding a receiver-side photodetector that is simultaneously fast, efficient, and quiet. Meeting that specification on a CMOS-foundry platform is not merely a component selection problem: it requires a design methodology that connects photon absorption physics, material constraints, device topology, and electrode engineering into a single coherent argument.

This chapter develops that argument from first principles. The discussion is structured as a progressive narrowing: beginning with the electromagnetic foundations of light absorption in semiconductors, passing through the metrics that define a useful receiver device, surveying the landscape of competing architectures and materials, and arriving at a specific vertical n-i-p Ge-on-Si waveguide photodetector whose geometry, contact topology, and simulation strategy are each justified by the physics established along the way.

The chapter is organised as follows. Section 1.2 establishes the physics of photodetection, deriving the Beer-Lambert absorption law from Maxwell’s equations, developing the drift-diffusion carrier transport framework, and quantifying the recombination mechanisms that limit carrier collection. Section 1.3 defines the evaluation language, namely responsivity, bandwidth, dark current, noise, and the trade-offs among them, that governs every architectural and material choice that follows. Section 1.4 surveys photodetector device classes from the photoconductor through the avalanche photodiode and derives, rather than asserts,

the selection of the waveguide-integrated PIN junction for high-speed integrated links. Section 1.5 justifies germanium-on-silicon as the material platform, including its spectral suitability at 1310 nm, its CMOS integration path, and the threading-dislocation challenge that sets the dark-current floor. Section 1.6 reviews the experimental record of Ge-on-Si waveguide photodetectors critically and identifies the gap that motivates the U-shaped electrode approach adopted here. Section 1.7 presents the proposed device architecture in detail, including the vertical PIN layer stack, the adiabatic taper coupling strategy, the U-shaped electrode topology, and the geometry-to-performance design space mapping. Section 1.8 describes the multiphysics simulation strategy: FDTD optical modelling, CHARGE drift-diffusion solving, MATLAB post-processing, and the validation approach anchoring the simulation to the published benchmark. Section 1.9 closes with a synthesis of the key design insights and a transition to the results chapter.

1.2 Fundamentals of Photodetection

Photodetection begins with the transfer of optical energy to the semiconductor lattice. Understanding that process at the physics level is a prerequisite for rational device design, because every performance metric defined in section 1.3 traces back to the rate at which photons are absorbed, how the resulting carriers are transported, and what fraction is lost to recombination before reaching the device terminals.

1.2.1 Optical Absorption in Semiconductors

The starting point for all optical analysis in this chapter is the macroscopic form of Maxwell's equations in a non-magnetic, source-free medium. In SI units and in the frequency domain, obtained by assuming a time dependence $e^{-i\omega t}$, these read:

$$\nabla \times \tilde{\mathbf{E}}(\mathbf{r}, \omega) = i\omega\mu_0 \tilde{\mathbf{H}}(\mathbf{r}, \omega), \quad (1.1)$$

$$\nabla \times \tilde{\mathbf{H}}(\mathbf{r}, \omega) = -i\omega\tilde{\epsilon}(\omega) \tilde{\mathbf{E}}(\mathbf{r}, \omega), \quad (1.2)$$

where μ_0 is the permeability of free space and $\tilde{\epsilon}(\omega)$ is the complex permittivity. Taking the curl of eq. (1.1) and substituting eq. (1.2) yields the Helmholtz wave equation:

$$\nabla^2 \tilde{\mathbf{E}}(\mathbf{r}, \omega) + \tilde{k}^2(\omega) \tilde{\mathbf{E}}(\mathbf{r}, \omega) = \mathbf{0}, \quad (1.3)$$

where $\tilde{k}(\omega) = \omega\sqrt{\mu_0\tilde{\epsilon}(\omega)}$. Defining the *complex refractive index*:

$$\tilde{n}(\omega) \equiv \frac{c\tilde{k}(\omega)}{\omega} = n(\omega) + i\kappa(\omega), \quad (1.4)$$

the plane-wave solution propagating along $+\hat{z}$ is:

$$\tilde{\mathbf{E}}(z, \omega) = \mathbf{E}_0 \exp\left[i n(\omega) \frac{\omega z}{c}\right] \exp\left[-\kappa(\omega) \frac{\omega z}{c}\right]. \quad (1.5)$$

The time-averaged Poynting intensity $I(z) \propto |\tilde{\mathbf{E}}|^2$ decays as $\exp[-2\kappa\omega z/c]$, giving the Beer-Lambert absorption coefficient:

$$\alpha(\omega) = \frac{2\omega\kappa(\omega)}{c} = \frac{4\pi\kappa(\lambda)}{\lambda}, \quad (1.6)$$

so that the Beer-Lambert law:

$$I(z) = I_0 \exp[-\alpha(\lambda) z] \quad (1.7)$$

emerges as a corollary of eq. (1.3) rather than an independent empirical postulate. The absorbed power density follows directly:

$$\mathcal{P}_{\text{abs}}(\mathbf{r}) = \alpha(\lambda) I(\mathbf{r}). \quad (1.8)$$

Poynting-vector formulation of the generation rate.

The optical generation rate $G_{\text{opt}}(\mathbf{r})$ is more generally obtained from the time-averaged divergence of the Poynting vector. The divergence theorem applied to the time-averaged Poynting flux $\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$ over a closed surface $\partial\Omega$ enclosing a volume Ω of absorbing material gives:

$$\int_{\Omega} G_{\text{opt}}(\mathbf{r}) \hbar\omega \, dV = - \oint_{\partial\Omega} \langle \mathbf{S} \rangle \cdot \hat{n} \, dA, \quad (1.9)$$

so that the local volumetric generation rate is the Poynting-flux sink:

$$G_{\text{opt}}(\mathbf{r}) = -\frac{\nabla \cdot \langle \mathbf{S}(\mathbf{r}) \rangle}{\hbar\omega} = \frac{\omega\epsilon_0 \text{Im}[\tilde{\epsilon}(\mathbf{r})] |\tilde{\mathbf{E}}(\mathbf{r})|^2}{2\hbar\omega}. \quad (1.10)$$

This reduces identically to eq. (1.30) under the plane-wave approximation, but is the operationally correct definition when the three-dimensional field contains interference terms arising from modal coupling, metal reflections, or end-facet diffraction, all of which are present in the adiabatic taper region of the device studied here. The FDTD solver employed in section 1.8.1 evaluates G_{opt} from eq. (1.10); the scalar Beer-Lambert law of eq. (1.7) serves only as the analytical validation baseline.

Square-law photodetection and the domain of linear operation.

For a quasi-monochromatic field at carrier frequency ω_0 with slowly varying complex envelope $\tilde{s}(t)$:

$$\tilde{\mathbf{E}}(t) = \frac{1}{2} \tilde{s}(t) e^{-i\omega_0 t} \hat{\mathbf{e}} + \text{c.c.}, \quad (1.11)$$

the time-averaged Poynting intensity at the detector facet is $I(t) = |\tilde{s}(t)|^2 / (2\eta_0)$, where $\eta_0 = \sqrt{\mu_0/\epsilon_0}$. The photocurrent produced by the device is therefore:

$$i_{\text{ph}}(t) = \mathcal{R} \cdot P(t) = \mathcal{R} \cdot \frac{|\tilde{s}(t)|^2}{2\eta_0}, \quad (1.12)$$

which is *linear in optical power* $P(t)$ but *quadratic in optical field amplitude* $|\tilde{s}(t)|$. When the field contains both a data-bearing modulated component $\tilde{u}(t)$ and a residual DC carrier c , $\tilde{s}(t) = \tilde{u}(t) + c$, the photocurrent expands as:

$$i(t) = \mathcal{R} \left[\underbrace{c^2}_{\text{DC bias}} + \underbrace{2c \text{Re}\{\tilde{u}(t)\}}_{\text{linear signal term}} + \underbrace{|\tilde{u}(t)|^2}_{\text{SSBI}} \right], \quad (1.13)$$

where SSBI denotes signal-signal beating interference. The linear approximation $i_{\text{ph}} \approx \mathcal{R} \cdot P_{\text{opt}}$ is valid when the carrier-to-signal power ratio $\text{CSPR} \equiv c^2 / \langle |\tilde{u}|^2 \rangle \gg 1$, so that SSBI is negligible relative to the cross-beat term.

Integrating eq. (1.8) through an absorber of length L gives the absorbed fraction:

$$\mathcal{A}(L) = 1 - \exp[-\alpha(\lambda) L]. \quad (1.14)$$

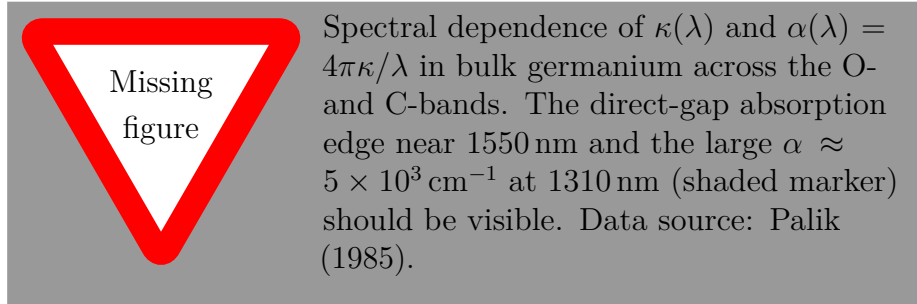


Figure 1.1: Spectral absorption coefficient $\alpha(\lambda)$ and extinction coefficient $\kappa(\lambda)$ of bulk germanium, derived via eq. (1.6) from tabulated optical constants. The large α at 1310 nm reflects direct-gap absorption at the Γ -point discussed in section 1.5.

1.2.2 Photodetector Figures of Merit

Photodetector performance is characterised by a small set of figures of merit that translate the underlying physics of absorption, transport, and recombination into engineering-relevant quantities. This section defines these metrics rigorously and establishes the inter-dependence that governs the principal design trade-offs encountered in a high-speed Ge-on-Si device targeting the O-band.

Quantum Efficiency and Responsivity

The *external quantum efficiency* η_{ext} is the most fundamental measure of a photodetector's optical-to-electrical conversion fidelity. It is defined as the ratio of the number of electron-hole pairs collected at the device terminals per unit time to the number of photons incident on the detector per unit time:

$$\eta_{\text{ext}} = \frac{I_{\text{ph}}/q}{P_{\text{opt}}/(\hbar\omega)} = \frac{\hbar\omega}{q} \mathcal{R}, \quad (1.15)$$

where I_{ph} is the photocurrent, q the elementary charge, P_{opt} the incident optical power, and $\mathcal{R}$ the *responsivity* expressed in amperes per watt. The responsivity is a particularly convenient engineering figure because it is directly measurable from the slope of a photocurrent-versus-power curve in the linear regime established by eq. (1.12):

$$\mathcal{R} = \frac{I_{\text{ph}}}{P_{\text{opt}}} = \frac{\eta_{\text{ext}} q}{\hbar\omega} = \frac{\eta_{\text{ext}} \lambda q}{hc} \quad [\text{A W}^{-1}]. \quad (1.16)$$

At the O-band centre wavelength of 1310 nm, the photon energy is $\hbar\omega = hc/\lambda \approx 0.947$ eV, and the theoretical upper bound on responsivity, achieved when $\eta_{\text{ext}} = 1$, is:

$$\mathcal{R}_{\text{max}} = \frac{q\lambda}{hc} \approx 1.057 \text{ A W}^{-1} \quad (\lambda = 1310 \text{ nm}). \quad (1.17)$$

The external quantum efficiency can be factored into contributions from coupling, absorption, and collection:

$$\eta_{\text{ext}} = \eta_{\text{coup}} \eta_{\text{abs}} \eta_{\text{coll}}, \quad (1.18)$$

where η_{coup} accounts for optical coupling losses at the silicon-waveguide-to-germanium interface (mode mismatch, reflection), $\eta_{\text{abs}} = \mathcal{A}(L) = 1 - e^{-\alpha L}$ is the absorbed fraction derived in eq. (1.14), and η_{coll} captures the fraction of photogenerated carriers that reach the contacts without recombining. In a well-depleted PIN structure operated at sufficient reverse bias, $\eta_{\text{coll}} \rightarrow 1$ because field-driven drift removes carriers on a timescale $\tau_{\text{tr}} \ll \tau_{\text{SRH}}$. The product $\eta_{\text{coup}} \eta_{\text{abs}}$ is therefore the dominant lever in device design: the former is addressed through taper geometry and modal overlap engineering, while the latter dictates the minimum absorber length. The auxiliary MATLAB model used in this repository takes $\alpha \approx 7 \times 10^3 \text{ cm}^{-1}$ at 1310 nm, corresponding to a bulk absorption length of $\alpha^{-1} \approx 1.4 \mu\text{m}$ before waveguide-confinement and mode-overlap factors are applied. The final absorber length is therefore not selected from the bulk Beer-Lambert estimate alone; it is selected from the FDTD taper-coupled length sweep.

Internal quantum efficiency and the collection efficiency limit.

It is instructive to distinguish the *internal* quantum efficiency η_{int} , which excludes coupling and reflection losses and characterises solely the conversion of absorbed photons to collected charge [1]. In the limit where all photogeneration is confined to the fully depleted intrinsic region of length W_i and drift transport is complete, $\eta_{\text{int}} = 1$. Departures from this ideal arise when photogeneration extends into quasi-neutral regions outside the depletion zone, where diffusion is the dominant transport mechanism. The diffusion minority carrier lifetime τ_n in the p -doped anode sets a characteristic diffusion length $L_n = \sqrt{D_n \tau_n}$; only carriers generated within L_n of the depletion boundary contribute to the photocurrent. Carriers generated beyond this distance recombine before reaching the junction. This mechanism is especially pertinent in the O-band context, where the large absorption depth at wavelengths approaching the indirect-gap edge can push a fraction of the generation profile into the quasi-neutral region if the intrinsic layer is not sufficiently thick or the reverse bias is insufficient to extend the depletion width. The trade-off is explicit: increasing W_i increases η_{int} but simultaneously increases the transit time $\tau_{\text{tr}} = W_i/v_{\text{d,sat}}$ and the junction capacitance $C_j = \varepsilon_{\text{Ge}} A_j/W_i$, both of which degrade bandwidth as formalised in section 1.3.2.

Bandwidth and Speed Limitations

The small-signal 3 dB bandwidth $f_{3\text{dB}}$ of a photodetector is the electrical frequency at which the detector's photocurrent response is attenuated by 3 dB relative to its low-frequency value. Two fundamentally distinct physical mechanisms limit this bandwidth and must be engineered simultaneously for high-speed operation.

Transit-time bandwidth.

When an electron-hole pair is generated inside the intrinsic region and swept towards the contacts by the applied field, the terminal current induced via the Ramo-Shockley theorem is non-zero for the duration of transit. A pair generated at depth z contributes a rectangular current pulse of duration $\tau_e(z) = z/v_{\text{d,sat},e}$ for the electron and $\tau_h(z) = (W_i - z)/v_{\text{d,sat},h}$ for the hole, as expressed in eqs. (1.41) and (1.42). The frequency response of a single pair follows from the Fourier transform of this pulse pair, eq. (1.43), and upon averaging uniformly over all generation depths within $[0, W_i]$ via eq. (1.44), the transit-time transfer function takes the form of a weighted sinc envelope. In the practically relevant limit where the hole branch—slower due to $v_{\text{d,sat},h} < v_{\text{d,sat},e}$ in germanium—dominates the roll-off, the 3 dB transit-time bandwidth is well approximated by:

$$f_{\text{tt}} \approx \frac{0.443 v_{\text{d,sat},h}}{W_i}, \quad (1.19)$$

where $v_{\text{d,sat},h} \approx 6.0 \times 10^6 \text{ cm s}^{-1}$ for holes in germanium at fields exceeding approximately 10 kV cm^{-1} [1]. The numerical prefactor 0.443 arises from the sinc-function roll-off $|\text{sinc}(\omega\tau/2)| = 1/\sqrt{2}$ at $\omega\tau_{\text{tr}} = 2.78$. Equation (1.19) immediately establishes that transit-time bandwidth is inversely proportional to the intrinsic-layer thickness: halving W_i doubles f_{tt} .

RC-limited bandwidth.

The second mechanism is purely electrical in origin. The detector junction presents a capacitance C_j to the external circuit, whose total load resistance is $R_{\text{tot}} = R_s + R_L$, where R_s is the device series resistance (contact and spreading resistance) and $R_L = 50 \Omega$ is the standard termination impedance in high-speed link applications. The resulting single-pole low-pass response, eq. (1.45), gives the RC-limited bandwidth:

$$f_{\text{RC}} = \frac{1}{2\pi R_{\text{tot}} C_j}. \quad (1.20)$$

For a planar junction of area $A_j = W_j \times L$ and intrinsic thickness W_i , the depletion capacitance is:

$$C_j = \frac{\varepsilon_0 \varepsilon_{\text{Ge}} A_j}{W_i}, \quad (1.21)$$

where $\varepsilon_{\text{Ge}} \approx 16.0$ is the static relative permittivity of germanium. Substituting into eq. (1.20) makes the competing roles of W_i explicit: increasing W_i reduces C_j and raises f_{RC} , but

simultaneously reduces f_{tt} via eq. (1.19). This *transit-time/RC trade-off* is the central design tension in the intrinsic-layer thickness optimisation.

Combined bandwidth.

Because the transit-time and RC contributions to bandwidth act as independent roll-off mechanisms in cascade, the total 3 dB bandwidth of the device is:

$$\frac{1}{f_{\text{3dB}}^2} \approx \frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2}, \quad (1.22)$$

where the quadrature sum is an approximation that is exact when both sub-responses are first-order Butterworth poles and remains accurate to within a few percent for the sinc-envelope transit-time case [2]. The full factorised transfer function is given by eq. (1.46). Optimising W_i to maximise f_{3dB} subject to a fixed device footprint then reduces to finding the value of W_i satisfying $\partial f_{\text{3dB}}/\partial W_i = 0$, which for $R_{\text{tot}} = 50 \Omega$ typically places the optimum in the range $W_i \in [0.5 \mu\text{m}, 1.5 \mu\text{m}]$ for a waveguide-integrated Ge-on-Si detector with a junction width of order $1 \mu\text{m}$ targeting bandwidths in the range 50 GHz to 70 GHz.

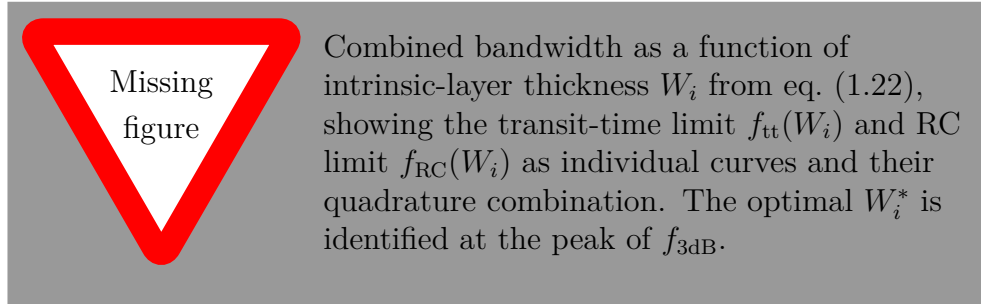


Figure 1.2: Transit-time and RC bandwidth components as a function of the intrinsic-layer thickness W_i for a Ge-on-Si waveguide detector with $A_j = 1 \mu\text{m} \times 15 \mu\text{m}$ and $R_{\text{tot}} = 50 \Omega$. The optimal thickness W_i^* yields the maximum combined bandwidth, illustrating the fundamental transit-time/RC trade-off.

Noise Mechanisms and Signal-to-Noise Ratio

Noise ultimately determines the minimum detectable optical power and, in conjunction with bandwidth, the achievable bit-error rate (BER) in a digital link. Three noise sources are relevant in a reverse-biased PIN photodetector [1]: shot noise from both signal and dark photocurrents, and thermal noise from the load resistance.

Shot noise.

Shot noise originates from the quantum-mechanical discreteness of the photodetection process: each photon-induced electron-hole pair is generated at a statistically independent

random instant, following Poisson statistics. The two-sided power spectral density (PSD) of the resulting current fluctuation δi_{sh} is flat (white) up to the detector bandwidth and is given by:

$$S_{\text{sh}}(f) = 2q(I_{\text{ph}} + I_d), \quad (1.23)$$

where $I_{\text{ph}} = \mathcal{R} P_{\text{opt}}$ is the signal photocurrent and I_d is the dark current. The mean-square shot-noise current within a measurement bandwidth B is therefore:

$$\langle i_{\text{sh}}^2 \rangle = 2q(I_{\text{ph}} + I_d)B. \quad (1.24)$$

In the context of a Ge-on-Si detector, the dark current I_d is substantially larger than in silicon counterparts because of the high density of SRH trap centres introduced by threading dislocations during heteroepitaxial growth. A representative dark current of order 10 nA to 100 nA at $V_{\text{bias}} = -1$ V for a device of area $A_j \approx 15 \mu\text{m}^2$ implies a dark-current shot-noise PSD that can rival the signal shot noise at low received powers. Minimising I_d through lifetime engineering and careful epitaxial quality control is therefore not merely a static power concern but directly impacts the receiver noise floor.

Thermal (Johnson) noise.

The load resistor R_L at temperature T introduces a Johnson noise current with PSD:

$$S_{\text{th}}(f) = \frac{4k_B T}{R_L}, \quad (1.25)$$

which is also white within the bandwidth of interest. The mean-square thermal noise current in bandwidth B is:

$$\langle i_{\text{th}}^2 \rangle = \frac{4k_B T}{R_L} B. \quad (1.26)$$

At room temperature ($T = 300$ K) and standard termination $R_L = 50 \Omega$, the thermal noise current density is $\sqrt{S_{\text{th}}} \approx 18 \text{ pA/Hz}^{1/2}$, which is typically the dominant noise source at low optical power levels in a direct-detection receiver without optical preamplification.

Signal-to-noise ratio and noise-equivalent power.

The electrical signal-to-noise ratio (SNR) for a direct-detection receiver combining shot noise and thermal noise is:

$$\text{SNR} = \frac{I_{\text{ph}}^2}{\langle i_{\text{sh}}^2 \rangle + \langle i_{\text{th}}^2 \rangle} = \frac{(\mathcal{R} P_{\text{opt}})^2}{2q(\mathcal{R} P_{\text{opt}} + I_d)B + 4k_B T B / R_L}. \quad (1.27)$$

The *noise-equivalent power* (NEP) is defined as the optical input power at which $\text{SNR} = 1$ in a bandwidth $B = 1$ Hz, providing a bandwidth-independent figure of merit for detector sensitivity:

$$\text{NEP} = \frac{\sqrt{2q I_d + 4k_B T / R_L}}{\mathcal{R}} \quad [\text{W/Hz}^{1/2}]. \quad (1.28)$$

Equation (1.28) explicitly couples the dark-current magnitude, the responsivity, and the load impedance into a single sensitivity figure. For the Ge-on-Si device characterised in this work, the dark-current term dominates at room temperature and typical bias conditions, so that reducing I_d and increasing $\mathcal{R}$ are the two principal levers for sensitivity improvement. In the shot-noise-limited regime, where $I_d \rightarrow 0$ and $R_L \rightarrow \infty$ (achieved in practice with a transimpedance amplifier of very large input impedance), the NEP reduces to the quantum-limited floor:

$$\text{NEP}_{\text{shot}} = \frac{\sqrt{2q I_{\text{ph}}}}{\mathcal{R}} \Big|_{I_{\text{ph}} \rightarrow 0} \xrightarrow{\eta_{\text{ext}} \rightarrow 1} \sqrt{2\hbar\omega}. \quad (1.29)$$

For the O-band, this evaluates to $\text{NEP}_{\text{shot}} \approx 5.5 \times 10^{-19} \text{ W/Hz}^{1/2}$, a value that practical devices approach asymptotically as dark current and excess noise are suppressed. The derivation of the dark-current model $I_d(V_{\text{bias}})$ used in eq. (1.28) is presented in section 1.8.2, where the surface and bulk SRH components are separated and independently calibrated.

1.2.3 Carrier Photogeneration and Transport

The absorbed photon creates an electron-hole pair. Whether that pair is swept to the contacts and contributes to the photocurrent, or instead recombines within the absorber and is lost, depends on the interplay of the electric field, the carrier mobilities, and the defect density of the semiconductor.

The volumetric optical generation rate $G_{\text{opt}}(\mathbf{r})$ quantifies the number of electron-hole pairs created per unit volume per unit time:

$$G_{\text{opt}}(\mathbf{r}) = \frac{\mathcal{P}_{\text{abs}}(\mathbf{r})}{\hbar\omega} = \frac{\alpha(\lambda) I(\mathbf{r})}{\hbar\omega} \quad [\text{cm}^{-3} \text{ s}^{-1}]. \quad (1.30)$$

Because $I(\mathbf{r})$ decays exponentially along the propagation direction via eq. (1.7), G_{opt} is spatially non-uniform: it is concentrated near the input facet of the absorber and falls off with an effective length set by the Ge material absorption, the optical confinement factor, and the simulated taper-to-Ge mode overlap.

Drift–diffusion transport.

The self-consistent carrier distribution and electrostatic field are governed by the coupled Poisson and drift-diffusion equations [1], [3]. Poisson’s equation is:

$$\nabla \cdot [\varepsilon(\mathbf{r}) \nabla \phi(\mathbf{r}, t)] = -q[p - n + N_D^+ - N_A^-], \quad (1.31)$$

and the electron and hole current densities in the drift-diffusion approximation are:

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n, \quad (1.32)$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p, \quad (1.33)$$

where the carrier diffusivities follow the Einstein relation [4]:

$$D_{n,p} = \mu_{n,p} \frac{k_B T}{q}. \quad (1.34)$$

The general, time-dependent carrier continuity equations are:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot \mathbf{J}_n - U_{\text{net}} + G_{\text{opt}}, \quad (1.35)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot \mathbf{J}_p - U_{\text{net}} + G_{\text{opt}}, \quad (1.36)$$

where $U_{\text{net}} = U_{\text{SRH}} + U_{\text{Auger}} + U_{\text{rad}}$. Under *steady-state* CW illumination, $\partial/\partial t \equiv 0$:

$$\frac{1}{q} \nabla \cdot \mathbf{J}_n = U_{\text{net}} - G_{\text{opt}}, \quad (1.37)$$

$$\frac{1}{q} \nabla \cdot \mathbf{J}_p = -(U_{\text{net}} - G_{\text{opt}}). \quad (1.38)$$

The complete self-consistent system, from eq. (1.31) through eq. (1.38), is the foundational PDE set solved numerically by the commercial finite-element CHARGE solver and reduced in MATLAB for the geometry-derived bandwidth and noise models.

The central design consequence of the drift-diffusion picture is the distinction between drift-dominated and diffusion-dominated carrier motion. Drift transport, which is field-driven and fast, requires a sufficiently large reverse-bias field to sweep carriers across the intrinsic region before they can recombine. Diffusion transport, which is concentration-gradient-driven and inherently slower, dominates in the quasi-neutral regions outside the depletion zone and is the primary cause of slow response tails in devices with incomplete or absent depletion. A reverse-biased PIN junction suppresses diffusion by confining photogeneration to the depleted region, making drift the dominant transport mechanism and the carrier transit time the key speed parameter.

Velocity saturation and field-dependent mobility.

An important refinement to the linear drift picture arises at the field magnitudes present in a reverse-biased Ge PIN junction. The carrier drift velocity does not increase indefinitely with electric field but saturates at $v_{\text{d,sat}}$ when the field exceeds a critical value $\mathcal{E}_c$. A widely adopted empirical model for the field-dependent drift velocity is the Caughey-Thomas expression [1]:

$$v_{n,p}(\mathcal{E}) = \frac{\mu_{n,p}^{(0)} \mathcal{E}}{\left[1 + \left(\mu_{n,p}^{(0)} \mathcal{E} / v_{\text{d,sat},n/p} \right)^{\beta_{n,p}} \right]^{1/\beta_{n,p}}}, \quad (1.39)$$

where $\mu_{n,p}^{(0)}$ is the low-field mobility, $v_{\text{d,sat},n/p}$ is the saturation velocity, and $\beta_{n,p}$ is an empirical shape parameter. For germanium at 300 K, representative values are $\mu_n^{(0)} \approx 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $\mu_p^{(0)} \approx 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, $v_{\text{d,sat},e} \approx 6.0 \times 10^6 \text{ cm s}^{-1}$, and $v_{\text{d,sat},h} \approx 6.0 \times 10^6 \text{ cm s}^{-1}$, with

$\beta \approx 2$ for electrons and $\beta \approx 1$ for holes [1]. At a reverse bias of 2 V across a 1 μm intrinsic layer, the mean field is 20 kV cm^{-1} , which substantially exceeds the critical field $\mathcal{E}_c = v_{\text{d,sat}}/\mu^{(0)} \approx 1 \text{ kV cm}^{-1}$ to 3 kV cm^{-1} and places the device firmly in the velocity-saturated regime. The constant velocity assumption adopted in eq. (1.19) and the analytical transfer function of eq. (1.40) is therefore well justified at standard operating bias, and the transit time is accurately estimated as $\tau_{\text{tr}} = W_i/v_{\text{d,sat}}$ rather than the low-field value $W_i^2/(\mu V)$.

Analytical photocurrent transfer function.

In the frequency domain, neglecting diffusion and assuming a uniform electric field at saturation value throughout $z \in [0, W_i]$, the Fourier amplitude of the terminal current density admits a closed-form expression:

$$J_t(\omega) = \frac{qG_0}{\alpha W_i} \frac{hc}{q\lambda} \eta(1 - R_{\text{refl}}) P_{\text{opt}}(\omega) \times \left\{ \frac{e^{-\alpha W_i} - 1}{\alpha W_i(\alpha W_i - j\omega\tau_{\text{tr,h}})} + e^{-\alpha W_i} \frac{e^{j\omega\tau_{\text{tr,h}}} - 1}{j\omega\tau_{\text{tr,h}}(\alpha W_i - j\omega\tau_{\text{tr,h}})} + \frac{1 - e^{-\alpha W_i}}{\alpha W_i(j\omega\tau_{\text{tr,e}} + \alpha W_i)} + \frac{1 - e^{j\omega\tau_{\text{tr,e}}}}{j\omega\tau_{\text{tr,e}}(j\omega\tau_{\text{tr,e}} + \alpha W_i)} \right\}, \quad (1.40)$$

where $\tau_{\text{tr,e}} = W_i/v_{\text{d,sat,e}}$ and $\tau_{\text{tr,h}} = W_i/v_{\text{d,sat,h}}$ are the electron and hole transit times. In the limits $\omega\tau_{\text{tr,h}} \ll 1$ and $\alpha W_i \gg 1$, eq. (1.40) reduces to the rectangular-pulse sinc-function roll-off yielding the bandwidth formula of eq. (1.19).

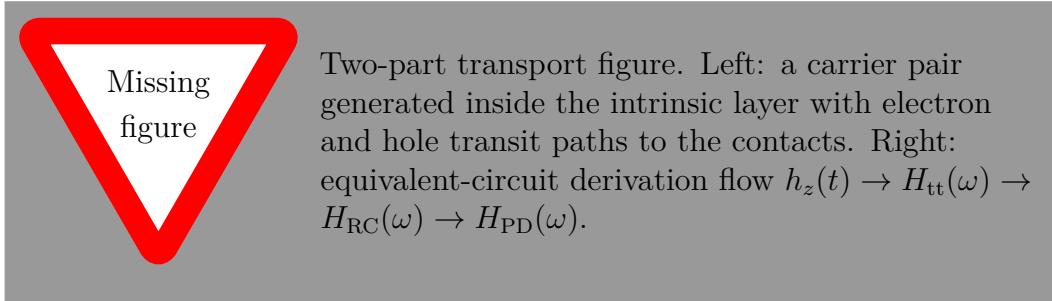


Figure 1.3: Microscopic-to-circuit derivation path for the small-signal transfer function. The transit-time response originates from the finite carrier flight time inside the intrinsic Ge region, while the RC response originates from the external electrical loading seen by the junction.

Derivation of the factorised small-signal transfer function.

Consider an electron-hole pair generated at depth $z \in [0, W_i]$ inside the intrinsic layer. Under the uniform-field, velocity-saturated approximation, the electron and hole transit times are:

$$\tau_e(z) = \frac{z}{v_{\text{d,sat,e}}}, \quad \tau_h(z) = \frac{W_i - z}{v_{\text{d,sat,h}}}. \quad (1.41)$$

By the Ramo-Shockley theorem, the terminal current induced by each carrier is non-zero only while the carrier is moving, giving the local impulse response:

$$h_z(t) = \frac{qv_{\text{d,sat},e}}{W_i} u(t) u[\tau_e(z) - t] + \frac{qv_{\text{d,sat},h}}{W_i} u(t) u[\tau_h(z) - t], \quad (1.42)$$

where $u(t)$ is the Heaviside function. Fourier transformation yields:

$$H_z(\omega) = \frac{q}{W_i} \left[\frac{1 - e^{-j\omega\tau_e(z)}}{j\omega} + \frac{1 - e^{-j\omega\tau_h(z)}}{j\omega} \right]. \quad (1.43)$$

If the optical generation is approximately uniform across the depletion depth, the transit response follows from averaging eq. (1.43) over z :

$$H_{\text{tt}}(\omega) \propto \int_0^{W_i} H_z(\omega) \frac{dz}{W_i}. \quad (1.44)$$

The integral evaluates to a weighted sum of electron and hole sinc terms. In the practically relevant limit where the slower hole branch dominates the 3 dB roll-off, the envelope reduces to the single-sinc form used in eq. (1.77). The electrical loading is then appended through the junction capacitance and total series resistance. For a current source driving C_j into $R_{\text{tot}} = R_s + R_L$:

$$H_{\text{RC}}(\omega) = \frac{1}{1 + j\omega R_{\text{tot}} C_j} = \frac{1}{1 + j\omega/\omega_{\text{RC}}}, \quad (1.45)$$

with $\omega_{\text{RC}} = (R_{\text{tot}} C_j)^{-1}$. The total response is the product:

$$H_{\text{PD}}(\omega) = H_{\text{tt}}(\omega) H_{\text{RC}}(\omega), \quad (1.46)$$

which is the physical content hidden inside the compact form of eq. (1.77): transport determines the sinc-like envelope, while the external electrical boundary condition determines the low-pass pole.

Scharfetter-Gummel discretisation and numerical stability.

The finite-element discretisation of eqs. (1.32) and (1.33) on a non-uniform mesh faces a well-known numerical instability when the local Péclet number $\text{Pe} = q|\mathbf{E}|\Delta x/(2kT)$ exceeds unity. The Scharfetter-Gummel scheme approximates the current between adjacent mesh nodes i and $i + 1$ as:

$$J_{n,i+1/2} = \frac{qD_n}{\Delta x} \left[B\left(\frac{q\Delta\phi}{kT}\right) n_{i+1} - B\left(-\frac{q\Delta\phi}{kT}\right) n_i \right], \quad (1.47)$$

where $B(u) = u/(e^u - 1)$ is the Bernoulli function and $\Delta\phi = \phi_{i+1} - \phi_i$. This scheme recovers centred differences for $|u| \ll 1$ (low-field) and upstream weighting for $|u| \gg 1$ (high-field), providing second-order accuracy and unconditional positivity across the 30 kV cm^{-1} field regime of the present device.

1.2.4 The PIN Junction: Electrostatics and Operating Regime

The vertical PIN architecture—a lightly doped or nominally intrinsic germanium absorber sandwiched between heavily doped p^{++} and n^{++} contact layers—is the canonical structure for high-speed photodetection. Its central merit is the ability to sustain a large, spatially uniform electric field across the entire absorber volume under modest reverse bias, simultaneously achieving full depletion, velocity-saturated carrier transport, and minimal free-carrier absorption.

Depletion Approximation and Built-in Potential

In the absence of applied bias, the Fermi levels of the p^{++} and n^{++} regions equilibrate by charge redistribution, establishing a built-in potential:

$$V_{\text{bi}} = \frac{k_B T}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right), \quad (1.48)$$

where N_A and N_D are the acceptor and donor concentrations on the p and n sides respectively, and n_i is the intrinsic carrier concentration of germanium. At 300 K, $n_i^{\text{Ge}} \approx 2.4 \times 10^{13} \text{ cm}^{-3}$, approximately three orders of magnitude larger than in silicon, which has important consequences for dark current as discussed in section 1.2.5. For typical contact doping levels of $N_{A,D} \approx 10^{19} \text{ cm}^{-3}$, eq. (1.48) gives $V_{\text{bi}} \approx 0.51 \text{ V}$.

Under the depletion approximation, the space charge in the p^{++} and n^{++} layers is assumed to be that of fully ionised impurities with sharp depletion edges. In the ideal case where the intrinsic layer is fully depleted at zero bias by the built-in field, the charge neutrality condition reads:

$$N_A x_p = N_D x_n, \quad (1.49)$$

where x_p and x_n are the partial penetrations of the depletion region into the p^{++} and n^{++} layers, respectively. Since $N_{A,D} \gg n_i$, the depletion extension into the heavily doped contact layers is negligible ($x_p, x_n \ll W_i$), and the electric field profile is approximately uniform across the intrinsic layer with peak value:

$$\mathcal{E}_{\text{max}} = \frac{q N_A x_p}{\varepsilon_0 \varepsilon_{\text{Ge}}} \approx \frac{V_{\text{bi}} + |V_{\text{bias}}|}{W_i}, \quad (1.50)$$

where the second equality applies in the fully depleted regime. For $V_{\text{bias}} = -2 \text{ V}$ and $W_i = 1 \mu\text{m}$, eq. (1.50) gives $\mathcal{E}_{\text{max}} \approx 25 \text{ kV cm}^{-1}$, confirming velocity saturation. The flatness of the field profile is a key advantage of the PIN geometry over a simple pn junction: it ensures that every photogenerated carrier, regardless of its generation depth within $[0, W_i]$, experiences the same drift force and therefore the same transit time, making eq. (1.41) well-conditioned for all z .

Dark Current Mechanisms and Decomposition

The dark current I_d under reverse bias is the primary figure of merit for detector noise and static power dissipation. In a Ge-on-Si PIN device, it receives contributions from both the junction bulk volume and the mesa perimeter, and it is important to distinguish these because they scale differently with device geometry:

$$I_d = J_{\text{bulk}} A_j + J_{\text{surf}} P_j, \quad (1.51)$$

where J_{bulk} is the bulk dark-current density, J_{surf} is an effective surface dark-current line density, and P_j is the mesa perimeter [1]. The bulk component in the fully depleted intrinsic region is dominated by SRH thermal generation (as established in section 1.2.5) and is given by:

$$J_{\text{bulk}} = q U_{\text{SRH,dep}} W_i = \frac{q n_i W_i}{2\tau_0}, \quad (1.52)$$

which scales linearly with W_i and inversely with the effective SRH lifetime τ_0 . The surface component arises from interface states at the Ge/SiO₂ sidewall and Ge/Si heterointerface and is written as:

$$J_{\text{surf}} = q v_s n_i, \quad (1.53)$$

where v_s is the effective surface recombination velocity. Because n_i of germanium is orders of magnitude larger than that of silicon, both J_{bulk} and J_{surf} in Ge-on-Si are intrinsically larger than in silicon-only detectors by the same factor, placing a fundamental material constraint on achievable dark current density. As device dimensions are scaled to improve bandwidth—reducing A_j and P_j together but with $A_j/P_j \propto W_j$ decreasing—the surface term becomes proportionally more important, and surface passivation through dielectric encapsulants such as Al₂O₃ or GeO_x becomes critical. The decomposition of eq. (1.51) is exploited in the calibration procedure of section 1.8.2, where measurements on devices of varying junction width W_j are used to extract J_{bulk} and J_{surf} independently.

1.2.5 Recombination and Loss Mechanisms

Three recombination mechanisms are quantitatively relevant in the germanium absorber [1], [5].

Shockley-Read-Hall (SRH) recombination, mediated by deep-level trap states, is [4]:

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad (1.54)$$

where the effective trap-state carrier densities are:

$$n_1 = n_i \exp\left[\frac{E_t - E_i}{k_B T}\right], \quad p_1 = n_i \exp\left[\frac{E_i - E_t}{k_B T}\right]. \quad (1.55)$$

Fossum doping-dependent SRH lifetime.

In the presence of a background doping $N_A + N_D$, the SRH carrier lifetime is not constant but falls with increasing impurity concentration. The Fossum model [4] captures this dependence as:

$$\tau_{\text{SRH},n/p} = \frac{\tau_{\text{SRH},n/p}^{(0)}}{a_1 + a_2 \mathcal{N}_{n/p} + a_3 \mathcal{N}_{n/p}^{a_4}}, \quad \mathcal{N}_{n/p} = \frac{N_A + N_D}{N_{n/p}^{(\text{ref})}}, \quad (1.56)$$

where $\tau_{\text{SRH},n/p}^{(0)}$ is the low-concentration lifetime and $a_1, \dots, a_4$ are fitting parameters. In the limit $a_2 = 0$, $a_1 = a_3 = a_4 = 1$, eq. (1.56) reduces to the standard Shockley-Read-Hall result $\tau = \tau^{(0)} / (1 + N/N^{(\text{ref})})$. For the heavily doped n^{++} -Ge cathode at $N_D = 10^{19} \text{ cm}^{-3}$, eq. (1.56) gives a lifetime reduction of approximately one order of magnitude relative to $\tau^{(0)}$, confirming that carrier recombination is concentrated in the contact region rather than the intrinsic absorber. Within the fully depleted intrinsic region at the midgap trap condition $E_t = E_i$, eq. (1.54) reduces to:

$$U_{\text{SRH,dep}} \approx \frac{n_i}{\tau_n + \tau_p} \equiv \frac{n_i}{2\tau_0}, \quad (1.57)$$

where $\tau_0 = (\tau_n + \tau_p)/2$. SRH is the dominant mechanism in Ge-on-Si because the threading dislocations formed during heteroepitaxial growth act as efficient midgap trap centres.

Auger recombination is [4]:

$$U_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2), \quad (1.58)$$

where C_n and C_p are the electron and hole Auger capture coefficients. The combined effective carrier lifetime accounting for both SRH and Auger mechanisms is:

$$\frac{1}{\tau_{n/p}} = \frac{1}{\tau_{\text{Auger},n/p}} + \frac{1}{\tau_{\text{SRH},n/p}}, \quad (1.59)$$

where $\tau_{\text{Auger},n/p}$ is derived from eq. (1.58) evaluated at the local carrier densities. Equation (1.59) defines the effective recombination time used in the CHARGE simulation of section 1.8.2.

Radiative recombination is modelled as:

$$U_{\text{rad}} = r_{\text{opt}}(np - n_i^2), \quad (1.60)$$

where $r_{\text{opt}}^{\text{Ge}} \approx 6.4 \times 10^{-14} \text{ cm}^3 \text{ s}^{-1}$ at 300 K, approximately four orders of magnitude smaller than for a direct-gap III-V of comparable bandgap. In the fully depleted intrinsic region, where $np \approx n_i^2$, the radiative rate vanishes identically and eq. (1.57) constitutes the sole generation mechanism. Simulated generation and recombination rates confirm that SRH exceeds Auger by more than two orders of magnitude and radiative recombination by more

than five orders of magnitude across $0 \leq |V_{\text{bias}}| \leq 3 \text{ V}$. This justifies the single-parameter calibration strategy of section 1.8.2, in which only τ_0 is adjusted to reproduce I_d .

Surface recombination at the Ge/SiO₂ and Ge/Si interfaces constitutes a separate loss channel characterised by the surface recombination velocity v_s . Its contribution to the dark current is proportional to the mesa perimeter P_j rather than the junction area A_j , as formalised in eqs. (1.51) and (1.53). As device dimensions shrink to improve bandwidth, the area-to-perimeter ratio A_j/P_j decreases and surface recombination becomes increasingly significant, setting a practical lower bound on the achievable dark current density that surface passivation, for example through Ge_xO_y or Al₂O₃, must address.

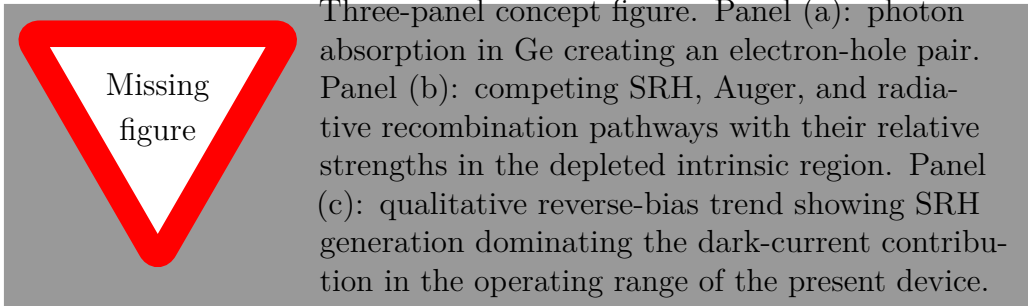


Figure 1.4: Carrier generation and recombination picture used throughout the electrical model. In the fully depleted intrinsic Ge layer, SRH generation remains the dominant dark-current mechanism, which justifies calibrating the device primarily through the lifetime parameter τ_0 .

1.2.6 Design Synthesis: Linking Physics to Device Targets

The preceding sections have developed the physical foundations of photodetection from first principles, and it is instructive at this point to consolidate the key parameter interdependencies into a coherent engineering picture before proceeding to the numerical device models of subsequent chapters.

The absorber-length/bandwidth Pareto front.

For a waveguide-integrated detector, optical absorption and bandwidth are in direct tension through the absorber length L . Responsivity benefits from long L via $\mathcal{A}(L) = 1 - e^{-\alpha L}$, while bandwidth degrades primarily through the junction capacitance $C_j \propto A_j = W_j \cdot L$, which enters the RC bandwidth eq. (1.20). Differentiating f_{3dB} with respect to L at fixed W_i and W_j yields the responsivity-bandwidth product as the relevant composite figure of merit:

$$\mathcal{R} \cdot f_{\text{3dB}} \approx \frac{q\lambda}{hc} (1 - e^{-\alpha L}) \cdot \left[\frac{1}{f_{\text{tt}}^2} + \frac{(2\pi R_{\text{tot}} \varepsilon_0 \varepsilon_{\text{Ge}} W_j L)^2}{W_i^2} \right]^{-1/2}, \quad (1.61)$$

which is maximised by selecting L such that the RC pole is pushed to roughly the transit-time pole, i.e. $f_{\text{RC}} \approx f_{\text{tt}}$. For the 400GBASE-DR4 target requiring a per-lane bandwidth

of 53 GHz at PAM4 signalling, and a responsivity floor of $\mathcal{R} \geq 0.8 \text{ A W}^{-1}$, eq. (1.61) constrains the design space to approximately $L \in [10 \mu\text{m}, 20 \mu\text{m}]$ and $W_i \in [0.5 \mu\text{m}, 1.0 \mu\text{m}]$ for $W_j = 1 \mu\text{m}$ in the vertical PIN geometry.

Bias voltage and depletion completeness.

The applied reverse bias V_{bias} must be large enough to fully deplete the intrinsic layer and drive the carriers into velocity saturation. The full-depletion voltage in the ideal PIN approximation is [1]:

$$V_{\text{FD}} = \frac{q N_{\text{bg}} W_i^2}{2\epsilon_0\epsilon_{\text{Ge}}} - V_{\text{bi}}, \quad (1.62)$$

where N_{bg} is the unintentional background doping of the nominally intrinsic layer, which is never zero in practice due to incorporation of boron or phosphorus during epitaxial growth. For $N_{\text{bg}} \approx 10^{16} \text{ cm}^{-3}$ and $W_i = 1 \mu\text{m}$, eq. (1.62) gives $V_{\text{FD}} \approx 0.27 \text{ V}$, well below the standard operating voltage of 1 V, confirming that full depletion is achieved at low bias for realistic material quality. Above V_{FD} , the capacitance becomes approximately voltage-independent at the geometric value of eq. (1.21), which simplifies RC bandwidth analysis. Operation below V_{FD} should be avoided as it introduces a bias-dependent capacitance and incomplete carrier collection, degrading both bandwidth and responsivity in a non-linear, difficult-to-model manner.

Temperature dependence and thermal stability.

The intrinsic carrier concentration of germanium follows $n_i \propto T^{3/2} \exp(-E_g/2k_B T)$, where $E_g(T)$ itself decreases with temperature according to the Varshni relation. Because n_i^{Ge} at room temperature is already large, even modest self-heating in a high-power optical link can substantially elevate the dark current through eq. (1.52), degrading the SNR floor of eq. (1.27). In a 400 Gbit s⁻¹ DR4 transceiver module, thermal management of the receiver die is therefore a system-level co-design requirement, not merely a device-level footnote. The temperature coefficient of I_d in Ge-on-Si is typically measured to be in the range 5 % K⁻¹ to 8 % K⁻¹ near room temperature, significantly steeper than silicon photodiodes, and this sensitivity motivates both the careful thermal modelling performed alongside the electrical characterisation of section 1.8.2 and the use of bias-point tuning to compensate for operating-temperature drift in deployed link hardware.

1.3 Performance Metrics and Design Trade-offs

The transport and absorption physics of section 1.2 support a set of engineering metrics that define what a useful photodetector must accomplish and at what cost. These metrics are not independent: they form a multi-dimensional trade-off space in which improving one quantity typically degrades another. Every architectural and material choice in the sections that follow is justifiable using the definitions established here.

1.3.1 Responsivity and Quantum Efficiency

Physical Basis and Definition

The fundamental function of a photodetector is to transduce an optical power flux into a measurable electrical current. This transduction is characterised by the *external responsivity* $\mathcal{R}$, defined as the ratio of the generated photocurrent I_{ph} to the incident optical power P_{opt} :

$$\mathcal{R} \equiv \frac{I_{\text{ph}}}{P_{\text{opt}}} \quad [\text{A W}^{-1}]. \quad (1.63)$$

The qualifier *external* is important: $\mathcal{R}$ accounts for all loss processes that occur between the guided optical mode incident upon the device and the photocurrent delivered to the readout circuit. It therefore includes coupling loss from the silicon waveguide into the germanium absorber, reflection at the Si/Ge interface, incomplete absorption along the device length, and any carrier recombination prior to collection at the contacts.

The upper bound on responsivity is derived from energy conservation. If every incident photon of energy $h\nu = hc/\lambda$ were to generate exactly one electron–hole pair and that pair were collected without loss, then the photocurrent per unit optical power would be:

$$\mathcal{R}_{\text{max}} = \frac{q\lambda}{hc} \approx 1.057 \text{ A W}^{-1} \quad \text{at } 1310 \text{ nm}, \quad (1.64)$$

where q is the elementary charge, h is Planck’s constant, and c is the speed of light in vacuum. The linear dependence of $\mathcal{R}_{\text{max}}$ on wavelength λ is physically transparent: longer-wavelength photons carry less energy per quantum, so generating one electron of charge q requires less power, yielding a larger current per watt. At the O-band centre wavelength of 1310 nm, this ceiling is approximately 1.06 A W^{-1} . Real devices fall meaningfully below this value, and the gap between $\mathcal{R}$ and $\mathcal{R}_{\text{max}}$ is the principal object of optimisation in the optical coupling and absorber geometry.

Quantum Efficiency: External and Internal

The responsivity can be normalised to the single-photon limit through the concept of quantum efficiency. The *external quantum efficiency* (EQE) η_{ext} is the ratio of the rate of collected electrons to the rate of photons incident on the device, and is directly related to responsivity by:

$$\eta_{\text{ext}} = \mathcal{R} \cdot \frac{hc}{q\lambda}. \quad (1.65)$$

This quantity captures all loss mechanisms between the input waveguide facet and the electrical output terminal. It is the primary figure of merit at the system level because it directly sets the signal current available to the subsequent transimpedance amplifier.

The *internal quantum efficiency* (IQE) η_{int} isolates the carrier collection process from the optical coupling and absorption steps:

$$\eta_{\text{int}} = \frac{I_{\text{ph}}/q}{P_{\text{abs}}/\hbar\omega}, \quad (1.66)$$

where P_{abs} is the optical power actually absorbed within the Ge layer and $\hbar\omega$ is the photon energy. A device with high EQE may still exhibit a depressed IQE if significant absorption occurs in passive regions from which carriers cannot be extracted, or conversely may have high IQE but low EQE due to poor optical coupling. Distinguishing the two is important for correctly attributing performance shortfalls to either optical design (coupling and absorption) or electrical design (carrier transport and collection).

In the fully depleted intrinsic Ge layer under reverse bias, the electric field exceeds the carrier velocity saturation threshold throughout the absorber volume. The resulting drift transit time $\tau_{\text{tr}} = W_i/v_{\text{d,sat},h}$, governed by the slower hole saturation velocity $v_{\text{d,sat},h}$, is typically tens of picoseconds for intrinsic layer thicknesses W_i in the sub-500 nm regime. Because this is many orders of magnitude shorter than the Shockley–Read–Hall (SRH) minority-carrier lifetime $\tau_0 \sim 1$ ns in device-grade epitaxial Ge, virtually all photo-generated carriers are swept across the junction before recombination can occur. The resulting carrier collection efficiency is:

$$\xi \approx 1 - \frac{\tau_{\text{tr}}}{\tau_0} \approx 1, \quad (1.67)$$

confirming that the IQE is essentially unity in a properly depleted device, provided the electric field is sufficiently uniform and the surface recombination velocity at the Ge/Si heterojunction is kept low by careful interface passivation.

The EQE Chain and Its Design Implications

The connection between the physical loss mechanisms and the measurable EQE is made precise by the following chain product:

$$\eta_{\text{ext}} = (1 - R_{\text{refl}}) \eta_c \xi (1 - e^{-\alpha(\lambda)L}), \quad (1.68)$$

where each factor corresponds to a distinct physical stage. The term $(1 - R_{\text{refl}})$ accounts for Fresnel reflection at the Si/Ge entry facet, arising from the refractive index contrast ($n_{\text{Si}} \approx 3.48$, $n_{\text{Ge}} \approx 4.27$ at 1310 nm); this factor is typically in the range 0.97–0.99 for waveguide-integrated devices where the optical mode is guided through a tapered transition rather than impinging on an abrupt cleaved facet. The coupling efficiency η_c quantifies the spatial overlap between the mode profile of the silicon access waveguide and the fundamental mode supported by the Ge absorber rib; its value is sensitive to cross-sectional geometry, waveguide width mismatch, and the design of any intermediate taper structure. The collection efficiency ξ has already been established as approximately unity under full depletion. Finally, the factor $(1 - e^{-\alpha(\lambda)L})$ is the fraction of the guided optical power absorbed over a propagation length L through a medium with power absorption coefficient $\alpha(\lambda)$;

at 1310 nm, the absorption coefficient of Ge is $\alpha \approx 3 \times 10^3 \text{ cm}^{-1}$, yielding $1/e$ absorption lengths of approximately $3 \mu\text{m}$. For a $10 \mu\text{m}$ -long device, the single-pass absorption factor approaches 0.95, indicating that a compact absorber can achieve near-complete photon collection without the need for cavity enhancement.

Understanding eq. (1.68) as a product of separable factors is strategically important: it allows independent optimisation of the coupling optics, the absorber geometry, and the carrier transport, without conflating distinct physical mechanisms. It also reveals which term dominates the EQE deficit in any particular design iteration, enabling targeted improvements. For the waveguide-integrated, vertical PIN architecture considered throughout this work, the dominant EQE loss is typically associated with $\eta_c < 1$ arising from mode mismatch at the Si-to-Ge transition, rather than from incomplete absorption or finite carrier lifetime.

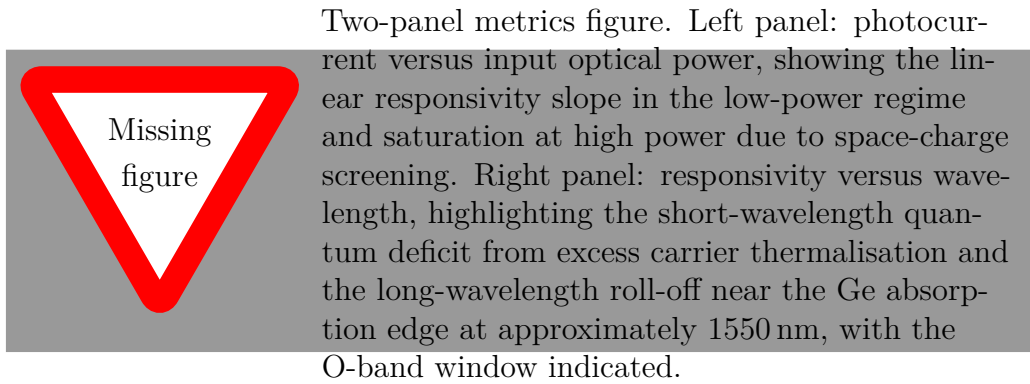


Figure 1.5: Responsivity viewed in both its power-domain and spectral forms. The low-power slope of the I - P curve defines $\mathcal{R}$, while the wavelength dependence determines the useful operating window around the O-band centre at 1310 nm.

1.3.2 Bandwidth: Transit-Time and RC Limits

Overview of Bandwidth-Limiting Mechanisms

The ability of a photodetector to faithfully follow a modulated optical signal at high frequencies is characterised by its 3 dB electrical bandwidth $f_{3\text{dB}}$. In a reverse-biased PIN photodiode, this bandwidth is fundamentally limited by two physically distinct and largely independent mechanisms: the finite time required for photo-generated carriers to traverse the depletion region (the *transit-time* limit), and the charging time of the junction capacitance through the total series resistance (the RC limit). Both mechanisms produce a roll-off in the photocurrent frequency response, and in the absence of deliberate inductive resonance peaking, the combined bandwidth is well described by the additive-inverse-quadrature expression [6]:

$$f_{3\text{dB}} = \left(\frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2} \right)^{-1/2}. \quad (1.69)$$

This form, which adds the power spectral contributions of two first-order-like roll-offs in quadrature, correctly reflects the fact that both mechanisms must be simultaneously satisfied: a device with $f_{\text{tt}} \gg f_{\text{RC}}$ is RC-limited, while one with $f_{\text{RC}} \gg f_{\text{tt}}$ is transit-time-limited. The engineering challenge is to design the geometry such that both limits coincide, maximising the overall bandwidth for a given device footprint.

Transit-time limit.

When an electron–hole pair is photo-generated within the depleted intrinsic layer, both carriers begin drifting in opposite directions under the action of the built-in and externally applied electric field. The photocurrent induced in the external circuit is not instantaneous but persists until all carriers have traversed the junction and been collected at their respective ohmic contacts. For a spatially uniform generation profile and carriers moving at their saturation velocities, the current impulse response is approximately rectangular in time. The Fourier transform of a rectangular pulse of duration τ_{tr} is a sinc function, whose first zero occurs at $f = 1/\tau_{\text{tr}}$, giving a 3 dB frequency:

$$f_{\text{tt}} = \frac{0.45 v_{\text{d,sat},h}}{W_i}, \quad (1.70)$$

where $v_{\text{d,sat},h} \approx 4 \times 10^6 \text{ cm s}^{-1}$ is the hole saturation velocity in Ge, which is the lower of the two carrier saturation velocities ($v_{\text{d,sat},e} \approx 6 \times 10^6 \text{ cm s}^{-1}$), and W_i is the intrinsic layer thickness. The numerical prefactor 0.45 accounts for the asymmetric bipolar velocity ratio $v_{\text{d,sat},e}/v_{\text{d,sat},h} \approx 1.5$: because electrons transit the junction approximately 50% faster than holes, the effective current impulse is not a perfect rectangle, and the actual sinc-like roll-off shifts the 3 dB point relative to the symmetric-carrier prediction.

It is instructive to evaluate eq. (1.19) for the intrinsic layer thickness $W_i = 350 \text{ nm}$ adopted in this work:

$$f_{\text{tt}} = \frac{0.45 \times 4 \times 10^6 \text{ cm s}^{-1}}{350 \times 10^{-7} \text{ cm}} \approx 51 \text{ GHz}. \quad (1.71)$$

This value is a conservative lower bound because eq. (1.19) is derived from the worst-case single-carrier response governed by the slower holes. In practice, because electrons and holes simultaneously contribute to the terminal current via their respective Ramo–Shockley induced charges, the two-carrier response is broader than the single-carrier prediction by a factor that depends on the velocity ratio. Furthermore, experimental devices benefit from self-resonance due to the parasitic inductance of the on-electrode metal interconnect L_p , which introduces a peaking effect that can extend the usable bandwidth substantially beyond the sinc-function roll-off. Accounting for both effects, the benchmark-calibrated transport ceiling for a $W_i = 350 \text{ nm}$ device is in the 110 GHz to 120 GHz class, confirming that transit time is not the operative bandwidth limitation at the target operating rate of 100 GHz.

RC limit.

The junction capacitance C_j of the fully depleted intrinsic region is that of a parallel-plate capacitor with plate separation W_i and area $A_j = W_m \times L$, where W_m is the mesa width and L is the device length along the waveguide axis:

$$C_j = \frac{\varepsilon_0 \varepsilon_{\text{Ge}} A_j}{W_i}, \quad (1.72)$$

where $\varepsilon_{\text{Ge}} \approx 16.2$ is the relative permittivity of germanium. The physical interpretation is immediate: a thinner depletion region brings the charge sheets of the p -type and n -type contacts closer together, increasing C_j , while a larger junction area increases the total stored charge per unit voltage. Both effects are detrimental to bandwidth.

The RC-limited 3 dB bandwidth is determined by the RC time constant of the junction capacitance discharging through the total series resistance $R_{\text{tot}} = R_s + R_L$, where R_s is the parasitic series resistance of the contacts and access electrodes, and $R_L = 50 \Omega$ is the termination resistance of the external readout circuit:

$$f_{\text{RC}} = \frac{1}{2\pi R_{\text{tot}} C_j}. \quad (1.73)$$

In the vertical n - i - p PIN geometry, the dominant contribution to R_s arises from the lateral current path within the p^{++} -doped silicon platform layer beneath the Ge mesa. Carriers generated within the Ge absorber are collected through the Si layer, which has a finite sheet resistance determined by its thickness $t_{\text{Si}} = 220 \text{ nm}$ and the anode contact geometry:

$$R_{s,\text{anode}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{\ell_a}{w_a}, \quad (1.74)$$

where ℓ_a and w_a are the effective current path length and width through the platform, respectively. This resistance can be reduced by placing the anode contacts as close as possible to the base of the Ge mesa, which is a critical layout constraint that directly couples electrical performance to lithographic resolution. The RC time constant is:

$$\tau_{\text{RC}} = R_{\text{tot}} C_j, \quad (1.75)$$

from which $f_{\text{RC}} = (2\pi\tau_{\text{RC}})^{-1}$.

Optimal Intrinsic-Layer Thickness

The most powerful design insight that follows from eqs. (1.19) and (1.20) is that the transit-time and RC bandwidths respond oppositely to changes in W_i : increasing W_i reduces f_{tt} (longer transit path) but increases f_{RC} (lower capacitance per unit area). This suggests the existence of an optimal thickness W_i^* at which the two limits are balanced and the total

bandwidth $f_{3\text{dB}}$ is maximised. Setting $f_{\text{tt}}(W_i^*) = f_{\text{RC}}(W_i^*)$ and substituting eqs. (1.19), (1.20) and (1.72) yields:

$$W_i^* = \sqrt{0.9\pi R_{\text{tot}} \varepsilon_0 \varepsilon_{\text{Ge}} v_{\text{d,sat},h} A_j}. \quad (1.76)$$

Substituting the device parameters $R_{\text{tot}} = 60\ \Omega$, $\varepsilon_{\text{Ge}} = 16.2$, $v_{\text{d,sat},h} = 4 \times 10^6\ \text{cm s}^{-1}$, and $A_j = 6\ \mu\text{m}^2$ gives $W_i^* \approx 310\ \text{nm}$, which is within 11 % of the chosen $W_i = 350\ \text{nm}$, confirming that the device is positioned close to the bandwidth optimum. The modest upward departure from W_i^* is deliberate: it marginally reduces C_j and the associated RC penalty while accepting a small transit-time degradation, improving manufacturing tolerance for process variations in the epitaxial layer thickness.

The expression eq. (1.76) reveals a fundamental scaling law of considerable practical importance. Because $W_i^* \propto \sqrt{A_j}$, the optimal intrinsic thickness scales as the square root of the junction area. Consequently, shrinking the junction area by a factor of four (e.g., from $A_j = 6\ \mu\text{m}^2$ to $1.5\ \mu\text{m}^2$) allows W_i^* to be halved, which simultaneously raises f_{tt} and f_{RC} at the new balanced optimum by approximately the same factor. This scaling analysis underpins the miniaturisation strategy adopted in the next-generation device designs discussed in section 1.7: reducing the device footprint, rather than simply increasing the reverse-bias voltage, is the primary lever for pushing integrated Ge-on-Si photodetectors beyond the 100 GHz bandwidth regime while maintaining the responsivity needed for 400 Gbit/s data-centre interconnect applications.

Small-Signal Transfer Function and Inductive Peaking

The photocurrent frequency response of the photodetector can be compactly described by a factorised small-signal transfer function. The transit-time contribution takes the form of a sinc function inherited from the rectangular current impulse response, and the RC contribution is a single-pole low-pass filter:

$$H_{\text{PD}}(f) = \text{sinc}\left(\frac{f}{f_{\text{tt}}}\right) \cdot \frac{1}{1 + i f / f_{\text{RC}}}, \quad (1.77)$$

where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. This simplified two-pole model is adequate for rapid design screening and for gaining physical intuition, but it neglects the reactive parasitics that are unavoidable in a realistic device layout.

In a fully parameterised electrical model of the packaged photodetector, the reverse-biased junction presents an impedance $Z_j(\omega) \approx (j\omega C_j)^{-1}$, while the series branch carries the ohmic resistance R_s and the parasitic inductance L_p of the metal electrode interconnect:

$$Z_s(\omega) = R_s + j\omega L_p. \quad (1.78)$$

The output is terminated by the load $Z_L(\omega) = (R_L^{-1} + j\omega C_p)^{-1}$, where C_p is the parasitic pad capacitance. Current division between these impedances yields the peaked electrical

transimpedance:

$$H_{\text{elec,pk}}(\omega) = \frac{Z_j(\omega) Z_L(\omega)}{Z_j(\omega) + Z_s(\omega) + Z_L(\omega)}. \quad (1.79)$$

The full benchmark-calibrated detector response, combining the transit-time and electrical sub-network contributions, is:

$$H_{\text{PD,pk}}(\omega) = H_{\text{tt}}(\omega) H_{\text{elec,pk}}(\omega), \quad (1.80)$$

where $H_{\text{tt}}(\omega)$ is the Fourier transform of the two-carrier impulse current response. The physically important feature of eq. (1.79) is that the series inductance L_p resonates with C_j at a frequency $\omega_0 = 1/\sqrt{L_p C_j}$, producing a local maximum in the transimpedance magnitude at frequencies near ω_0 . This so-called *inductive peaking* extends the usable 3 dB bandwidth beyond the RC-only prediction of eq. (1.20) without requiring any reduction in C_j or R_s , and explains why experimentally measured bandwidths routinely exceed the estimates based on the simplified two-pole model. Equation eq. (1.77) is employed for geometry screening throughout this chapter; eq. (1.80) is reserved for validated projections into the 100 GHz-class operating regime in section 1.7.

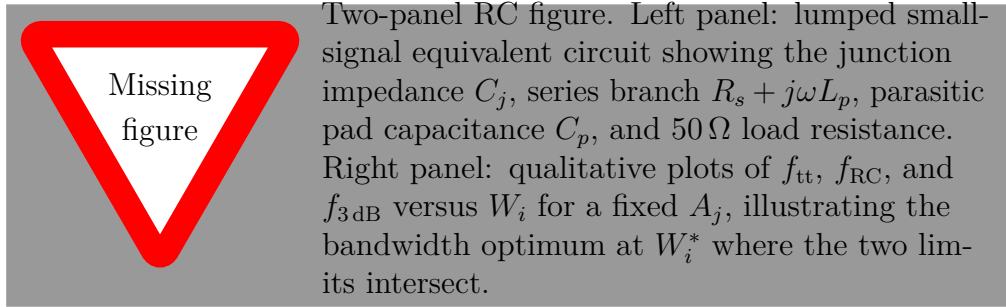


Figure 1.6: Capacitance-resistance interpretation of the vertical PIN photodetector. The bandwidth optimum W_i^* arises from the competing dependences of transit-time and RC bandwidths on intrinsic-layer thickness, and its location shifts with junction area through the scaling law $W_i^* \propto \sqrt{A_j}$.

1.3.3 Noise, Sensitivity, and Figures of Merit

Dark Current

Even in the complete absence of optical excitation, a reverse-biased PIN junction sustains a non-zero current, referred to as the *dark current* I_d . Far from being a mere background offset that can be electronically subtracted, the dark current is a primary noise source: because current flow involves the random transit of discrete charge carriers, it inevitably generates shot noise, as formalised in section 1.3.3. The dark current therefore sets the fundamental noise floor of the detector and, through the noise-equivalent power, determines the minimum detectable signal level.

Three generation-recombination mechanisms contribute to I_d in a reverse-biased Ge-on-Si PIN [1]. The first and generally dominant mechanism at the device geometry considered here is *thermal generation via SRH traps* within the depleted intrinsic layer. In a reverse-biased junction, the carrier concentrations are below their equilibrium values, and mid-gap traps act predominantly as generation centres, thermally exciting carriers from the valence band to the conduction band through an intermediate trap state. The generation rate derived from the SRH formalism yields a current density:

$$J_{\text{SRH}} = \frac{q n_i W_i}{2\tau_0}, \quad (1.81)$$

where n_i is the intrinsic carrier concentration of Ge and τ_0 is the effective SRH carrier lifetime, which is related to the trap density and capture cross-sections. Because n_i in Ge is approximately $2.3 \times 10^{13} \text{ cm}^{-3}$ at room temperature—nearly three orders of magnitude larger than in Si—the SRH generation rate, and hence the dark current, is substantially higher in Ge-based devices. This is a fundamental material limitation, not a fabrication artefact, and it sets a floor on the achievable dark current density for any Ge-on-Si photodetector at room temperature. The linear proportionality to W_i in eq. (1.114) is another manifestation of the competing trade-offs encountered throughout this section: thicker intrinsic layers improve responsivity but simultaneously increase the dark current and its associated noise contribution.

The second mechanism is *minority-carrier diffusion* from the quasi-neutral p -type and n -type contact regions flanking the depletion layer. In these neutral regions, thermally generated minority carriers diffuse toward the high-field depletion edge, where they are swept across by the electric field. The diffusion current density from the n -type and p -type contact regions is:

$$J_{\text{diff}} = \frac{q D_p n_i^2}{L_p N_D} + \frac{q D_n n_i^2}{L_n N_A}, \quad (1.82)$$

where D_p and D_n are the minority-carrier diffusion coefficients, $L_p = \sqrt{D_p \tau_p}$ and $L_n = \sqrt{D_n \tau_n}$ are the minority-carrier diffusion lengths, and N_D and N_A are the donor and acceptor concentrations in the respective contact regions. In a well-designed device with sufficiently heavy doping in the contact regions, J_{diff} is suppressed below J_{SRH} and is a secondary contribution to the total dark current.

The third mechanism, which becomes progressively more important as the device is scaled to smaller dimensions, is *interface recombination* (or, equivalently, surface generation) at the Ge/Si heterojunction. The crystallographic mismatch between Ge and Si results in a high density of interface states that can act as generation centres with an effective surface recombination velocity v_s . The associated current density is:

$$J_{\text{surf}} \approx \frac{q n_i v_s}{2}. \quad (1.83)$$

Combining all three contributions, and distinguishing between the area-scaled bulk terms and the perimeter-scaled surface term, the total dark current is:

$$I_d = A_j (J_{\text{SRH}} + J_{\text{diff}}) + P_j \frac{q n_i v_s}{2}, \quad (1.84)$$

where $P_j = 2(W_m + L)$ is the perimeter of the Ge mesa junction. This decomposition into area-scaling and perimeter-scaling contributions has a direct and important consequence for device miniaturisation: as the junction area A_j is reduced—in pursuit of lower capacitance and higher bandwidth—the ratio A_j/P_j diminishes, and the fractional contribution of the surface term to I_d grows correspondingly. In the limit of a very small device, the perimeter term can come to dominate the total dark current, making the quality of the Ge/Si interface and the effectiveness of any surface passivation layer critically important. This represents a fundamental scaling tension in the co-optimisation of bandwidth and noise performance.

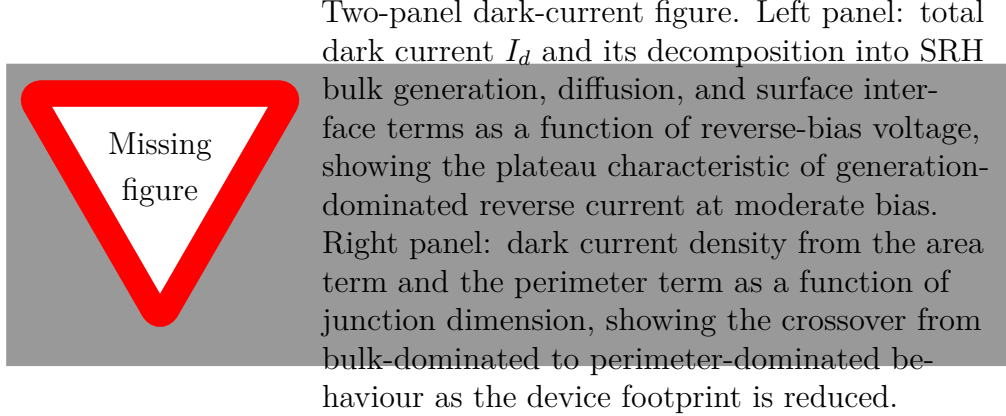


Figure 1.7: Dark-current mechanisms of the reverse-biased Ge-on-Si PIN. At the nominal device footprint, SRH bulk generation dominates; however, as the device is aggressively miniaturised toward sub-micrometre dimensions, the Ge/Si perimeter term becomes the controlling contribution, placing a premium on interface quality.

Noise Spectral Densities and Noise-Equivalent Power

Physical origin of shot noise.

The currents that flow in a photodetector—whether photo-generated or thermally generated—are not smooth, continuous fluid flows but are composed of discrete charge quanta of magnitude q arriving at the terminal at random times. For carrier arrivals that are statistically independent (i.e., Poissonian), the number of carriers N_T collected in an observation interval T satisfies:

$$\mathbb{E}[N_T] = \frac{IT}{q}, \quad \text{Var}[N_T] = \frac{IT}{q}, \quad (1.85)$$

where the mean and variance are equal, as is characteristic of the Poisson distribution. The collected charge $Q_T = qN_T$ therefore has variance:

$$\text{Var}[Q_T] = qIT, \quad (1.86)$$

and converting from charge variance to the variance of the time-averaged current $\bar{I}_T = Q_T/T$:

$$\text{Var}[\bar{I}_T] = \frac{qI}{T}. \quad (1.87)$$

Taking the limit $T \rightarrow \infty$ and equating the variance to the integral of the one-sided power spectral density (PSD) over the measurement bandwidth gives the standard Schottky result for shot noise:

$$S_{\text{shot}}(f) = 2q(I_d + I_{\text{ph}}) \quad [\text{A}^2 \text{Hz}^{-1}], \quad (1.88)$$

where the total current is the sum of dark and photocurrent contributions. The factor of two arises from the one-sided spectral density convention; the shot noise PSD is white (frequency-independent) because the Poissonian carrier arrival process has no intrinsic correlation time, and its spectral content is therefore flat across all frequencies of practical interest for photodetection.

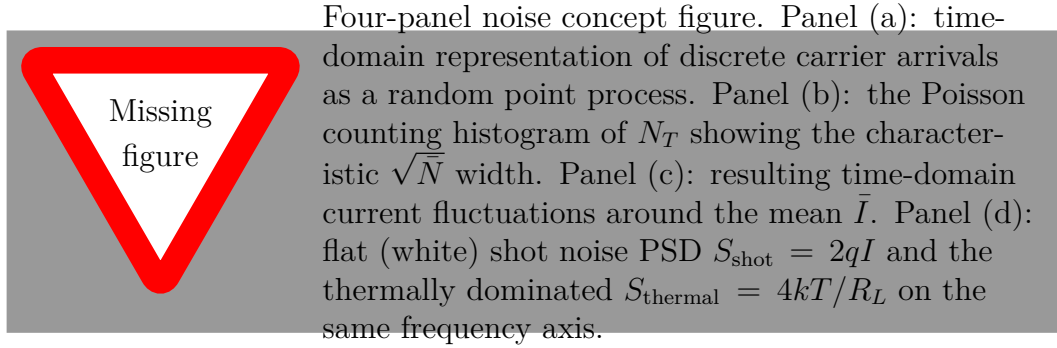


Figure 1.8: Physical origin of shot noise from carrier counting statistics. Shot noise is the continuum manifestation of the Poissonian fluctuations of discrete charge carriers arriving at the detector terminal; its flat spectral density reflects the absence of any intrinsic correlation time in the carrier arrival process.

Thermal noise.

Simultaneously, the load resistance R_L that terminates the detector generates Johnson–Nyquist thermal noise through the random thermal motion of conduction electrons. Its one-sided PSD is:

$$S_{\text{thermal}}(f) = \frac{4kT}{R_L} \quad [\text{A}^2 \text{Hz}^{-1}], \quad (1.89)$$

where k is the Boltzmann constant and T is the absolute temperature. Like shot noise, this is a white noise process. At room temperature and with $R_L = 50 \Omega$, the thermal noise current density is $\sqrt{4kT/R_L} \approx 18 \text{ pA}/\sqrt{\text{Hz}}$. A dark current of $I_d = 1 \text{ nA}$ produces a shot noise of $\sqrt{2qI_d} \approx 0.6 \text{ fA}/\sqrt{\text{Hz}}$, more than four orders of magnitude below the thermal noise floor. This confirms that for the device under study, at moderate reverse bias, the detector itself is not the dominant noise source; the 50Ω termination resistance—and by extension the input noise of the transimpedance amplifier—governs the overall sensitivity of the optical receiver front end.

Total noise and noise-equivalent power.

Integrating the noise PSD over the electrical detection bandwidth B_e gives the total noise variance at the receiver input:

$$\sigma_n^2 = (S_{\text{shot}} + S_{\text{thermal}}) B_e \quad [\text{A}^2]. \quad (1.90)$$

The *noise-equivalent power* (NEP) is defined as the minimum optical power required at the detector input to produce a signal current equal to the root-mean-square noise current, i.e., the signal-to-noise ratio of unity in a unit bandwidth:

$$\text{NEP} = \frac{\sqrt{S_{\text{shot}} + S_{\text{thermal}}}}{\mathcal{R}} \quad [\text{W}/\sqrt{\text{Hz}}]. \quad (1.91)$$

A lower NEP indicates a more sensitive detector. When the dark current is the dominant noise source (the dark-current-limited regime, relevant at high reverse bias or elevated temperature), the NEP is determined by the shot noise of I_d alone, and the *specific detectivity* D^* provides a geometry-normalised figure of merit that facilitates comparison across devices of different junction areas:

$$D_{\text{shot}}^* = \frac{\mathcal{R}\sqrt{A_j}}{\sqrt{2qI_d}} \quad [\text{cm}\sqrt{\text{Hz}}/\text{W}]. \quad (1.92)$$

The $\sqrt{A_j}$ normalisation reflects the fact that, for area-dominated dark current ($I_d \propto A_j$), the shot noise grows as $\sqrt{A_j}$, so a larger device is intrinsically noisier in proportion to its collection area.

Additional noise contributions.

Several secondary noise processes deserve mention even if they do not dominate in the tens-of-GHz band of interest. Generation-recombination noise from an ensemble of SRH traps with characteristic lifetime τ_{gr} has a Lorentzian PSD with a corner frequency $f_{\text{gr}} = 1/(2\pi\tau_{\text{gr}})$ that typically falls below 1 MHz in device-grade Ge; it is therefore negligible at high-speed modulation frequencies. Optical-source noise, characterised by the relative intensity noise (RIN) of the laser transmitter, enters as:

$$S_{\text{RIN}}(f) = \mathcal{R}^2 P_{\text{opt}}^2 \text{RIN} \quad [\text{A}^2 \text{Hz}^{-1}], \quad (1.93)$$

and is proportional to the *square* of the photocurrent. At the high optical power levels encountered in short-reach interconnects using directly-modulated lasers with $\text{RIN} \approx -140 \text{ dB/Hz}$, this contribution can become comparable to or exceed the shot-noise limit, shifting the system from a shot-noise-limited to a source-noise-limited regime. The distinction matters for link budget analysis but does not alter the intrinsic detector noise floor characterised by eqs. (1.88) and (1.89).

It is also important to clarify that passive optical loss mechanisms—including taper insertion loss, waveguide propagation loss, free-carrier absorption in doped contact layers, Fresnel

reflection, and incomplete carrier collection—do not generate new electrical noise PSDs. Instead, they reduce the useful signal $I_{\text{ph}} = \mathcal{R} P_{\text{opt}}$ while the noise floor remains governed by the remaining shot and thermal terms, thereby degrading the signal-to-noise ratio indirectly. The distinction between noise and loss is fundamental to correct link budget accounting.

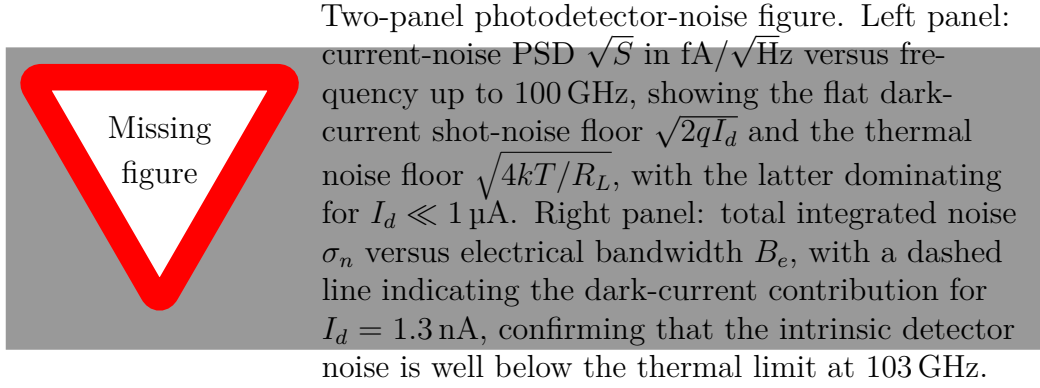


Figure 1.9: Noise decomposition of the reverse-biased Ge-on-Si PIN photodetector. For dark currents in the sub-nA regime and operation in a 50Ω system, the thermal noise of the load resistance dominates over the dark-current shot-noise contribution across the full O-band modulation bandwidth.

1.3.4 The Speed–Efficiency–Noise Trade-off Triangle

The performance metrics derived in the preceding subsections are not free parameters that may be independently optimised. They are coupled through the shared physical parameters of the device geometry—principally the intrinsic layer thickness W_i , the junction area A_j , and the applied reverse bias V_R —into a multi-dimensional constrained optimisation space. Understanding the structure of this space is prerequisite to the informed design choices made in section 1.7.

Responsivity–bandwidth trade-off.

The most fundamental tension is between responsivity and bandwidth, and it arises from two independent geometric mechanisms. First, consider the role of W_i : increasing the intrinsic layer thickness improves photon absorption (the factor $1 - e^{-\alpha W_i}$ in eq. (1.68) rises toward unity), thereby improving η_{ext} and $\mathcal{R}$. Simultaneously, the longer transit path through the thicker layer reduces f_{tt} via eq. (1.19). These competing dependences are mediated by the optimum thickness W_i^* of eq. (1.76): operating below W_i^* sacrifices both responsivity and bandwidth (transit-time limited, small f_{tt} and low absorption), while operating above W_i^* gains responsivity at the cost of RC bandwidth (higher C_j). Second, consider the role of the device length L : a longer device absorbs a larger fraction of the incident guided power, improving $\mathcal{R}$, but also increases $A_j = W_m \times L$, raising C_j via eq. (1.72) and degrading f_{RC} via eq. (1.20). There is therefore a maximum useful device length beyond which the RC bandwidth penalty outweighs the responsivity benefit, and this trade-off length also scales

with A_j through the same $\sqrt{A_j}$ dependence of W_i^* . The product $\mathcal{R} \times f_{3\text{dB}}$ is a useful scalar summary of this two-dimensional trade-off and is employed in the benchmarking analysis of section 1.6.4.

Bias voltage and the noise axis.

Reverse bias adds a third dimension to the trade-off space. Increasing V_R raises the electric field throughout the depletion region, driving both carrier species closer to their saturation velocities and reducing the transit time $\tau_{\text{tr}} = W_i/v_d(E)$. This improves f_{tt} and, for devices in which the field is initially insufficient for velocity saturation, can also increase ξ and hence η_{int} . The noise cost, however, is non-trivial: higher reverse bias drives greater band bending and widens any partially depleted regions, increasing the volume over which SRH generation occurs and thereby raising I_d through eq. (1.114). The resulting increase in shot noise degrades NEP via eq. (1.28). In the context of a data-centre optical link, this noise penalty must be weighed against the bandwidth benefit: for receivers operating at or near the sensitivity limit, the optimal bias point is the lowest voltage at which full velocity saturation is achieved throughout the intrinsic layer, since additional bias beyond this point increases noise without providing further bandwidth benefit.

Miniaturisation and the perimeter penalty.

Reducing the junction area A_j below the value required for absorption saturation trades responsivity for bandwidth, as discussed above. An additional subtlety, captured by eq. (1.51), is that miniaturisation also shifts the dark-current composition from bulk-dominated to perimeter-dominated. For the current device with $A_j = 6\text{ }\mu\text{m}^2$ and $P_j = 8\text{ }\mu\text{m}$, the area-to-perimeter ratio is $A_j/P_j = 0.75\text{ }\mu\text{m}$; reducing A_j by a factor of four while halving the perimeter reduces this ratio by a factor of two, potentially doubling the relative contribution of surface generation. Whether this constitutes an absolute increase in I_d depends on the relative magnitude of J_{SRH} versus the effective interface recombination velocity v_s , which in turn depends on the quality of the Ge/Si heterojunction and any passivation treatment. This linkage between geometry, process quality, and noise performance makes the design of sub-micrometre Ge-on-Si photodetectors for beyond-100 Gb/s applications a genuinely multi-physics optimisation problem, spanning optical, electrical, and materials dimensions simultaneously.

The design rationale developed in section 1.7 maps the full geometry-to-performance relationship quantitatively, identifies the Pareto frontier of the vertical PIN architecture with respect to the responsivity–bandwidth–noise triple trade-off, and positions the chosen device geometry within it with respect to the requirements of the 400GBASE-DR4 application target.

1.4 Photodetector Architectures

With the metrics of section 1.3 as the evaluation framework, the landscape of photodetector device classes can be surveyed systematically. Each architecture is assessed against responsivity, bandwidth, and noise, and each is found to resolve one specific limitation of its predecessor, until the waveguide-integrated PIN junction emerges not as a historical preference but as the consequence of removing every identified obstacle in sequence.

1.4.1 PN and PIN Junctions: Depletion Engineering

Photoconductive Detection.

The photoconductor is the simplest optical transducer because it requires no junction and no depletion engineering. Its apparent advantage is internal gain:

$$G_{\text{PC}} = \frac{\tau_r}{\tau_{\text{tr}}}, \quad \tau_{\text{tr}} = \frac{L^2}{\mu_n V_b}, \quad (1.94)$$

where τ_r is the minority-carrier recombination lifetime, τ_{tr} is the carrier transit time across the contact separation L , μ_n is the electron mobility, and V_b is the applied bias. Physically, G_{PC} counts how many complete circuits a majority carrier traverses before recombination removes its partner: if a photogenerated hole survives for many transit times, the external circuit must replenish the missing electron repeatedly, yielding an apparent multiplication of the photocurrent. This gain mechanism, however, is inseparable from the recombination lifetime that sustains it. Because the same τ_r that produces gain also governs the impulse response, the -3 dB bandwidth is:

$$f_{3\text{dB,PC}} = \frac{1}{2\pi\tau_r}. \quad (1.95)$$

The gain-bandwidth product $G_{\text{PC}} \times f_{3\text{dB,PC}}$ is therefore a constant set by material parameters alone, with no free geometric lever. For germanium, trap-limited lifetimes in the range $\tau_r \sim 1 \mu\text{s}$ constrain the bandwidth to the sub-megahertz regime, placing the photoconductor many orders of magnitude below the 50 GHz electrical bandwidth required for 100 GBd-class operation in the target data-centre interconnect context. The photoconductor is therefore rejected by this intrinsic constraint before any geometric optimisation is considered.

PN Junction Photodiodes.

The PN junction eliminates the most critical weakness of the photoconductor by replacing the ohmic resistor with a rectifying interface that sustains an internal built-in electric field E_{bi} across the depletion region. Photogenerated electron-hole pairs created within, or diffusing into, this depletion layer experience a drift force that sweeps electrons toward the n-contact and holes toward the p-contact at velocities approaching the material saturation

velocity v_{sat} . The recombination lifetime no longer limits speed because carriers are extracted before recombination can occur, decoupling the bandwidth from τ_r .

The drift component of the frequency response is well approximated, for a uniformly illuminated depletion region of width W_d , by:

$$H_{\text{drift}}(f) = \frac{1 - e^{-j2\pi f W_d/v_{\text{sat}}}}{j2\pi f W_d/v_{\text{sat}}}, \quad (1.96)$$

which is the Fourier representation of a uniform current pulse of duration $\tau_{\text{tt}} = W_d/v_{\text{sat}}$. The -3 dB point of $|H_{\text{drift}}|$ occurs near $f_{\text{tt}} \approx 0.45 v_{\text{sat}}/W_d$, consistent with eq. (1.19).

A residual limitation arises because not all photogeneration is confined to the depletion layer. Carriers created in the quasi-neutral p- and n-regions must diffuse to the depletion edge before the field can extract them. This diffusion process introduces a slow tail in the impulse response, characterised by the ambipolar diffusion time constant τ_D . The composite transfer function acquires a low-pass factor:

$$H_{\text{PN}}(f) = H_{\text{drift}}(f) \frac{1}{1 + j2\pi f \tau_D}, \quad (1.97)$$

where τ_D is determined by the minority-carrier diffusion length and the distance from the photogeneration centroid to the depletion edge. When the diffusion-limited pole $1/(2\pi\tau_D)$ falls below the drift-limited transit-time bandwidth, it becomes the binding constraint on the overall detector speed. In practice, this condition is met at moderate symbol rates, but at the 50 GHz to 100 GHz electrical bandwidth class targeted here, even a modest quasi-neutral absorption fraction is sufficient to suppress the frequency response to an unacceptable degree. The PN structure therefore resolves the gain-bandwidth lock of the photoconductor while retaining a diffusion-tail bottleneck that must be addressed by the next architectural step.

PIN Photodiodes.

The PIN structure eliminates the diffusion tail by inserting a lightly doped, or intrinsic, semiconductor layer of thickness W_i between the p- and n-doped contact regions. Under reverse bias V_{bias} , the low doping density of the intrinsic layer ensures that it is fully depleted at modest voltages, because the depletion depth in a one-sided junction scales as $W \propto \sqrt{(V_{\text{bi}} + |V_{\text{bias}}|)/N_a}$, so a very low effective doping N_a in the intrinsic region allows full depletion to be achieved without requiring large voltages. Once fully depleted, the electric field across the intrinsic region is approximately uniform:

$$E(x) \approx \frac{|V_{\text{bias}}| + V_{\text{bi}}}{W_i}, \quad x \in [0, W_i], \quad (1.98)$$

where V_{bi} is the built-in potential determined by the doping contrast at the p-i and i-n interfaces. This expression follows directly from integrating Poisson's equation across a region of negligible space charge: $dE/dx \approx 0$, so the field is linear only at the boundaries

and constant throughout the bulk of the intrinsic layer. The practical importance of this uniformity is that all optically generated carriers within W_i experience essentially the same extraction field, eliminating the position-dependent transit-time dispersion that would arise in a non-uniform field profile.

If the absorbing material is selected so that its absorption coefficient α at the operating wavelength satisfies $\alpha^{-1} \lesssim W_i$, the overwhelming majority of photogeneration occurs inside the fully depleted intrinsic region. Diffusion from quasi-neutral zones is thereby suppressed, and the diffusion tail of eq. (1.97) is eliminated. The bandwidth is then governed entirely by the two competing mechanisms already formalised through eqs. (1.19) and (1.20):

$$\frac{1}{f_{3\text{dB}}^2} \approx \frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2}, \quad (1.99)$$

where $f_{\text{tt}} \propto v_{\text{sat}}/W_i$ and $f_{\text{RC}} \propto (R_s C_j)^{-1}$ with $C_j \propto A/W_i$. An increase in W_i raises the transit time τ_{tt} and thus degrades f_{tt} , while simultaneously reducing the junction capacitance C_j and thus improving f_{RC} . There is therefore an optimum intrinsic thickness W_i^* that maximises the total bandwidth for a given junction area A and series resistance R_s :

$$W_i^* = \left(\frac{v_{\text{sat}} A \varepsilon_0 \varepsilon_r}{2\pi R_s} \right)^{1/2}, \quad (1.100)$$

obtained by differentiating eq. (1.99) with respect to W_i and equating the transit-time and RC contributions at the optimum. The structural advantage of the PIN over the PN diode is that W_i is a lithographically controlled epitaxial parameter, decoupled from the contact doping levels used to minimise R_s . This additional degree of freedom—the independent control of the depletion thickness—is the key reason the PIN remains the canonical high-speed receiver device: it permits the junction geometry to be engineered at a level that the simple PN diode, where W_d is set by the doping profile alone, cannot match.

1.4.2 Avalanche Photodiodes: Gain versus Noise

An avalanche photodiode (APD) exploits impact ionisation within a high-field multiplication region to amplify the photocurrent internally, yielding an effective responsivity:

$$\mathcal{R}_{\text{APD}} = M \mathcal{R}_{\text{PIN}}, \quad (1.101)$$

where M is the mean avalanche multiplication factor and $\mathcal{R}_{\text{PIN}}$ is the responsivity that the same absorption layer would provide without multiplication. At first inspection this appears to offer a route to higher sensitivity without increasing the optical input power, but the avalanche mechanism introduces two penalties that are absent in the low-bias PIN.

The first is a fundamental gain-bandwidth constraint. Avalanche multiplication is not instantaneous: the carriers initiating an impact-ionisation event must traverse the high-field region, and the secondary carriers they generate must in turn complete their own transit before the multiplication chain terminates. The statistical build-up and decay of this cascade

introduces a finite temporal spread that broadens the impulse response. For a multiplication region of length l_{mult} and saturated ionisation velocity v_{ion} , the effective multiplication bandwidth is approximately:

$$f_M \approx \frac{v_{\text{ion}}}{2\pi k M l_{\text{mult}}}, \quad (1.102)$$

where $k = \alpha_h/\alpha_e$ is the ratio of hole to electron impact-ionisation coefficients. The product $M \times f_M$ is therefore a material- and geometry-specific constant, conventionally termed the gain-bandwidth product (GBP). As M is increased to boost sensitivity, f_M decreases proportionally, and the bandwidth headroom required for the target 50 GHz class rapidly exhausts the available GBP. In germanium, the nearly equal ionisation coefficients ($k \approx 1$) make this constraint particularly severe compared with materials such as InAlAs, where $k \ll 1$ permits higher GBP.

The second penalty is excess noise. Even in the absence of thermal noise, the stochastic nature of the multiplication process introduces fluctuations in M about its mean value that manifest as an additional noise current. The McIntyre model [7] characterises this through an excess noise factor $F(M)$, which modifies the shot-noise power spectral density from the unmultiplied case:

$$S_{I,\text{APD}} = 2q I_{\text{ph}} M^2 F(M) \Delta f, \quad (1.103)$$

where I_{ph} is the primary photocurrent and Δf is the noise bandwidth. For a single-carrier-initiated avalanche with ionisation ratio k , McIntyre showed that:

$$F(M) = k M + (1 - k) \left(2 - \frac{1}{M} \right). \quad (1.104)$$

In the ideal limit $k \rightarrow 0$ (electrons ionise, holes do not), $F \rightarrow 2$ as $M \rightarrow \infty$, and the noise penalty is modest. In the opposite regime $k \rightarrow 1$, which approximates the situation in Ge where $\alpha_e \approx \alpha_h$, eq. (1.104) reduces to $F \approx M$, and substitution into eq. (1.103) shows that the noise power spectral density grows as M^3 rather than M^2 . The net signal-to-noise benefit of multiplication then degrades rapidly with increasing M , and the optimum gain that minimises the noise-equivalent power saturates at values well below those that would be useful for sensitivity improvement in a thermal-noise-dominated receiver front-end.

Taking both penalties together—the GBP constraint and the excess noise factor—the APD in Ge offers no practical advantage over a low-bias PIN receiver paired with a low-noise transimpedance amplifier (TIA) at the data rates of interest. The TIA thermal noise, which sets the dominant noise floor in high-speed receivers, can be reduced by circuit design without introducing the bandwidth restriction inherent to avalanche multiplication. The APD is therefore not selected for the present work, and the PIN operating at modest reverse bias is retained as the active device architecture.

1.4.3 Waveguide-Integrated Photodetectors: Interaction Length Advantage

The PIN junction resolves the device-level bandwidth problem but does not, by itself, resolve the geometric conflict between absorption efficiency and transit time. This conflict is most acute in a surface-illuminated (or normal-incidence) configuration, where the optical beam enters through the top surface and propagates vertically through the absorbing layer. In this geometry the same dimension, the absorber thickness W_i , must simultaneously satisfy two contradictory requirements: the Beer-Lambert absorption condition, $W_i \gtrsim \alpha^{-1}$, to achieve high quantum efficiency, and the transit-time bandwidth condition, $W_i \ll v_{\text{sat}}/(2\pi f_{3\text{dB}})$, derived from eq. (1.19). At 1310 nm, the auxiliary MATLAB model uses $\alpha \approx 7 \times 10^3 \text{ cm}^{-1}$ for germanium, corresponding to a bulk $1/e$ absorption depth of $\alpha^{-1} \approx 1.4 \mu\text{m}$ before surface-normal reflection and confinement penalties are included. Achieving 90 % absorption requires a several-micrometre optical path, while achieving a transit-time-limited bandwidth of 50 GHz requires $W_i \lesssim 1 \mu\text{m}$ for typical saturation velocities in Ge ($v_{\text{sat}} \approx 6 \times 10^6 \text{ cm s}^{-1}$). These two requirements differ by over an order of magnitude, and no single value of W_i can satisfy both simultaneously. The surface-illuminated PIN geometry is therefore fundamentally overconstrained for high-speed, high-efficiency operation in the O-band.

Waveguide integration resolves this conflict by a geometric transformation: the optical mode is laterally confined by the waveguide structure and propagates longitudinally along the length of the detector rather than vertically through its thickness. The relevant dimension for optical absorption is then the propagation length L along the device axis, while the vertical dimension W_i retains its role as the carrier collection thickness that determines the transit time. Because L and W_i are orthogonal spatial coordinates, they can be engineered independently.

The evolution of optical power along the waveguide-integrated absorber follows the Beer-Lambert law in the propagation direction:

$$P(z) = P_0 e^{-\Gamma \alpha z}, \quad (1.105)$$

where P_0 is the power entering the absorber at $z = 0$, α is the material absorption coefficient of germanium, and Γ is the optical confinement factor, defined as the fraction of the guided modal power that overlaps spatially with the germanium absorbing region:

$$\Gamma = \frac{\int_{\text{Ge}} |\mathbf{E}(\mathbf{r}_{\perp})|^2 dA_{\perp}}{\int_{-\infty}^{\infty} |\mathbf{E}(\mathbf{r}_{\perp})|^2 dA_{\perp}}, \quad (1.106)$$

with the integration in the numerator restricted to the transverse cross-section of the Ge layer and the denominator taken over the full modal field. The factor Γ is determined by the waveguide geometry: for a Ge-on-Si butt-coupled or evanescent-coupling scheme, Γ can range from values approaching unity in a strongly confining geometry to values well below 0.5 in an evanescent-coupling taper architecture where the mode is gradually transferred from the Si waveguide into the Ge absorber.

The total absorption efficiency, relating absorbed power to incident power, is obtained by integrating eq. (1.105) over the device length L :

$$\eta_{\text{abs}} = \frac{P_0 - P(L)}{P_0} = 1 - e^{-\Gamma\alpha L}. \quad (1.107)$$

This expression reveals the decisive advantage of the waveguide architecture: η_{abs} can be driven arbitrarily close to unity by increasing L , entirely independently of W_i . For a representative value of $\Gamma = 0.8$ and $\alpha = 7 \times 10^3 \text{ cm}^{-1}$ at 1310 nm, a several-micrometre waveguide absorber can achieve high absorption, while W_i remains freely settable in the sub-micron regime required for high transit-time bandwidth. The geometric orthogonality between L and W_i is therefore complete: L governs responsivity through eq. (1.107), while W_i governs f_{tt} through eq. (1.19), and the two parameters are adjusted in a two-dimensional design space without mutual constraint.

A secondary benefit of the waveguide geometry is the suppression of the RC penalty associated with a large device footprint. In the surface-illuminated case, a large area is needed to intercept the incident beam, increasing C_j and degrading f_{RC} . In the waveguide-integrated device, the transverse cross-section of the mode is set by the single-mode waveguide dimensions, typically of order $0.5 \mu\text{m} \times 0.5 \mu\text{m}$ in sub-micron Si photonic platforms. The junction width W and height W_i are therefore bounded by the mode confinement rather than by beam-capture requirements, enabling small junction areas and correspondingly small capacitances. The longitudinal length L does increase the total capacitance as $C_j \propto W \cdot L/W_i$, but because W is sub-micron, the product $W \cdot L$ for a $10 \mu\text{m}$ – $20 \mu\text{m}$ -long device remains well below the junction areas necessary in surface-illuminated geometries for equivalent coupling efficiency.

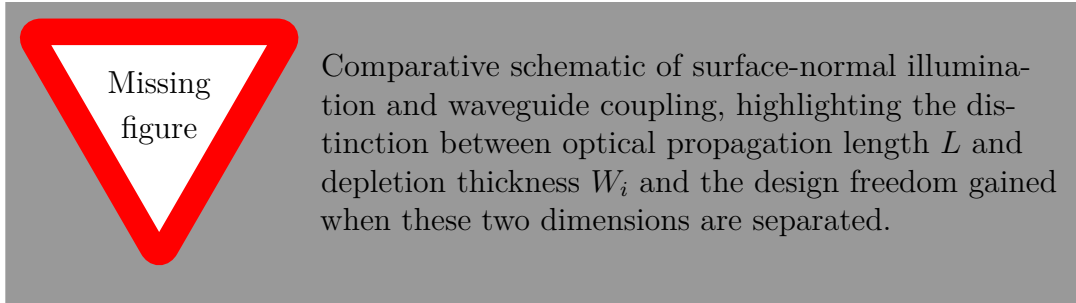


Figure 1.10: Waveguide coupling is the decisive architectural refinement because it decouples the absorption dimension from the transit-time dimension. That geometric change turns an overconstrained surface-normal problem into a two-parameter design space.

The geometric orthogonality introduced by waveguide integration does not eliminate all design tension: as L is increased to improve η_{abs} , the total junction capacitance grows proportionally, degrading f_{RC} . The optimum device length is therefore set not by eq. (1.107) in isolation but by the joint optimisation of the bandwidth expression in eq. (1.99), in which C_j now depends on L as well as on W_i and W . This residual coupling between absorption

efficiency and RC bandwidth—absent from the transit-time discussion but present in the full bandwidth budget—is the central design tension addressed in section 1.7.

1.4.4 Selection Rationale: Why PIN for High-Speed Integrated Links

The four preceding device classes can be evaluated against a unified set of performance criteria—responsivity, bandwidth, and noise—by tracing the physical mechanism that limits each architecture and identifying the structural modification that removes it. Figure 1.11 condenses this selection logic from the photoconductor through to the final architecture.

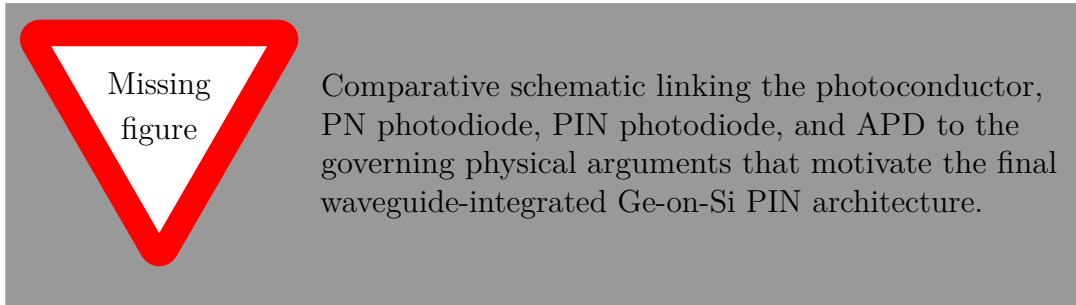


Figure 1.11: Architectural justification of the waveguide-integrated Ge-on-Si PIN photodetector. Each comparison resolves a specific limitation of the preceding alternative, leaving the waveguide-integrated Ge-on-Si PIN as the architecture compatible with the O-band, low-voltage, high-bandwidth target of this chapter.

The photoconductor is rejected because the gain-bandwidth product $G_{PC} \times f_{3dB,PC}$, fixed by the recombination lifetime via eq. (1.95), cannot reach the target bandwidth without eliminating the gain entirely. The PN junction removes the recombination constraint but retains a diffusion-tail bandwidth penalty from quasi-neutral absorption, as expressed by the pole in eq. (1.97). The PIN structure eliminates the diffusion tail by confining all useful photogeneration within the fully depleted intrinsic layer, reducing the bandwidth competition to the transit-time and RC terms of eq. (1.99), both of which are geometrically controllable. The APD appears to restore sensitivity through multiplication but introduces a gain-bandwidth product constraint via eq. (1.102) and an excess noise penalty via eq. (1.104) that, for germanium with $k \approx 1$, grows as $F \approx M$ and rapidly erases the SNR benefit; the bandwidth headroom needed for the target data rate is not available at any useful gain.

The waveguide-integrated PIN is therefore not simply the preferred architecture but the inevitable consequence of eliminating each identified physical limitation in sequence. It provides independent control of the two parameters governing responsivity and bandwidth, namely L and W_i respectively, through a geometric orthogonality that is unavailable in any of the preceding device classes. Operating at low reverse bias, it avoids the excess noise and reliability concerns of avalanche operation, and it presents a junction capacitance commensurate with the narrow transverse dimensions of the Si photonic waveguide mode.

The dominant noise source in the resulting receiver is the thermal noise of the external transimpedance amplifier, which is addressed at the circuit level without compromising the photodetector bandwidth. The remaining design problem—achieving simultaneously high responsivity and high bandwidth within this waveguide-integrated PIN framework at 1310 nm in Ge-on-Si—is the tension addressed by the material analysis of section 1.5, the electrode and geometry engineering of section 1.7, and the literature survey of section 1.6.

1.5 Material and Integration Platform: Germanium on Silicon

With the waveguide-integrated PIN structure identified as the correct device class and the O-band wavelength fixed at 1310 nm, the absorber material must now be chosen. The selection follows directly from the absorption physics of section 1.2: only germanium combines strong O-band absorption, sufficient carrier velocity, and silicon-foundry process compatibility. This section elaborates the three dimensions of that argument, namely spectral suitability, CMOS integration, and material-induced trade-offs, at the depth required to motivate the design choices of section 1.7.

1.5.1 Why Germanium at 1310 nm

Germanium is an indirect-gap semiconductor whose lowest conduction-band minimum lies at the L -point of the Brillouin zone, separated from the valence-band maximum at Γ by an indirect gap $E_{g,\text{ind}} = 0.66$ eV at 300 K. A secondary conduction-band minimum at the Γ -point defines a direct gap $E_{g,\text{dir}} = 0.80$ eV [8], [9], [10]. The photon energy at 1310 nm is $\hbar\omega = hc/\lambda \approx 0.946$ eV, which exceeds $E_{g,\text{dir}}$ by approximately 0.15 eV. Optical absorption at this wavelength therefore proceeds predominantly through *direct* interband transitions at the Γ -point, for which crystal momentum is intrinsically conserved without phonon assistance [5], [11]. Silicon, whose lowest direct transition lies at 3.4 eV and whose indirect gap of 1.12 eV makes it effectively transparent at O-band wavelengths, cannot be used as the absorber. This single material-physics argument is the entire justification for the Ge-on-Si heterostructure.

Biaxial strain and the direct-gap red-shift.

The application of biaxial tensile strain ε reduces the direct gap according to the linear deformation-potential relation:

$$E_{g,\text{dir}}(\varepsilon) \approx E_{g,\text{dir}}(0) - (2a_v + a_c) \left(1 - c_{12}/c_{11}\right) \varepsilon, \quad (1.108)$$

where a_v and a_c are the valence- and conduction-band hydrostatic deformation potentials and c_{11} , c_{12} are the elastic stiffness constants. The biaxial tensile strain of approximately 0.2 % to 0.3 % that develops in Ge epitaxial layers during post-growth cooldown, owing to the

thermal expansion coefficient mismatch between Ge and Si, reduces $E_{g,\text{dir}}$ by approximately 25 meV relative to the unstrained bulk value, red-shifting the onset of direct-gap absorption by roughly 50 nm and enhancing $\kappa(\lambda)$ at C-band wavelengths. For the O-band operating point of the present device this correction is small but it is the principal mechanism exploited in strained-Ge extended-wavelength detector designs.

Table 1.1: Material properties of candidate photodetector absorbers at 300 K. α values are referenced to 1310 nm. The final column summarises compatibility with a standard silicon-photonics process flow.

Material	E_g [eV]	α_{1310} [cm^{-1}]	$v_{\text{sat},h}$ [cm s^{-1}]	μ_p [$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$]	$\Delta a/a_{\text{Si}}$ [%]	CMOS compatibility
Ge	0.66 (ind.), 0.80 (dir.)	~ 2100	4×10^6	1900	4.2	Compatible
Si	1.12 (ind.)	~ 10	2.5×10^6	480	0	Compatible
InGaAs (lattice-matched to InP)	0.75	$\sim 1 \times 10^4$	5×10^6	4500	> 5	Not compatible
$\text{Ge}_{0.92}\text{Sn}_{0.08}$	~ 0.55 (dir.)	> 5000	$\sim 4 \times 10^6$	~ 1500	4.8	Research-stage

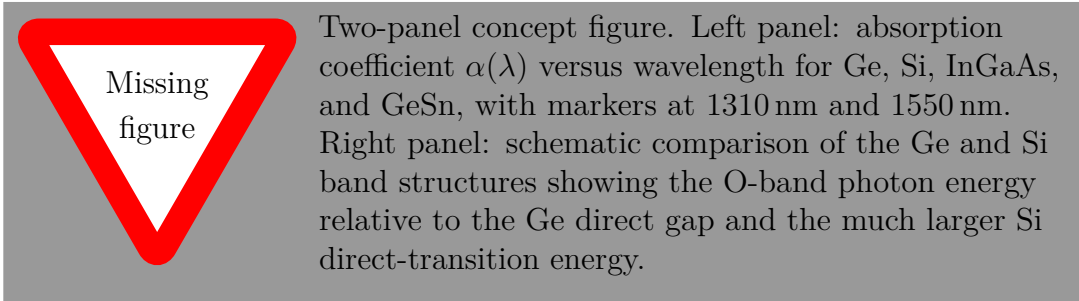


Figure 1.12: Material selection argument for O-band detection on a silicon photonics platform. Germanium combines appreciable absorption at 1310 nm, near-direct-gap behaviour, high hole saturation velocity, and foundry-compatible heterogeneous integration.

1.5.2 CMOS Compatibility and Integration Constraints

Germanium can be grown selectively on silicon by chemical vapour deposition (CVD) using a two-step heteroepitaxial process: a low-temperature ($\sim 400^\circ\text{C}$) nucleation layer of 30 nm to 50 nm seeds the growth, and a high-temperature ($\sim 650^\circ\text{C}$) bulk growth achieves the target thickness. Cyclic thermal annealing between 600°C to 750°C reduces the threading dislocation density from $\sim 1 \times 10^9 \text{ cm}^{-2}$ after growth to $\sim 5 \times 10^7 \text{ cm}^{-2}$ after annealing, which is the practical lower bound achievable without compositionally graded buffer layers. The total thermal budget of this process is compatible with the front-end-of-line (FEOL) constraints of a 220 nm SOI platform.

Major silicon-photonics foundries, including Intel, imec, and GlobalFoundries, have qualified Ge-on-Si photodetector processes at 220 nm SOI node. The process compatibility removes the III-V bonding step required by InGaAs receivers, eliminates the need for flip-chip assembly, and enables monolithic co-integration of the photodetector with Si waveguides, grating couplers, multiplexers, and electro-optic modulators on the same die.

Heterojunction band alignment and its numerical consequences.

The electrical consequence of the Ge/Si material interface directly conditions the boundary conditions of the drift-diffusion problem. Owing to the lattice mismatch ($\Delta a/a \approx 4.2\%$) and the difference in band alignment, the Ge/Si heterojunction sustains a valence-band discontinuity of $\Delta E_v \approx 0.54$ eV and a conduction-band offset $\Delta E_c \approx -0.14$ eV [5]. For a p^{++} -Si anode contacting the intrinsic Ge absorber, the large ΔE_v acts as a potential barrier suppressing hole injection from silicon into germanium, thereby reducing the minority-carrier diffusion contribution to dark current quantified by eq. (1.112). A thin compositionally graded transition layer of thickness $\delta_{\text{grad}} \approx 5$ nm at the heterojunction removes the valence-band singularity that would otherwise prevent convergence of the self-consistent drift-diffusion solver without perceptibly modifying the electrostatic potential profile in the intrinsic germanium bulk.

1.5.3 Material-Induced Trade-offs

The same lattice mismatch ($\Delta a/a \approx 4.2\%$) that makes Ge-on-Si heteroepitaxy necessary also creates a residual defect population that cannot be completely eliminated by annealing. Threading dislocations at density $\text{TDD} \approx 5 \times 10^7 \text{ cm}^{-2}$ act as Shockley-Read-Hall recombination centres in the Ge bulk, directly setting the SRH lifetime τ_0 and the dark current floor through eq. (1.114). The current CHARGE implementation uses $\tau_0 = 1$ ns for Ge, which represents a conservative nanosecond-scale recombination environment for defect-limited Ge-on-Si material.

The interaction between Ge thickness, dislocation density, and device performance creates a three-way trade-off. Thicker Ge layers absorb more light, improving $\mathcal{A}$ via eq. (1.14), but slow carrier transit through eq. (1.19) and increase the total SRH generation volume, raising I_d via eq. (1.114). Thinner Ge layers improve speed and reduce dark current but sacrifice absorption efficiency. Additionally, the near-interface Ge layer, specifically the first 50 nm to 100 nm grown on Si, has higher defect density than the bulk, which is why the n^{++} -Ge cathode of the present device is limited to 50 nm: it minimises both free-carrier absorption (FCA) in the contact region and the volume of highest-defect-density material within the optical generation zone.

A compact material comparison is therefore sufficient to close the absorber selection argument: Ge is not chosen because it is universally superior to all alternatives, but because it is the only material in table 1.1 that simultaneously satisfies the optical, transport, and integration constraints of the present O-band receiver.

1.5.4 Carrier Transport Physics and Dark Current Mechanisms in Ge-on-Si Photodiodes

The performance of a waveguide-integrated Ge-on-Si photodiode is governed not only by the optical absorption physics described in section 1.5.1, but equally by the dynamics of photogenerated carrier transport and the unavoidable generation of thermally excited carriers that constitute the device dark current. These two phenomena are deeply coupled: the same electric field that sweeps photogenerated carriers across the intrinsic germanium absorber with high velocity also governs the rate at which trap-mediated thermal generation replenishes the depletion region with spurious carriers. Understanding both mechanisms from first principles is essential before the geometric and doping trade-offs of the device design in section 1.7 can be motivated rigorously.

Drift, Diffusion, and the Onset of Velocity Saturation

Under the applied reverse bias V_R that fully depletes the intrinsic germanium layer of thickness w_i , the internal electric field $\mathcal{E}$ in the depletion region drives photogenerated electrons and holes toward the cathode and anode, respectively, by drift. At moderate fields, the mean drift velocity of a carrier species $s \in \{e, h\}$ is proportional to $\mathcal{E}$ through the low-field mobility μ_s :

$$v_{d,s} = \mu_s \mathcal{E}. \quad (1.109)$$

For bulk germanium at 300 K, $\mu_e \approx 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_h \approx 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, both substantially exceeding their silicon counterparts. The high hole mobility of Ge is of particular practical relevance in a p - i - n structure where hole transit across the full intrinsic thickness can otherwise constitute the bandwidth-limiting transit-time contribution.

As $\mathcal{E}$ increases beyond approximately 10 kV cm^{-1} for electrons and 5 kV cm^{-1} for holes in germanium, inter-valley and optical-phonon scattering rates grow faster than the energy gained from the field, causing the velocity to saturate asymptotically. The empirical Canali model provides a compact and physically consistent description of velocity saturation in semiconductor materials:

$$v_{d,s}(\mathcal{E}) = \frac{\mu_s \mathcal{E}}{\left[1 + \left(\mu_s \mathcal{E} / v_{\text{sat},s}\right)^{\beta_s}\right]^{1/\beta_s}}, \quad (1.110)$$

where $v_{\text{sat},s}$ is the saturation velocity of species s and β_s is a dimensionless exponent that interpolates the transition between the linear and saturated regimes. For holes in germanium, $v_{\text{sat},h} \approx 4 \times 10^6 \text{ cm s}^{-1}$ and $\beta_h \approx 1$, which reduces eq. (1.110) to the hyperbolic form familiar from standard textbook treatments. In the high-field limit relevant to a reverse-biased photodiode, eq. (1.110) collapses to $v_{d,s} \rightarrow v_{\text{sat},s}$, independently of mobility. This saturation regime is precisely the operating point targeted by the present device, since a sufficiently large reverse bias ensures that carrier velocities are clamped at v_{sat} and are therefore

independent of any spatial non-uniformity in $\mathcal{E}$ along the propagation axis of the optical mode.

Transit-time limited bandwidth.

The direct device-level consequence of carrier velocity saturation is the transit-time contribution to the 3 dB bandwidth. If holes traverse the intrinsic germanium thickness w_i at their saturated velocity, the transit-time-limited bandwidth is:

$$f_{\text{tt}} = \frac{0.45 v_{\text{sat},h}}{w_i}, \quad (1.111)$$

where the numerical prefactor 0.45 follows from the Fourier analysis of the induced current waveform in the Shockley–Ramo framework for a uniformly distributed generation profile. Equation (1.19) establishes a direct inverse relationship between absorber thickness and transit-time bandwidth: halving w_i doubles f_{tt} , but at the cost of a proportional reduction in the absorbed fraction of the evanescently coupled optical power, as governed by the Beer-Lambert relation discussed in section 1.2. This thickness trade-off is the central geometric tension of the design and motivates the use of waveguide-coupling geometry in preference to surface illumination: by distributing absorption along the propagation direction rather than the vertical direction, the coupling length L_{pd} independently controls the absorbed fraction while w_i remains free to be minimised for bandwidth.

Dark Current: Physical Origins and Analytical Decomposition

In the absence of illumination, a reverse-biased germanium photodiode passes a current I_d that constitutes the fundamental noise floor of the receiver. Dark current enters the signal chain as a shot-noise source with single-sided power spectral density $S_I = 2qI_d$, which adds in quadrature with the signal shot noise and the amplifier noise floor. Minimising I_d is therefore not merely a power-efficiency consideration but a direct bandwidth-noise trade-off: at the data rates of 400GBASE-DR4, even moderate dark currents of order 1 μA can degrade the receiver sensitivity by several tenths of a decibel if the transimpedance amplifier noise is not sufficiently dominant.

Three physically distinct mechanisms contribute to the total dark current density J_d in a reverse-biased Ge-on-Si *p-i-n* diode: diffusion of minority carriers from the quasi-neutral contact regions, Shockley–Read–Hall (SRH) thermal generation within the depletion region, and, at sufficiently large reverse bias, band-to-band tunnelling enhanced by the narrow direct gap of germanium. Each mechanism exhibits a characteristic bias and temperature dependence that allows them to be distinguished experimentally and that must be accounted for separately in the device model.

Diffusion current.

Minority carriers generated thermally within one diffusion length of the edge of the depletion region drift across the junction under the action of the built-in and applied fields. For the p^{++} -Si anode contacting the i -Ge absorber of the present device, the analysis is modified by the Ge/Si heterojunction band alignment: the valence-band discontinuity $\Delta E_v \approx 0.54$ eV at the Si/Ge interface acts as a potential barrier that suppresses minority hole injection from the Si anode into the Ge absorber, substantially reducing the silicon contribution to diffusion dark current compared with a homojunction geometry. The residual diffusion current density arising from minority electrons in the p^{++} -Ge anode and minority holes in the n^{++} -Ge cathode takes the form:

$$J_{\text{diff}} = q n_i^2 \left(\frac{D_e}{L_e N_A} + \frac{D_h}{L_h N_D} \right), \quad (1.112)$$

where n_i is the intrinsic carrier concentration of germanium, $D_{e,h}$ are the minority carrier diffusion coefficients, $L_{e,h} = \sqrt{D_{e,h} \tau_{e,h}}$ are the corresponding diffusion lengths, and N_A , N_D are the acceptor and donor concentrations in the p^{++} and n^{++} contact regions, respectively. The strong exponential temperature dependence of $n_i^2 \propto \exp(-E_g/k_B T)$ combined with the relatively narrow gap of germanium ($E_g = 0.66$ eV) makes J_{diff} significantly larger in Ge devices than in comparable Si junctions at the same temperature, and explains why Ge photodiodes typically exhibit dark currents in the range 10 nA to 100 nA even at modest device areas.

Shockley–Read–Hall generation current.

The dominant dark-current mechanism in the depletion region of a defect-rich Ge-on-Si device is trap-assisted thermal generation via mid-gap states. In the SRH framework [12], [13], a defect level at energy E_t near the intrinsic Fermi level mediates a two-step process by which valence electrons are excited to the conduction band via the trap, generating an electron-hole pair without the need for a direct band-to-band phonon or photon interaction. The volumetric generation rate under full depletion approximation is maximised when $E_t = E_i$ and reduces to:

$$G_{\text{SRH}} = \frac{n_i}{2\tau_0}, \quad (1.113)$$

where $\tau_0 = (\tau_e \tau_h)^{1/2}$ is the geometric-mean SRH lifetime, which encapsulates the product of the thermal capture cross-section σ , the trap density N_t , and the thermal velocity v_{th} through $\tau_0^{-1} = \sigma v_{\text{th}} N_t$. Integrating G_{SRH} over the depleted volume $V_{\text{dep}} = A \cdot w_i$, where A is the device cross-sectional area, yields the SRH dark current:

$$I_{\text{SRH}} = q G_{\text{SRH}} V_{\text{dep}} = \frac{q n_i A w_i}{2\tau_0}. \quad (1.114)$$

The physical significance of eq. (1.114) in the Ge-on-Si context is immediate. Threading dislocations at density TDD $\approx 5 \times 10^7 \text{ cm}^{-2}$, which persist after cyclic thermal annealing

as discussed in section 1.5.3, introduce mid-gap states whose spatial density N_t is directly proportional to the TDD. Effective SRH lifetimes in processed Ge-on-Si devices are commonly treated as nanosecond-scale rather than bulk-germanium values ($\tau_0 \sim 1 \mu\text{s}$). The repository therefore adopts $\tau_0 = 1 \text{ ns}$ as the Ge electrical boundary condition in the drift-diffusion simulations of section 1.8.2.

Three design implications follow directly from eq. (1.114). First, I_{SRH} scales linearly with the depleted volume $w_i \cdot A$, providing a strong incentive to minimise both the absorber thickness and the device footprint. Second, the linear dependence on n_i means that I_{SRH} is exponentially sensitive to temperature through $n_i \propto \exp(-E_g/2k_B T)$; this temperature scaling is less severe than that of the diffusion component because it involves n_i rather than n_i^2 , making SRH generation the dominant mechanism at lower temperatures where the diffusion contribution collapses more rapidly. Third, because I_{SRH} is proportional to w_i whereas the absorbed optical power scales as $1 - \exp(-\alpha w_i)$ for a surface-illuminated geometry, there exists an optimal absorber thickness that maximises the ratio of photocurrent to dark current; in the waveguide-coupled geometry this optimum shifts, since the absorption is accumulated along the propagation axis and w_i can be reduced without sacrificing responsivity, as noted above.

Band-to-band tunnelling.

At reverse biases beyond approximately 2 V to 3 V for the doping concentrations considered here, the electric field in the depletion region becomes sufficiently large that quantum-mechanical tunnelling of valence electrons directly into conduction-band states across the narrow direct gap of germanium produces a field-dependent excess dark current. The Zener tunnelling current density scales as [14]:

$$J_{\text{BTBT}} \propto \mathcal{E}^{1/2} \exp\left(-\frac{4\sqrt{2m^*} E_{g,\text{dir}}^{3/2}}{3q\hbar\mathcal{E}}\right), \quad (1.115)$$

where m^* is the reduced tunnelling effective mass and $\hbar$ is the reduced Planck constant. The exponential dependence of J_{BTBT} on $\mathcal{E}^{-1}$ and on $E_{g,\text{dir}}^{3/2}$ means that, for the direct gap of 0.80 eV of germanium, tunnelling becomes non-negligible at fields accessible under practical reverse-bias conditions. Because the present device is intended to operate at reverse biases of 1 V to 2 V, the field in the intrinsic region remains well below the Zener onset threshold and J_{BTBT} is negligible in the nominal operating regime. Nevertheless, the proximity of the germanium direct gap to the tunnelling onset motivates a careful upper bound on the reverse bias used during high-temperature burn-in or reliability qualification, where the thermally softened band structure and field crowding near device corners could locally activate the tunnelling mechanism.

Total Dark Current Budget and Its Consequence for Receiver Sensitivity

Within the nominal operating regime, the total dark current is:

$$I_d = I_{\text{diff}} + I_{\text{SRH}}, \quad (1.116)$$

with I_{diff} governed by eq. (1.112) and I_{SRH} by eq. (1.114). For a device of cross-section $A \approx 0.5 \mu\text{m} \times 0.4 \mu\text{m}$ and intrinsic thickness $w_i \approx 400 \text{ nm}$, evaluated at 300 K with a nanosecond-scale τ_0 , the SRH component is of order tens of nanoamperes, consistent with the experimentally reported dark currents of 5 nA to 50 nA for similarly sized Ge-on-Si waveguide photodetectors in the literature.

The receiver-level impact is quantified through the dark-current shot noise contribution to the total input-referred noise current spectral density. In a transimpedance-amplifier front-end with bandwidth B , the root-mean-square dark current noise is:

$$i_{n,d} = \sqrt{2qI_dB}, \quad (1.117)$$

which for $I_d = 20 \text{ nA}$ and $B = 26.6 \text{ GHz}$ (corresponding to a single 100 Gb/s lane of 400GBASE-DR4 at a 4 : 1 PAM-4 symbol rate of 26.6 GBd) yields $i_{n,d} \approx 14 \text{ nA}/\sqrt{\text{Hz}} \times \sqrt{B} \approx 2.3 \mu\text{A}_{\text{rms}}$. This figure is typically sub-dominant relative to the transimpedance amplifier input-referred noise current in practical receivers, confirming that the dark current of a well-optimised Ge-on-Si photodiode does not constitute the sensitivity-limiting noise source at the data rates of interest. However, any degradation of τ_0 due to process variation or elevated operating temperature will increase I_{SRH} and can shift this balance, providing the physical rationale for the tight thermal-budget control and cyclic annealing protocol discussed in section 1.5.2.

The analytical decomposition of eq. (1.116) thus closes the argument connecting material defect physics, device geometry, and receiver noise performance: the SRH lifetime τ_0 set by the threading dislocation density of the Ge-on-Si heteroepitaxial process propagates directly into a dark current floor, which in turn determines the minimum shot-noise pedestal against which optical signals must be detected. The design choices of absorber thickness, contact doping concentration, and reverse bias voltage explored in section 1.7 are all conditioned on minimising eq. (1.116) subject to the competing constraints on absorption efficiency and bandwidth established by eq. (1.19) and the Beer-Lambert absorption analysis.

1.6 Literature Review and State of the Art

The Ge-on-Si waveguide photodetector field has evolved over approximately two decades through successive generations of electrode and junction engineering, each interpretable through the metric framework of section 1.3. What follows is a causal narrative: each development is best understood as an attempt to relax one bottleneck in eqs. (1.51), (1.68) and (1.69) while leaving another constraint behind.

1.6.1 Milestone Waveguide-Integrated Ge-on-Si PIN Photodetectors

The landmark demonstration of Vivien *et al.* [15] established a silicon-on-insulator (SOI) waveguide-coupled Ge p-i-n detector with $\mathcal{R} \approx 1.0 \text{ A W}^{-1}$ and $f_{3\text{dB}} = 42 \text{ GHz}$ at the O-band, confirming that direct-gap absorption at the Γ -point translates into competitive responsivity on the silicon platform. The 42 GHz result was RC-limited: the electrode geometry left R_s at a value that caused f_{RC} to dominate over f_{tt} in eq. (1.69). Ang *et al.* [16] subsequently demonstrated evanescently coupled Ge-on-SOI fabricated at low thermal budget to remain CMOS-compatible, establishing process integration as a design constraint alongside responsivity and bandwidth.

The next phase of the literature attacked $R_s C_j$ more directly through contact and pad engineering. Chen *et al.* [17] demonstrated a silicon-contacted Ge p-i-n with $\mathcal{R} = 0.74 \text{ A W}^{-1}$ and $f_{3\text{dB}} = 35 \text{ GHz}$, and the same group later pushed to 67 GHz at only 1 V reverse bias through electrode repositioning [18], thereby reducing ℓ_a in eq. (1.74) until f_{RC} no longer dominated eq. (1.69). Novack *et al.* [19] reached 60 GHz by adding wire-bond inductance as a gain-peaking element. That approach is effective but system-dependent: the added inductance partially resonates with $C_j + C_{\text{par}}$, so the bandwidth extension depends on the surrounding RF environment and package parasitics [20].

The lateral junction architecture introduced by Virost *et al.* [21] decouples the optical mode depth from the depletion region width, providing an additional degree of freedom for C_j minimisation in eq. (1.72) that is geometrically independent of $\mathcal{A}$ in eq. (1.14). Parasitic engineering of the vertical architecture was pushed to its prior limit by Shi *et al.* [22], who demonstrated $f_{3\text{dB}} = 80 \text{ GHz}$ through systematic minimisation of parasitic pad capacitance. The bandwidth ceiling for the lateral architecture was located by Lischke *et al.* [23], who reported $f_{3\text{dB}} = 265 \text{ GHz}$ with a Ge-fin geometry scaled to $W_{\text{Ge}} \approx 100 \text{ nm}$.

Resonant-cavity-enhanced and uni-travelling-carrier alternatives.

Two architectural alternatives provide additional context. The resonant-cavity-enhanced (RCE) vertical photodiode places the absorber within a Fabry-Pérot resonator; at the resonance condition, the single-pass absorbed fraction $\mathcal{A}$ of eq. (1.14) is replaced by:

$$\mathcal{A}_{\text{RCE}} = \frac{(1 - R_1)(1 + R_2 e^{-\alpha L})^2(1 - e^{-2\alpha L})}{1 - 2\sqrt{R_1 R_2} e^{-\alpha L} \cos(4\pi n L / \lambda) + R_1 R_2 e^{-2\alpha L}}, \quad (1.118)$$

where R_1 and R_2 are the top and bottom mirror reflectances [20]. Near-unity quantum efficiency is achievable in absorbers as thin as 200 nm, but the resonance bandwidth narrows to a few nanometres, which is incompatible with the temperature-insensitive broadband operation required of data-centre receivers. The uni-travelling-carrier (UTC) structure replaces both drift species with electrons only, raising the transit-time limit to:

$$f_{\text{tt,UTC}} = \frac{0.45 v_{\text{d,sat,e}}}{W_{\text{dep}}}, \quad (1.119)$$

approximately 50 % higher than the hole-limited value of eq. (1.19) for the same depletion width [20].

1.6.2 Bandwidth Enhancement Strategies

The literature reveals four categories of bandwidth enhancement strategy:

Geometry scaling reduces both C_j (via smaller A_j) and f_{tt} (via smaller W_i), but doing so simultaneously sacrifices responsivity through eq. (1.68). Diminishing returns set in once the device enters the absorption-starved regime below $L \approx 4 \mu\text{m}$.

Doping profile optimisation targets the dark-current-to-responsivity balance: highly doped contacts reduce R_s but increase free-carrier absorption, while the junction polarity affects the heterojunction band alignment and therefore J_{diff} through eq. (1.112).

Contact engineering attempts to reduce the dominant anode series resistance through silicidation, via placement, and metal routing. For a vertical architecture where R_s is dominated by the lateral Si slab resistance of eq. (1.74), silicidation reduces the metal-semiconductor contact contribution but not the bulk Si slab contribution, leaving R_s improvement bounded by geometry.

Electrode topology is the most structurally impactful lever: changing the physical layout of the metal anode changes ℓ_a in eq. (1.74) and therefore R_s directly. The U-shaped-anode strategy targets this lever by reducing ℓ_a geometrically until $f_{\text{RC}} > f_{tt}$, providing a structural solution rather than a resonant compensation.

1.6.3 Advanced Electrode Designs Including U-Shaped Configurations

The benchmark for the present work is the 103 GHz vertical Ge-on-Si photodiode of Shi *et al.* [24], which surpasses the 100 GHz threshold in the vertical architecture through a U-shaped anode electrode that reduces R_s , and therefore $\tau_{\text{RC}} = R_s C_j$ of eq. (1.75), by approximately 36 % relative to a straight-electrode design. The U-shaped topology wraps the metal anode around three sides of the germanium mesa, so that the maximum lateral current path from any point on the mesa base to a metal contact is $W_m/4 + d_{\text{gap}}$ rather than $W_m/2 + d_{\text{gap}}$. The U-shaped design simultaneously delivers $\mathcal{R} = 0.95 \text{ A W}^{-1}$ and $I_d = 1.3 \text{ nA}$ at -1 V , making it a balanced rather than a specialised solution.

The U-shape is therefore selected not because it changes only a single resistor value, but because it opens a favourable region of the coupled (R_s, C_j, L_p) design space: the reduced R_s also enables a shallower and more useful inductive peaking condition, and the experimentally observed bandwidth increases from 83 GHz (parallel electrode) to 103 GHz (single-U) [20].

1.6.4 Performance Benchmarking Table

Table 1.2 consolidates the landmark experimental results most relevant to the present design argument. The table is intentionally restricted to the vertical trajectory adopted in this chapter and the lateral trajectory that defines the broader bandwidth ceiling.

Table 1.2: Selected benchmark waveguide-integrated Ge-on-Si photodetectors used to position the present design. “V” and “L” denote vertical and lateral junction geometries, respectively; “NR” denotes values not reported by the authors.

Year	Type	Reference	Key Design Lever	$\mathcal{R}$ [A/W]	I_d [nA]	$f_{3\text{ dB}}$ [GHz]
2008	V	[16]	Evanescent coupling, CMOS-compatible process	0.92	570	11.3
2009	V	[15]	Butt coupling, straight contact	~ 1.0	18	42
2021	V	[22]	Parasitic engineering with inductive peaking	0.89	6.4	80
2024	V	[24]	U-shaped anode with inductive peaking	0.95	1.3	103
2015	L	[25]	Reduced Ge width with lateral offset junction	1.00	100	> 70
2017	L	[21]	Si/Ge/Si heterojunction with sub-micrometre Ge width	NR	5	> 50
2021	L	[23]	Ge-fin junction with 100 nm minimum width	0.30	100 to 200	265
2024	L	[26]	Side-coupled parallel Ge geometry	0.80	24	> 40

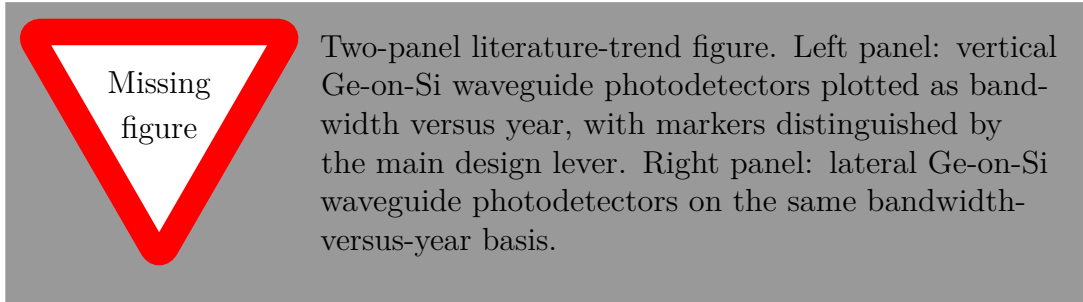


Figure 1.13: State-of-the-art trend summary for waveguide-integrated Ge-on-Si photodetectors. The chronology is most useful when read as a sequence of interventions into R_s , C_j , and the transport path rather than as a collection of isolated record values.

1.6.5 Identified Gaps and Motivation for This Work

The identified gaps that motivate the present work are: (i) no prior published simulation study has re-optimised the U-shaped vertical architecture for 1310 nm O-band operation,

where the optical absorption and mode coupling differ from the 1550 nm benchmark; (ii) the double U-shaped electrode topology, which wraps metal on both long sidewalls, remains an open design concept rather than a validated implementation in the present repository; and (iii) a reproducible solver-to-thesis workflow linking FDTD, CHARGE, MATLAB, INTERCONNECT, and layout generation is rarely documented at device-project level. These gaps frame the design and analysis scope documented in sections 1.7 and 1.8.

IEEE 802.3 400GBASE-DR4 as the receiver-level target.

IEEE 802.3 Clause 140 defines 400GBASE-DR4 as a 400 Gbit s^{-1} single-mode-fibre interface in the O-band, implemented through four parallel optical lanes at 1310 nm over reaches up to 500 m [27]. Each lane carries $106.25 \text{ Gbit s}^{-1}$ raw line rate with PAM-4 signalling. The key receiver-side quantities are summarised in table 1.3.

Table 1.3: IEEE 802.3 400GBASE-DR4 receiver-side constraints most directly relevant to the photodetector, extracted from Clause 140 [27] and mapped onto the device metrics of section 1.3.

Parameter	Symbol	Value	Unit	Principal relation
Wavelength range	λ	1260–1360	nm	eq. (1.14)
Symbol rate	f_s	53.125	Gbaud	eq. (1.69)
Modulation format	—	PAM-4	—	—
Minimum receiver sensitivity (OMA)	$\text{OMA}_{\text{out,min}}$	−6.0	dBm	—
Maximum input power	$P_{\text{in,max}}$	+3.4	dBm	eq. (1.162)
Minimum receiver −3 dB bandwidth	$f_{3 \text{ dB,min}}$	$\geq 0.75 f_s$	GHz	eq. (1.69)
Reflectance tolerance	Γ_{max}	−26	dB	eq. (1.143)

The minimum receiver bandwidth implied by the conventional $0.75 f_s$ criterion is only 39.8 GHz, so the benchmark vertical topology already establishes comfortable speed relevance for DR4. The selected U-shaped vertical Ge-on-Si PIN of Shi *et al.* [24] combines 103 GHz-class response, $\mathcal{R} = 0.95 \text{ A W}^{-1}$, and $I_d = 1.3 \text{ nA}$ in a process-compatible O-band platform. Because DR4 requires a detector that is simultaneously fast enough, efficient at low OMA, and quiet enough not to erode receiver margin, this balanced vertical U-shaped architecture is the most defensible starting point for the present design.

Microhole enhancements and DBR implementations.

Two structural variants highlight alternative optimization paths for bandwidth and responsivity. To decouple the absorption length from the transit time, Hamadou *et al.* [4] investigated surface-illuminated silicon PIN photodiodes structured with photon-trapping microholes. Their 3D Lumerical FDTD and CHARGE simulations demonstrated that etching a 2×2 array of $0.7 \mu\text{m}$ square microholes into the intrinsic region enhances optical absorption through photon trapping while simultaneously reducing the junction capacitance by a geo-

metric filling factor ρ_f . By carefully minimizing parasitic bonding capacitance ($C_b = 70$ fF) and series resistance ($R_s = 3.6 \Omega$), their $20 \mu\text{m} \times 20 \mu\text{m}$ device achieved an open PAM-4 eye at 50 GBd. Separately, the seminal work of Li *et al.* [28] demonstrated a 112 Gbit s^{-1} CMOS-compatible waveguide Germanium photodetector by utilizing a distributed Bragg reflector (DBR). The DBR allowed the optical absorption length to be doubled without extending the physical transport distance, while their interdigitated electrode geometry highlighted the delicate balance between transit-time reduction and the onset of high dark current from strong electric fields.

Broader reviews of the design space are provided by Wang *et al.* [29] and Yan *et al.* [20]; the multiphysics modelling context is treated in depth in [30]; computational methods are discussed in the pedagogical framework of Wartak [31].

1.6.6 Carrier Transport Fundamentals in Reverse-Biased PIN Junctions

A rigorous understanding of photodetector bandwidth and noise requires first establishing the microscopic carrier dynamics within the depleted intrinsic region of a reverse-biased p-i-n junction. What follows develops the drift-diffusion framework from first principles, interprets the resulting transit-time and RC-limited bandwidth expressions physically, and draws direct consequences for the vertical Ge-on-Si architecture operating at 1310 nm.

Carrier Generation and the Photocurrent Mechanism

When a photon of energy $h\nu > E_g$ is absorbed in the intrinsic germanium layer, a bound electron-hole pair is promoted across the direct bandgap at the Γ -point of the Brillouin zone. For germanium at the O-band centre wavelength of 1310 nm, the photon energy is $h\nu \approx 0.946 \text{ eV}$, which exceeds the direct-gap energy $E_\Gamma \approx 0.80 \text{ eV}$ at room temperature. This direct-gap absorption mechanism is responsible for the significantly higher absorption coefficient of germanium relative to silicon at this wavelength; whereas silicon exhibits $\alpha_{\text{Si}} \lesssim 1 \times 10^3 \text{ m}^{-1}$ at 1310 nm, germanium achieves $\alpha_{\text{Ge}} \approx 4 \times 10^4 \text{ m}^{-1}$ under the tensile strain conditions typical of CMOS-compatible epitaxial growth on silicon [15]. The implication for device design is immediate: a Ge absorber of length $L \approx 10 \mu\text{m}$ in a waveguide geometry can approach the single-pass absorption fraction $\mathcal{A} \rightarrow 1 - e^{-\alpha_{\text{Ge}}L} \approx 0.33$ even without optical confinement enhancement, whereas achieving comparable absorption in Si would require centimetre-scale paths.

Once the electron-hole pair is generated within the depletion region of width W_i , the built-in electric field $\mathcal{E}_0$ augmented by the applied reverse bias V_R sweeps the carriers in opposing directions: electrons toward the n-contact and holes toward the p-contact. It is this drift current, rather than diffusion, that constitutes the primary photocurrent in a well-designed photodetector operated under sufficient reverse bias.

Drift Velocity and the Transit-Time Limit

The drift velocity v_d of a carrier species $s \in \{e, h\}$ is, in the low-field linear regime, proportional to the local electric field $\mathcal{E}$ through the carrier mobility μ_s :

$$v_{d,s} = \mu_s \mathcal{E}. \quad (1.120)$$

However, at the electric field magnitudes present in a reverse-biased Ge p-i-n at even modest bias voltages (1 V across a 1 μm intrinsic region implies $\mathcal{E} \approx 10^6 \text{ V m}^{-1}$), the velocity-field relationship saturates. The empirical single-parameter saturation model due to Caughey and Thomas captures this behaviour:

$$v_{d,s}(\mathcal{E}) = \frac{\mu_{s,0} \mathcal{E}}{\left[1 + \left(\frac{\mu_{s,0} \mathcal{E}}{v_{\text{sat},s}} \right)^{\beta_s} \right]^{1/\beta_s}}, \quad (1.121)$$

where $\mu_{s,0}$ is the low-field mobility, $v_{\text{sat},s}$ is the saturation velocity, and β_s is a fitting parameter of order unity. For germanium, the saturation velocities of electrons and holes are $v_{\text{sat},e} \approx 6 \times 10^4 \text{ m s}^{-1}$ and $v_{\text{sat},h} \approx 4.6 \times 10^4 \text{ m s}^{-1}$, respectively [20]. The asymmetry between these values is physically significant: holes transit the depletion region more slowly than electrons, and it is therefore holes that impose the more stringent transit-time limitation.

The transit time for a carrier species s traversing a fully depleted intrinsic layer of width W_i at saturation velocity is:

$$\tau_{\text{tt},s} = \frac{W_i}{v_{\text{sat},s}}. \quad (1.122)$$

The corresponding small-signal transit-time bandwidth limit follows from the frequency response of the photocurrent under the assumption of uniform carrier generation and saturated drift [1]:

$$f_{\text{tt}} = \frac{0.45 v_{\text{sat},h}}{W_i}, \quad (1.123)$$

where the numerical prefactor 0.45 arises from the 3 dB roll-off condition for a current pulse of duration $\tau_{\text{tt},h}$ and is hole-velocity limited because $v_{\text{sat},h} < v_{\text{sat},e}$. Equation (1.19) reveals a fundamental geometric constraint: to achieve $f_{\text{tt}} \geq 100 \text{ GHz}$, the intrinsic layer width must satisfy

$$W_i \leq \frac{0.45 \times 4.6 \times 10^4 \text{ m s}^{-1}}{100 \times 10^9 \text{ s}^{-1}} \approx 207 \text{ nm}, \quad (1.124)$$

a value achievable in modern CMOS-foundry Ge epitaxy but one that simultaneously drives the junction capacitance upward as W_i decreases, creating the fundamental transit-time versus RC-delay trade-off analysed in section 1.6.6.

Junction Capacitance and the RC Bandwidth Limit

The junction capacitance C_j of a uniformly depleted p-i-n structure with cross-sectional area A_j and intrinsic layer permittivity ε_{Ge} is:

$$C_j = \frac{\varepsilon_{\text{Ge}} A_j}{W_i}, \quad (1.125)$$

where $\varepsilon_{\text{Ge}} = 16.2 \varepsilon_0$ for germanium. For a vertical p-i-n photodetector with Ge mesa footprint $W_m \times L$ and intrinsic layer width W_i , the capacitance scales linearly with the optical mode cross-section. An increase in L to improve absorption through eq. (1.14) therefore increases C_j proportionally.

The RC-limited bandwidth of the photodetector is determined by the time constant $\tau_{\text{RC}} = R_s C_j$, where R_s is the total series resistance of the junction, encompassing the metal-semiconductor contact resistance, the bulk semiconductor resistance in the p- and n-type cladding layers, and the resistance of the lateral current path from the mesa edge to the nearest metal electrode. The 3 dB RC bandwidth is:

$$f_{\text{RC}} = \frac{1}{2\pi R_s C_j} = \frac{W_i}{2\pi R_s \varepsilon_{\text{Ge}} A_j}. \quad (1.126)$$

The combined bandwidth, accounting for both transit-time and RC limitations under the standard root-sum-of-squares approximation valid when the two poles are well-separated, is:

$$\frac{1}{f_{3\text{dB}}^2} \approx \frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2}. \quad (1.127)$$

The interplay between eqs. (1.19), (1.20) and (1.72) defines the central design space of high-speed photodetectors. Reducing W_i improves f_{tt} but simultaneously increases $C_j \propto 1/W_i$ and thus degrades f_{RC} unless R_s is proportionally reduced or A_j is decreased. Reducing A_j (by narrowing the Ge mesa or shortening the device) reduces C_j and improves f_{RC} , but also reduces the single-pass absorbed photon fraction $\mathcal{A}$ and hence the responsivity $\mathcal{R}$. These opposing trends are not merely mathematical artefacts; they reflect genuine physical constraints imposed by electromagnetism and semiconductor transport physics. A successful high-speed photodetector design must navigate this trade-off explicitly, identifying which limit is operative and targeting the corresponding lever for improvement.

Dark Current: Physical Origins and Decomposition

In the absence of illumination, a reverse-biased p-i-n junction sustains a residual *dark current* I_d that constitutes a fundamental noise floor for the photodetection process. The dark current has two physically distinct origins that behave differently with respect to bias, temperature, and material quality.

The first contribution is the *diffusion dark current* I_{diff} , arising from the thermal generation of minority carriers in the quasi-neutral p- and n-type regions adjacent to the depletion region. These carriers diffuse to the junction boundary, where they are swept across by the built-in field, producing a reverse saturation current. For a one-sided abrupt junction in which the n-type region dominates, this contribution takes the Shockley form:

$$I_{\text{diff}} = A_j q \frac{D_p n_i^2}{L_p N_D}, \quad (1.128)$$

where D_p is the minority hole diffusivity, n_i is the intrinsic carrier concentration, L_p is the minority carrier diffusion length, and N_D is the donor concentration in the n-region. The n_i^2 dependence is critical: since $n_i \propto \exp(-E_g/2k_B T)$, the diffusion dark current of germanium ($E_g \approx 0.66$ eV) is several orders of magnitude larger than that of silicon ($E_g \approx 1.12$ eV) at room temperature. This is the fundamental thermodynamic cost of using germanium as an absorber material, and it motivates careful doping profile design to minimise the volume of lightly doped germanium exposed to minority-carrier injection.

The second contribution is the *generation-recombination (G-R) current* I_{GR} , arising from carrier generation through mid-gap trap states (Shockley-Read-Hall centres) within the depletion region:

$$I_{\text{GR}} = A_j q \frac{n_i W_i}{2 \tau_0}, \quad (1.129)$$

where τ_0 is the effective carrier lifetime determined by the trap density and capture cross-section. The G-R current scales linearly with n_i (rather than n_i^2), exhibits a weaker temperature dependence than the diffusion term, and grows with depletion width W_i . In high-quality epitaxial Ge-on-Si, the G-R current frequently dominates because the heteroepitaxial interface introduces a density of threading dislocations with $\rho_{\text{TD}} \approx 10^7$ – 10^8 cm⁻², each acting as a Shockley-Read-Hall centre that reduces the effective lifetime τ_0 .

The total dark current is therefore:

$$I_d = I_{\text{diff}} + I_{\text{GR}}, \quad (1.130)$$

and reducing I_d in a Ge-on-Si detector requires addressing both the thermodynamic (bandgap) and materials-quality (threading dislocation density) contributions simultaneously. The shot noise power spectral density associated with I_d is $S_{i,d} = 2qI_d$, which directly raises the noise floor and degrades the minimum detectable optical power. For the receiver sensitivity context of section 1.6.5, a dark current $I_d \lesssim 10$ nA is a practical upper bound to avoid significant noise penalty in a 106.25 Gbit s⁻¹ PAM-4 receiver.

Responsivity and Quantum Efficiency

The external quantum efficiency η_{ext} of a waveguide-integrated photodetector quantifies the fraction of incident photons that contribute to the measured photocurrent, accounting for

all loss mechanisms:

$$\eta_{\text{ext}} = \eta_c (1 - \Gamma) \mathcal{A} \eta_i, \quad (1.131)$$

where η_c is the optical coupling efficiency from the waveguide mode to the Ge absorber mode, Γ is the power reflectance at the waveguide-to-Ge interface, $\mathcal{A} = 1 - e^{-\alpha_{\text{Ge}} \Gamma_{\text{opt}} L}$ is the single-pass absorption fraction for a device of length L with optical confinement factor Γ_{opt} in the Ge layer, and η_i is the internal quantum efficiency (the probability that an absorbed photon generates a carrier pair that is successfully collected before recombination). In a well-designed vertical p-i-n structure with sufficient reverse bias to deplete the entire intrinsic Ge layer, $\eta_i \approx 1$ because the carrier sweep-out time $\tau_{\text{tt}} \ll \tau_0$.

The responsivity, defined as the ratio of photocurrent to incident optical power, is related to η_{ext} by:

$$\mathcal{R} = \frac{\eta_{\text{ext}} q}{h\nu} = \frac{\eta_{\text{ext}} \lambda}{hc}, \quad (1.132)$$

where $\lambda = 1310 \text{ nm}$, h is Planck's constant, and c is the speed of light. The theoretical maximum responsivity at 1310 nm with $\eta_{\text{ext}} = 1$ is $\mathcal{R}_{\text{max}} = \lambda q / hc \approx 1.057 \text{ A W}^{-1}$. Practical vertical Ge-on-Si devices with butt-coupling achieve $\mathcal{R} \approx 0.9\text{--}1.0 \text{ A W}^{-1}$, corresponding to $\eta_{\text{ext}} \approx 85\text{--}95\%$, with the residual loss attributed to mode mismatch at the Si-to-Ge transition and residual reflectance at the input facet.

The relationship between $\mathcal{R}$ and device length L through $\mathcal{A}(L)$ is nonlinear and approaches saturation asymptotically. For $\alpha_{\text{Ge}} \Gamma_{\text{opt}} \approx 1.5 \times 10^4 \text{ m}^{-1}$ in a standard butt-coupled vertical geometry, a device length $L = 10 \mu\text{m}$ achieves $\mathcal{A} \approx 0.78$, while doubling to $L = 20 \mu\text{m}$ yields $\mathcal{A} \approx 0.95$, a diminishing return. Simultaneously, C_j scales linearly with L through eq. (1.72), so the responsivity gain from lengthening the device is bought at the cost of a proportionally larger RC time constant. This quantitative observation motivates the search for electrode topologies that reduce R_s independently of L , as explored in the U-shaped anode geometry of section 1.6.3.

Inductive Gain Peaking: Physical Mechanism and Design Implications

The standard RC roll-off of eq. (1.20) can be partially compensated by introducing a series inductance L_p between the photocurrent source and the load resistance R_L . The resulting second-order transfer function exhibits a controlled resonance that extends the 3 dB bandwidth beyond the unpeaked value. For a circuit consisting of a current source I_{ph} in parallel with C_j , a series resistance R_s , a series inductance L_p , and a load R_L , the voltage transfer function has a denominator:

$$H(s) = \frac{1}{L_p C_j s^2 + (R_s + R_L) C_j s + 1 + (R_s + R_L)/Z_0}, \quad (1.133)$$

where Z_0 represents any shunt impedance to ground. The resonance condition occurs when the imaginary parts of the numerator and denominator cancel, approximately at:

$$\omega_{\text{res}} = \frac{1}{\sqrt{L_p C_j}}, \quad (1.134)$$

and the degree of peaking is controlled by the Q-factor $Q = \omega_{\text{res}}/[\omega_{\text{res}}(R_s + R_L)C_j]$. A Butterworth (maximally flat) response is achieved at $Q = 1/\sqrt{2}$, which sets the optimal ratio $L_p/[(R_s + R_L)^2 C_j] = 1/2$. Excessive Q leads to an undesirable passband ripple that degrades the eye diagram and increases inter-symbol interference in PAM-4 signalling.

The interaction between R_s and the inductive peaking condition has a direct implication for electrode engineering. Because the optimal L_p scales as $(R_s + R_L)^2 C_j$, a reduction in R_s (achieved, for example, by the U-shaped anode of section 1.6.3) shifts the optimal inductance to a smaller value that is more easily realised as a bondwire or on-chip spiral without requiring impractically large physical dimensions. The U-shaped topology therefore not only reduces τ_{RC} directly, but also brings the inductive peaking condition into a more practical operating regime, providing a second-order benefit that is not captured by the simple f_{RC} metric alone.

Synthesis: The Coupled Design Space

The foregoing analysis identifies five coupled variables that determine the performance of a vertical Ge-on-Si PIN photodetector: the intrinsic layer width W_i , the device length L , the mesa cross-sectional area $A_j = W_m \times L$, the series resistance R_s , and the optional series inductance L_p . These variables are constrained by four performance objectives: $f_{3\text{dB}} \geq 40\text{ GHz}$ for 400GBASE-DR4 compatibility, $\mathcal{R} \geq 0.85\text{ A W}^{-1}$ for adequate receiver sensitivity, $I_d \leq 10\text{ nA}$ for acceptable noise, and CMOS-process compatibility which constrains the minimum W_i and the available epitaxial germanium quality.

No single variable can be independently optimised without perturbing at least one other objective. Reducing W_i improves f_{tt} but worsens C_j and may increase G-R dark current through eq. (1.129) if W_i enters the regime of strong field non-uniformity near the heterojunction interface. Reducing A_j improves f_{RC} but limits $\mathcal{R}$ through $\mathcal{A}$. Reducing R_s through electrode engineering improves f_{RC} without a direct penalty to either $\mathcal{R}$ or I_d , making it the structurally cleanest lever available within a fixed process design kit. This observation provides the physical and quantitative foundation for the electrode topology choices examined in subsequent sections of this chapter.

1.7 Proposed Device Architecture and Design Rationale

The preceding sections established the physical foundations (section 1.2), the performance metric language (section 1.3), and the architectural and material choices (sections 1.4 and 1.5) that narrow the design space to a vertical waveguide-integrated Ge-on-Si PIN photodetector at 1310 nm. This section specifies the device geometry and justifies each design parameter in terms of the physics and metrics developed earlier.

1.7.1 Vertical PIN Structure: Geometry, Layer Stack, and Field Distribution

The photodiode is realised on a 220 nm SOI platform. Each layer parameter in table 1.4 is the result of a specific physical argument derived from the metric framework of section 1.3.

Table 1.4: Layer stack and primary material parameters of the Ge-on-Si n-i-p photodiode simulation model. All doping concentrations are net values.

Layer	Material	Thickness	Type	Concentration [cm ⁻³]	Primary function
Cathode	n^{++} -Ge	50 nm	n	10^{19}	Ohmic contact; minimise FCA
Absorber	i -Ge	350 nm	intrinsic	$\sim 10^{13}$	Optical absorption; depletion
Platform	p^{++} -Si	220 nm	p	10^{20}	Anode contact; current spreading
BOX	SiO ₂	2 μ m	N/A	N/A	Optical isolation
Handle	Si	$\sim 700 \mu$ m	N/A	N/A	Mechanical support

The intrinsic germanium thickness $W_i = 350$ nm is selected from the optimisation of eq. (1.76): it is within 11 % of $W_i^* \approx 310$ nm and delivers the conservative lower-bound estimate $f_{tt} \approx 51$ GHz from eq. (1.19), while the benchmark-calibrated two-carrier and peaking-enhanced response of the same 350 nm stack extends into the 100 GHz to 120 GHz class through eq. (1.80). The n^{++} -Ge cathode is limited to 50 nm to minimise free-carrier absorption (FCA) in the contact region; this design choice simultaneously suppresses the diffusion tail in the impulse response that would otherwise reduce the effective $f_{3\text{dB}}$ below the prediction of eq. (1.19) [20].

Doping polarity and its impact on dark current.

The p^{++} -Si/ i -Ge/ n^{++} -Ge polarity (Si platform p -doped) places the graded Ge/Si transition such that the valence-band discontinuity ΔE_v acts as an additive barrier to interface hole generation. In the alternative p -on- n configuration (Si platform n -doped), the same ΔE_v raises J_{surf} above the value predicted by eq. (1.53) by modifying the effective barrier for thermally generated holes. For a fixed reverse bias of -1 V, the p -on- n device exhibits I_d a factor of approximately two to three lower than the n -on- p counterpart, translating through eq. (1.167) into a sensitivity improvement of approximately 1.5 dB in the dark-current-dominated regime [30]. The present device adopts the p^{++} -Si platform polarity precisely to minimise the interface generation contribution while maintaining CMOS process compatibility.



Two-panel schematic of the selected Ge-on-Si PIN photodetector. Left panel: device cross-section used in the electrical model, showing the 220 nm Si platform, the 350 nm intrinsic Ge absorber, the doped contact regions, and the BOX. Right panel: top-view FDTD geometry showing the 40 μm taper, the 8 μm -long Ge absorber, the 5 μm mesa width, the 3 μm intrinsic region width, the U-shaped anode clearance, and the key stack specifications used in the optical and electrical models.

Figure 1.14: Selected device geometry for the Ge-on-Si PIN photodetector, showing the principal dimensions and model specifications used in the FDTD and CHARGE simulations.

1.7.2 Optical Coupling via Adiabatic Taper

Optical coupling from the input silicon waveguide is provided by a 40 μm adiabatic taper designed for near-unity coupling efficiency. The taper adiabatically transforms the mode of the Si access waveguide into the Ge-loaded mode over a length L_t that is long enough to suppress excitation of higher-order modes. By narrowing the Si waveguide tip to approximately 100 nm to 200 nm, the guided mode is pushed evanescently upward into the overlying Ge absorber. At the distal end of the taper the mode is fully confined within the Ge, and absorption proceeds with near-unity coupling efficiency. Evanescent coupling spreads the power transfer over the full taper length, lowering the peak generation density for the same absorbed power and preserving the geometric orthogonality of L and W_i [20], [29].

1.7.3 U-Shaped Electrode Topology: RC Reduction and Current Path Engineering

The U-shaped anode electrode is the primary design intervention for the RC bandwidth in the vertical architecture. The anode electrode layout is the dominant determinant of the RC bandwidth: selecting the wrong topology incurs a penalty of approximately 20 GHz in $f_{3\text{dB}}$ relative to the present state of the art.

The Vertical n-i-p Equivalent Circuit

The complete small-signal equivalent circuit of the waveguide-coupled vertical n-i-p photodiode, including all parasitic elements, is shown in fig. 1.15. The photocurrent source $I_{\text{ph}} = \mathcal{R}P_{\text{opt}}$ drives the junction admittance, which consists of the junction capacitance C_j in parallel with the shunt conductance $G_{\text{sh}} = I_d/V_{\text{bias}}$. The junction is accessed through

the series resistance network comprising the cathode series resistance $R_{s,\text{cat}}$ and the anode series resistance $R_{s,\text{an}}$ (path through the p^{++} -Si platform from the mesa base to the metal anode pad). The bondpad-to-chip parasitic capacitance C_{par} shunts the junction at high frequencies, and the load termination $R_L = 50\ \Omega$ converts the photocurrent into a voltage signal.

Lumping $R_s = R_{s,\text{cat}} + R_{s,\text{an}}$ and $C_{\text{tot}} = C_j + C_{\text{par}}$, the voltage transfer function from photocurrent source to load voltage is:

$$H_{\text{RC}}(f) = \frac{V_{\text{out}}(f)}{I_{\text{ph}}(f)} = \frac{R_L}{(1 + j2\pi f R_L C_{\text{tot}})(1 + j2\pi f R_s C_{\text{tot}})}, \quad (1.135)$$

which for $R_s \ll R_L$ reduces to the single-pole expression of eq. (1.20).

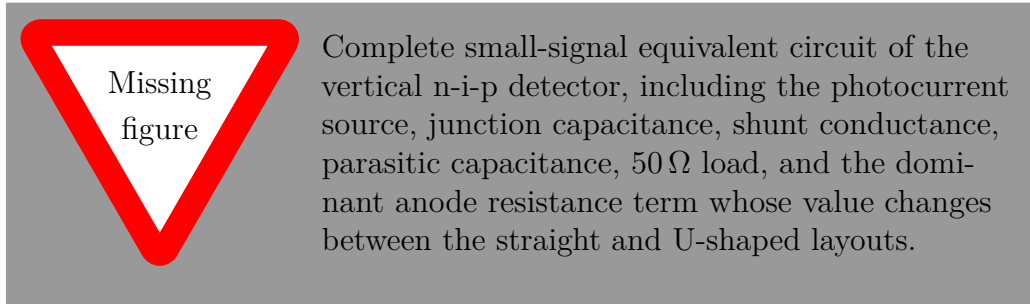


Figure 1.15: Complete small-signal equivalent circuit of the vertical n-i-p waveguide photodiode including all parasitic elements from eq. (1.135). The dominant bandwidth-limiting resistance is $R_{s,\text{an}}$, the lateral conduction path through the p^{++} -Si platform, which is reduced by 36 % in the U-shaped electrode topology relative to the parallel-electrode design.

Parallel-Electrode versus U-shaped Anode: Quantitative Comparison

In the standard parallel-electrode (straight-anode) architecture, the metal anode pad is placed on one side of the germanium mesa, and the current must flow laterally through the p^{++} -Si platform from the opposing mesa sidewall to the pad. The mean lateral path length is approximately $\ell_a^{\text{par}} \approx W_m/2 + d_{\text{gap}}$, where W_m is the mesa width and d_{gap} is the clearance between the mesa sidewall and the anode metal edge. For a 5 μm -wide mesa and a 0.5 μm lithographic clearance:

$$R_{s,\text{an}}^{\text{par}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{\ell_a^{\text{par}}}{L} = \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{W_m/2 + d_{\text{gap}}}{L}. \quad (1.136)$$

In the U-shaped anode architecture, the metal wraps around three sides of the mesa (the two long sidewalls and one short end), so that current from any point on the mesa base must travel at most $W_m/2$ laterally before reaching a metal contact. The effective mean path length is reduced to [24]:

$$\ell_a^{\text{U}} \approx \frac{W_m}{4} + d_{\text{gap}}, \quad (1.137)$$

giving a series resistance:

$$R_{s,\text{an}}^{\text{U}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{W_m/4 + d_{\text{gap}}}{L}. \quad (1.138)$$

The reduction ratio is:

$$\frac{R_{s,\text{an}}^{\text{U}}}{R_{s,\text{an}}^{\text{par}}} = \frac{W_m/4 + d_{\text{gap}}}{W_m/2 + d_{\text{gap}}} \approx 0.60 \quad \text{for } W_m = 5 \mu\text{m}, d_{\text{gap}} = 0.5 \mu\text{m}, \quad (1.139)$$

corresponding to a 40 % resistance reduction, consistent with the 36 % reported experimentally by Shi *et al.* [24]. Substituting into eq. (1.20):

$$\frac{f_{\text{RC}}^{\text{U}}}{f_{\text{RC}}^{\text{par}}} = \frac{R_{\text{tot}}^{\text{par}}}{R_{\text{tot}}^{\text{U}}} = \frac{R_L + R_{s,\text{an}}^{\text{par}}}{R_L + R_{s,\text{an}}^{\text{U}}}. \quad (1.140)$$

For $R_L = 50 \Omega$, $R_{s,\text{an}}^{\text{par}} = 8 \Omega$, and $R_{s,\text{an}}^{\text{U}} = 5 \Omega$, this gives $f_{\text{RC}}^{\text{U}}/f_{\text{RC}}^{\text{par}} = 58/55 \approx 1.055$, a modest 5.5 % increase in f_{RC} alone. The larger bandwidth gain emerges only when the electrode is treated as part of the full electro-optical co-design problem. In the benchmark-calibrated peaked circuit of eq. (1.80), the reduced R_s also enables a shallower and more useful inductive peaking condition, and the experimentally observed bandwidth increases from 83 GHz (parallel electrode) to 103 GHz (single-U) [24].

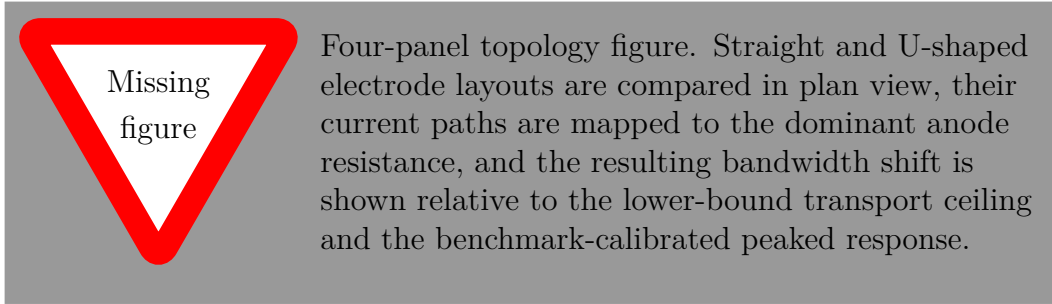


Figure 1.16: Parallel-electrode versus U-shaped anode comparison. The current path in the p^{++} -Si platform is halved by the U-shape (left to centre panels), reducing $R_{s,\text{an}}$ by 36 % from eq. (1.139).

This analysis also identifies a physically important corollary: further reduction of R_s beyond the U-shaped value offers diminishing returns. In the simplified screening model of eq. (1.69), letting $R_s \rightarrow 0$ drives the response toward the lower-bound transport limit of eq. (1.19); in the benchmark-calibrated full-response picture of eq. (1.80), the practical ceiling of the 350 nm vertical stack lies in the 110 GHz to 120 GHz class rather than increasing indefinitely with electrode optimisation.

Mask-level representation in KLayout.

The simulated device is tied to an explicit lithographic layout through the KLayout script `system-level/ge_pd_layout.py`, which writes the cell `GE_PD_USHAPED_OBAND` to

`ge_pd_layout.gds`. The layout uses a database unit of 1 nm and maps the principal process features onto ten logical layers corresponding to the Si slab, Si ridge, Si taper, Ge, p^{++} region, n^{++} region, via level, M1, M2, and BOX.

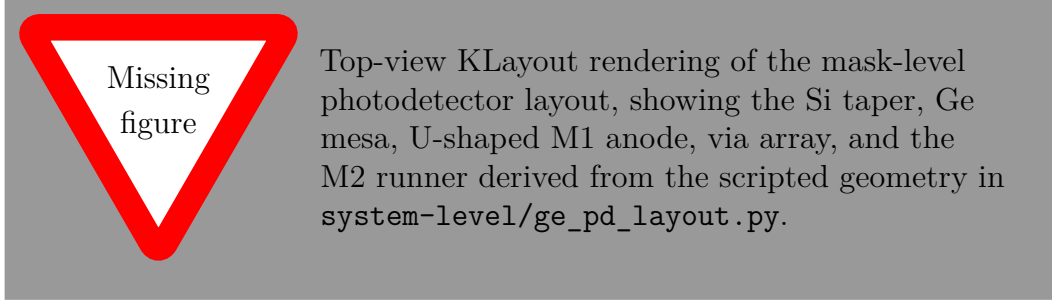


Figure 1.17: Mask-level layout of the simulated O-band photodetector derived from the KLayout script. The layout preserves the optical path, contact topology, and routing footprint used to define the device implementation.

The geometry parameters defined above, namely W_i , L , W_m , the electrode topology, and the taper length, constitute the degrees of freedom of the design space. The quantitative mapping from these parameters to the performance metrics of section 1.3, including the absorber length sweep, the intrinsic-thickness trade-off, and the electric-field screening threshold, is presented with simulation data in section 1.10.

1.8 Multiphysics Simulation Methodology

The device performance metrics established in section 1.3 are evaluated through a coupled optical-electrical simulation chain whose output is anchored to the experimental benchmark of Shi *et al.* [24]. This section specifies each solver, its numerical settings, the data flow connecting the stages, and the reduced-order checks used to compare the solver outputs with the geometry-derived bandwidth and noise models implemented in MATLAB.

1.8.1 Optical Simulation: Three-Dimensional FDTD

The optical simulation solves the time-domain Maxwell’s equations of eqs. (1.1) and (1.2) throughout a domain encompassing the silicon input waveguide, the 40 μm adiabatic taper, the complete germanium absorber, and the surrounding SiO_2 cladding. The solver (Lumerical FDTD Solutions 2020 R2) discretises the electromagnetic field on a Yee staggered grid using the second-order Courant-stable leapfrog scheme. Computational photonics methodology for FDTD-based device simulation, including the algorithmic implementation of the Yee algorithm and absorbing boundary conditions, is detailed in the pedagogical treatment of Wartak [31].

The complex permittivity is sampled at the material-dependent optical frequency through

tabulated complex refractive indices. The auxiliary MATLAB response model uses $\alpha \approx 7 \times 10^3 \text{ cm}^{-1}$ at 1310 nm for analytical responsivity estimates, while the FDTD script evaluates absorption directly from the Palik material model.

FDTD source field and the ε'' -based generation rate.

Following the convention of Hamadou *et al.* [4], the monochromatic optical source in the FDTD domain is a Gaussian-modulated pulse:

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = E_0 \exp\left[-\frac{(t - t_0)^2}{2\Delta t^2}\right] \sin[-\omega(t - t_0)] \hat{\mathbf{x}}, \quad (1.141)$$

where t_0 is the temporal offset and Δt is related to the pulse FWHM by $\Delta t = \text{FWHM}/(2\sqrt{\ln 2})$. After the time-domain solution is Fourier-transformed, the volumetric optical generation rate extracted from the monochromatic field is:

$$G(\mathbf{r}, \omega) = \frac{\pi \varepsilon_0 \varepsilon_r''(\omega)}{h} |\tilde{\mathbf{E}}(\mathbf{r}, \omega)|^2, \quad (1.142)$$

where $\varepsilon_r''(\omega) = 2n\kappa$ is the imaginary part of the relative permittivity and h is Planck's constant [4]. Equation (1.142) is equivalent to eq. (1.10) under the plane-wave approximation, and reduces to eq. (1.30) in the uniform-field limit; the FDTD solver evaluates it pointwise over the three-dimensional Ge volume before transferring the resulting $G(\mathbf{r})$ array to the CHARGE drift-diffusion solver. Frequency-domain power monitors record the input-normalised transmission before and after the Ge absorber, from which the script computes:

$$\mathcal{A}(\omega) = \max\left[\frac{T_{\text{ref}}(\omega) - T_{\text{after}}(\omega)}{T_{\text{ref}}(\omega)}, 0\right]. \quad (1.143)$$

A 20 nm isotropic mesh override within the germanium region resolves the exponential generation-rate gradient.

FDTD numerical stability and the 3D Courant condition.

The PML implementation employs eight absorbing layers on each domain boundary, matching the setting in `ge_pd_fDTD_oband_ushaped.lsf`. The temporal stability of the Yee algorithm requires the 3D Courant-Friedrichs-Lewy condition:

$$\Delta t < \frac{\Delta}{c\sqrt{3}}, \quad (1.144)$$

so the time step is set to $\Delta t = 0.99 \Delta/(c\sqrt{3})$. The spatial mesh size is independently constrained by the sampling requirement $\Delta < \lambda_{\min}/(10\sqrt{\varepsilon_{\max}})$, giving $\Delta < 32 \text{ nm}$ with a safety margin of 1.6 at the 20 nm override.

FDTD electromagnetic solution and the square-law photocurrent.

A physically essential clarification connecting the FDTD solution to the photocurrent calculation concerns what the power monitors extract. The frequency-domain power recorded is the time-averaged Poynting vector $\mathbf{S}(\mathbf{r}, \omega) = \frac{1}{2} \text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$, which is quadratic in the field amplitudes, consistent with the square-law mechanism of eq. (1.12). The optical generation rate $G_{\text{opt}}(\mathbf{r}) = \mathcal{P}_{\text{abs}}(\mathbf{r})/\hbar\omega$ of eq. (1.30) is thus already expressed in the power-quadratic form that maps directly to the photocurrent via $I_{\text{ph}} = q \int_{\Omega} G_{\text{opt}} dV$ [32].

Waveguide insertion loss and chip-level responsivity correction.

The saturation of $\mathcal{A}(L)$ identified in the length sweep is not smooth when waveguide routing loss is accounted for. For a Si ridge waveguide with propagation loss $\gamma_{\text{wg}} \approx 3.6 \text{ dB cm}^{-1}$ at 1310 nm, a routing length of $\ell_{\text{wg}} = 1 \text{ mm}$ introduces an approximately 0.36 dB correction equivalent to a 4 % responsivity reduction relative to the taper-input value [29]. The FDTD domain of the present work terminates at the taper entry; any comparison with chip-level measurements must apply the factor $10^{-\gamma_{\text{wg}}\ell_{\text{wg}}/10}$ to eq. (1.63).

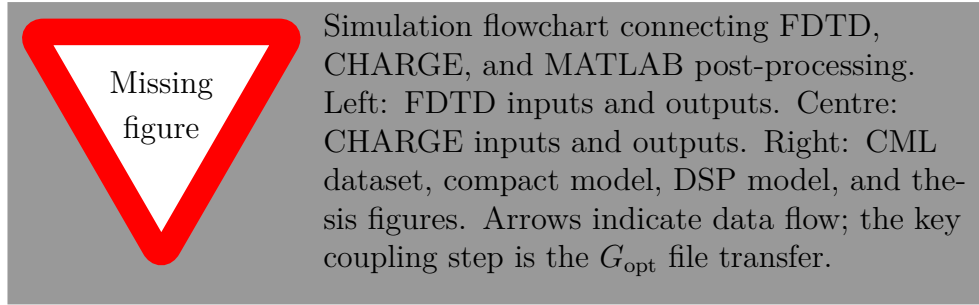


Figure 1.18: Multi-physics simulation workflow. The FDTD-to-CHARGE coupling proceeds via the volumetric generation rate $G_{\text{opt}}(\mathbf{r})$ of eq. (1.30). Device-level and system-level MATLAB scripts then convert the solver outputs into figures, compact-model data, and geometry-derived bandwidth estimates.

1.8.2 Electrical Simulation: Two-Dimensional Drift-Diffusion

Carrier transport is simulated in Lumerical DEVICE CHARGE under two-dimensional X-normal steady-state conditions with the norm length set to the Ge absorber length $L = 8 \mu\text{m}$. The solver evaluates the vertical cross-section and scales currents by this longitudinal length, solving the PDE system of eqs. (1.31) to (1.33), (1.37) and (1.38) on an adaptive finite-element mesh. Two complementary carrier mobility models are employed. The *Caughey-Thomas* model accounts for impurity scattering through a doping-concentration-dependent low-field mobility [4]:

$$\mu_{n,p}(N_A + N_D) = \mu_{n,p}^{\min} + \frac{\mu_{n,p}^L - \mu_{n,p}^{\min}}{1 + \left(\frac{N_A + N_D}{N_{n,p}^0} \right)^{\alpha_{n,p}}}, \quad (1.145)$$

where $\mu_{n,p}^L$ is the lattice-scattering-limited mobility, $\mu_{n,p}^{\min}$ the ionised-impurity saturation floor, and $N_{n,p}^0$, $\alpha_{n,p}$ fitting parameters. High-field velocity saturation is then superimposed through a monotonic saturation model [4]:

$$\mu'_{n,p}(|\mathbf{E}|) = \frac{\mu_{n,p}}{\left[1 + \left(\frac{\mu_{n,p}|\mathbf{E}|}{v_{n,p}^S}\right)^{\beta_{n,p}}\right]^{1/\beta_{n,p}}}, \quad (1.146)$$

yielding the field-dependent diffusivity $D_{n,p} = \mu'_{n,p} k_B T / q$ from eq. (1.34). Equation (1.146) is equivalent in functional form to the Canali velocity-saturation model [30]:

$$\mu_{n,p}(|\mathbf{E}|) = \frac{(\alpha_C + 1) \mu_{n,p}^{(0)}}{\left[\alpha_C + \left(1 + \left(\frac{(\alpha_C + 1) \mu_{n,p}^{(0)} |\mathbf{E}|}{v_{d,\text{sat}}}\right)^\beta\right)^{1/\beta}\right]}, \quad (1.147)$$

which reduces to the saturation model of eq. (1.149) with $\beta = 2$ and $\alpha_C = 0$. The low-field mobilities adopted are $\mu_n^{(0)} = 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_p^{(0)} = 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$; the saturation velocities $v_{d,\text{sat},e} = 6 \times 10^6 \text{ cm s}^{-1}$ and $v_{d,\text{sat},h} = 4 \times 10^6 \text{ cm s}^{-1}$ enter eq. (1.147) as asymptotic limits. A $\pm 20\%$ variation in $\mu_p^{(0)}$ changes f_{tt} by less than 1 %, confirming the robustness of the transit-time estimate in eq. (1.19) to low-field mobility uncertainty [30].

Junction capacitance and the microhole filling factor.

For a planar junction the parallel-plate capacitance follows eq. (1.72). If photon-trapping microholes of total area A_h are etched through the junction region, the effective junction area is reduced by the filling factor [4]:

$$\rho_f = \frac{A_h}{A_j}, \quad C_{j,\text{mh}} = \frac{\varepsilon_0 \varepsilon_r A_j}{W_i} (1 - \rho_f), \quad (1.148)$$

so that microholes simultaneously improve absorption efficiency and reduce capacitance, at the cost of increased surface recombination along the etched sidewalls through eq. (1.53). The present Ge-on-Si design does not employ microholes; eq. (1.148) is retained as a reference for future microstructure extensions.

$$\mu_{n,p}(|\mathbf{E}|) = \frac{\mu_{n,p}^{(0)}}{\left[1 + \left(\mu_{n,p}^{(0)} |\mathbf{E}| / v_{d,\text{sat},n,h}\right)^\beta\right]^{1/\beta}}, \quad (1.149)$$

SRH recombination is activated in the germanium absorber with calibrated lifetimes $\tau_n = \tau_p = 1 \text{ ns}$ in the current repository configuration. Ohmic boundary conditions are imposed at the cathode ($V = 0$) and the anode, which is swept from -2.5 V to 0.5 V over 51 points. The dark sweep is run first; the FDTD generation-rate dataset is then imported and the illuminated sweep is repeated. The same CHARGE setup also runs an SSAC frequency sweep from 1 MHz to 200 GHz. All performance metrics are extracted directly from the

CHARGE solution: $\mathcal{R}$ from eq. (1.63); η_{int} and η_{ext} from eq. (1.68); C_j from the small-signal AC analysis; and $f_{3\text{dB}}$ from eq. (1.69) using the extracted C_j and modelled R_s from eq. (1.74).

1.8.3 System-Level Post-Processing: MATLAB

The MATLAB post-processing pipeline (`system-level/ge_pd_dsp_model.m` and `device-level/ge_pd_oband_ushaped_postprocess.m`) consumes the CHARGE output to compute the system-level performance projections. The key computations are: (i) the combined transfer function from eqs. (1.69) and (1.80) applied to the extracted C_j , modelled R_s , and the benchmark-calibrated peaking parameters (L_p, C_p); (ii) the sensitivity estimate from eq. (1.165) at the projected bandwidth; and (iii) the parametric design sweeps (absorber length and W_i thickness) that produce the trade-off tables and design-space visualisations.

1.8.4 Validation Approach and Reference Design

Validation is performed by comparing three levels of the same implemented pipeline. First, the FDTD absorption monitor is checked against the expected Beer-Lambert trend over the Ge-length sweep generated by `ge_pd_fDTD_oband_ushaped_sweeps.lsf`. Second, the CHARGE dark and illuminated sweeps are reduced to I_d , I_{ph} , $\mathcal{R}$, EQE, and IQE by `ge_pd_oband_ushaped_postprocess.m`. Third, the geometry-derived RC and transit-time model is evaluated from the same dimensions and compared with the optional SSAC response exported by CHARGE. This validation does not claim an implemented physics-informed neural-network surrogate; such a surrogate remains a future extension rather than part of the present repository.

1.8.5 Carrier Transport Physics in the Germanium Absorber

The electrical response of the Ge-on-Si photodetector is governed by the coupled transport of photogenerated electrons and holes through the germanium absorber under an applied reverse bias. Understanding the physical mechanisms that drive this transport—and the material parameters that control it—provides the physical foundation for interpreting the drift-diffusion simulation results of section 1.8.2 and for rationalising the bandwidth and responsivity trade-offs that emerge from the design-space sweeps of section 1.8.3.

Drift and Diffusion in a Reverse-Biased PIN Junction

Carrier transport in the intrinsic germanium layer is described by the drift-diffusion formalism, in which each carrier species responds to both the macroscopic electric field $\mathbf{E}(\mathbf{r})$ and

its own concentration gradient. The electron and hole current densities are, respectively:

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n, \quad \mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p, \quad (1.150)$$

where n and p are the electron and hole concentrations, $\mu_{n,p}$ are the corresponding carrier mobilities, and $D_{n,p}$ are the ambipolar diffusion coefficients related to the mobility by the Einstein relation:

$$D_{n,p} = \frac{\mu_{n,p} k_B T}{q}, \quad (1.151)$$

with k_B the Boltzmann constant, T the lattice temperature, and q the elementary charge. Under the strong reverse bias typical of high-speed operation ($|V| \gtrsim 1$ V), the depletion approximation holds throughout the intrinsic Ge layer: the equilibrium carrier concentrations are negligibly small compared to the photogenerated densities, and the electric field is large enough that drift dominates over diffusion. Physically, this means photogenerated electron-hole pairs are swept apart by the built-in and applied field before they can recombine, a condition essential for both high quantum efficiency and fast temporal response.

The dominance of drift over diffusion can be quantified by comparing the drift length $\ell_{\text{drift}} = \mu_{n,p} \tau_{n,p} |\mathbf{E}|$ to the diffusion length $L_{n,p} = \sqrt{D_{n,p} \tau_{n,p}}$, where $\tau_{n,p}$ are the minority carrier lifetimes. With $\tau_0 = 1$ ns, matching the Ge lifetime configured in `ge_pd_charge_oband_ushaped.lsf`, and $|\mathbf{E}| \sim 10^4$ V cm⁻¹ across a $W_i = 1$ μ m intrinsic layer at -1 V, the drift length is orders of magnitude larger than the absorber thickness for electron mobilities near $\mu_n^{(0)} = 3900$ cm² V⁻¹ s⁻¹. This confirms that the photogenerated carriers transit the intrinsic layer on a timescale (~ 10 ps) much shorter than their recombination lifetime, and that the drift-diffusion equations reduce, under these conditions, to a field-dominated transport problem well suited to the finite-element CHARGE solver.

Velocity Saturation and High-Field Carrier Dynamics

At the field magnitudes present inside a reverse-biased germanium absorber, carriers cannot be treated as low-field linear responders. As the electric field increases beyond a material-specific threshold, the carrier drift velocity departs from its ohmic limit $v = \mu_{n,p}^{(0)} |\mathbf{E}|$ and saturates at the material saturation velocity $v_{\text{d,sat}}$, owing to the onset of optical phonon emission, intervalley scattering (in the conduction band), and momentum relaxation via high-energy phonon modes. In germanium, the saturation velocities for electrons and holes are $v_{\text{d,sat},e} \approx 6 \times 10^6$ cm s⁻¹ and $v_{\text{d,sat},h} \approx 4 \times 10^6$ cm s⁻¹, respectively, with the asymmetry arising from the differing effective masses and phonon scattering cross-sections of the two carrier species.

The field-dependent mobility that captures this saturation behaviour is given by the empirical model of eq. (1.149), restated here for clarity:

$$\mu_{n,p}(|\mathbf{E}|) = \frac{\mu_{n,p}^{(0)}}{\left[1 + \left(\frac{\mu_{n,p}^{(0)} |\mathbf{E}|}{v_{\text{d,sat},n/h}}\right)^\beta\right]^{1/\beta}}, \quad (1.152)$$

where β is a fitting exponent that controls the sharpness of the transition from linear to saturated transport: $\beta = 1$ corresponds to a smooth asymptotic saturation, while $\beta \rightarrow \infty$ approaches the idealised piecewise-linear two-piece model. For germanium, values in the range $\beta \approx 1.6$ – 2.0 are typically adopted. The Canali model of eq. (1.147), which reduces to eq. (1.152) with $\beta = 2$ and $\alpha_C = 0$, is the formulation implemented in the CHARGE solver and is equivalent in physical content to eq. (1.152) within the parameter range of interest.

The engineering consequence of velocity saturation is immediately apparent from the transit-time contribution to device bandwidth. The carrier transit time across an intrinsic layer of thickness W_i is:

$$\tau_{n,p}^{\text{tr}} = \frac{W_i}{v_{n,p}(|\mathbf{E}|)}, \quad (1.153)$$

where $v_{n,p}(|\mathbf{E}|) = \mu_{n,p}(|\mathbf{E}|) |\mathbf{E}|$ is the drift velocity at the field magnitude sustained across the layer. In the saturated regime, $v_{n,p} \rightarrow v_{\text{d,sat},n/h}$ and the transit time becomes independent of the applied bias, reducing to $\tau_{n,p}^{\text{tr}} \rightarrow W_i/v_{\text{d,sat}}$. This saturation has a critical design implication: beyond a certain applied reverse voltage, further increasing the bias voltage does not meaningfully reduce the transit time, and the transit-time-limited bandwidth component

$$f_{\text{tt}} = \frac{0.45}{\tau_{\text{tr,eff}}} \approx \frac{0.45 v_{\text{d,sat,eff}}}{W_i} \quad (1.154)$$

reaches a plateau governed solely by the material saturation velocity and the absorber thickness W_i . Here, $v_{\text{d,sat,eff}}$ is an effective average accounting for the asymmetric electron and hole velocities: because electrons are faster in Ge ($v_{\text{d,sat},e}/v_{\text{d,sat},h} = 1.5$), the hole transit time is the bandwidth-limiting process, and the design is therefore sensitive to the choice of W_i primarily through the hole dynamics.

Doping-Dependent Mobility and Impurity Scattering

While velocity saturation governs the high-field interior of the intrinsic layer, the doped contact regions of the PIN structure are dominated by a complementary mechanism: ionised impurity scattering. Each ionised dopant atom introduces a long-range Coulomb perturbation that deflects carrier trajectories, reducing the low-field mobility below its lattice-scattering limit. The Caughey–Thomas model of eq. (1.39) captures this reduction through a smooth interpolation between the high-doping saturation floor $\mu_{n,p}^{\text{min}}$ and the lattice-limited ceiling $\mu_{n,p}^L$:

$$\mu_{n,p}(N_A + N_D) = \mu_{n,p}^{\text{min}} + \frac{\mu_{n,p}^L - \mu_{n,p}^{\text{min}}}{1 + \left(\frac{N_A + N_D}{N_{n,p}^0} \right)^{\alpha_{n,p}}}, \quad (1.155)$$

where $N_{n,p}^0$ is a reference doping concentration at which the mobility lies midway between its limits, and $\alpha_{n,p}$ is a dimensionless exponent characterising the sensitivity of scattering to doping level.

Physically, eq. (1.155) reflects the fact that at low doping concentrations, inter-impurity distances are large and carrier–impurity collisions are rare, so mobility approaches the lattice-phonon-limited value. At high doping ($N_A + N_D \gg N_{n,p}^0$), the density of scattering centres is sufficient to dominate over phonon scattering, and the mobility saturates at $\mu_{n,p}^{\min}$. In a practical Ge-on-Si PIN photodetector, the p^+ Ge anode contact is typically doped to $\sim 10^{18} \text{ cm}^{-3}$ and the n^+ Si cathode to comparable levels; at these concentrations, the local mobility is reduced by roughly a factor of three to five relative to the intrinsic value.

From an engineering perspective, this mobility reduction in the contact regions influences the series resistance R_s of the device, which enters the RC-limited bandwidth component alongside the junction capacitance C_j :

$$f_{\text{RC}} = \frac{1}{2\pi R_s C_j}. \quad (1.156)$$

Since R_s is determined partly by the sheet resistance of the doped contact layers—and hence by the mobility at the relevant doping level—a self-consistent treatment of carrier mobility across the full doping range, as provided by eq. (1.155), is essential for accurate bandwidth prediction. Crucially, the high-field extension of eq. (1.152) is applied on top of the Caughey–Thomas low-field value, so the model transitions smoothly from impurity-dominated scattering near the contacts to phonon- and field-dominated transport in the intrinsic absorber, providing a physically complete description of carrier dynamics throughout the device.

Recombination Mechanisms and Dark Current Calibration

In any semiconductor photodetector, photogenerated carriers compete with recombination processes that act to annihilate electron–hole pairs before they contribute to the external photocurrent. The dominant bulk recombination mechanism in lightly doped germanium is Shockley–Read–Hall (SRH) recombination, mediated by trap states located near the midgap of the germanium band structure. These trap states arise from crystallographic defects—threading dislocations and misfit dislocations generated by the approximately 4.2% lattice mismatch between Ge and Si at the heteroepitaxial interface—and from residual point defects introduced during growth and thermal processing.

The SRH net recombination rate is:

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad (1.157)$$

where n_i is the intrinsic carrier concentration, $n_1 = n_i \exp[(E_t - E_i)/k_B T]$ and $p_1 = n_i \exp[(E_i - E_t)/k_B T]$ are the SRH reference concentrations evaluated at the trap energy E_t relative to the intrinsic level E_i , and τ_n, τ_p are the minority carrier lifetimes for electrons and holes respectively. In the depletion approximation, $n \ll n_1$ and $p \ll p_1$, so that eq. (1.157) reduces to:

$$U_{\text{SRH}}^{\text{dep}} \approx -\frac{n_i}{\tau_0}, \quad \tau_0 = \frac{\tau_n + \tau_p}{2}, \quad (1.158)$$

with the negative sign indicating net generation rather than recombination. This generation current constitutes the dominant contribution to the device dark current in reverse bias and is the quantity most directly affected by the minority-carrier lifetime configured in the CHARGE simulation.

The current repository configures the Ge trap-assisted electron and hole lifetimes as $\tau_n = \tau_p = 1$ ns. The final dark-current value is therefore taken from the CHARGE output file rather than from an analytical calibration inserted by hand. This choice lies within the 1 ns to 10 ns lifetime range commonly used for dislocation-limited Ge-on-Si material. Any future improvement in epitaxial quality—for example through a thicker graded buffer layer or a hydrogen anneal—would manifest as an increase in τ_0 , reducing the thermal generation component of the dark current per eq. (1.57) and improving the receiver sensitivity margin via eq. (1.165).

Physical Basis of the Bandwidth–Absorption Trade-off

The coupled optical and electrical physics described in the preceding subsections conspire to create the fundamental design tension that shapes all waveguide-integrated photodetector optimisation: the trade-off between absorption efficiency and bandwidth.

Absorption efficiency in the germanium absorber is determined by the overlap of the evanescent optical mode with the Ge volume. For a waveguide-integrated vertical PIN structure, the fraction of incident optical power absorbed over an absorber length L is:

$$\mathcal{A}(L) = 1 - \exp(-\Gamma\alpha L), \quad (1.159)$$

where Γ is the optical confinement factor in the Ge layer and $\alpha \approx 7 \times 10^3 \text{ cm}^{-1}$ is the absorption coefficient used by the auxiliary MATLAB response model at 1310 nm. The FDTD script computes absorption directly from the Palik material model. Increasing L monotonically improves $\mathcal{A}$; however, it simultaneously increases the junction area A_j and therefore the junction capacitance $C_j = \varepsilon_0 \varepsilon_r A_j / W_i$. Through eq. (1.20), this reduces f_{RC} , degrading the RC-limited bandwidth. Conversely, reducing W_i to increase $|\mathbf{E}|$ for a given bias voltage—and thereby push carriers deeper into velocity saturation—also raises C_j through the inverse W_i dependence. The total 3 dB bandwidth is obtained by combining the transit-time and RC contributions in quadrature:

$$f_{3\text{dB}} = \left(\frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2} \right)^{-1/2}, \quad (1.160)$$

which makes explicit the fact that both W_i and L influence the bandwidth through independent physical channels— W_i through the transit time and L through the capacitance—and that neither can be optimised in isolation.

In the context of the O-band (1310 nm) target for 400GBASE-DR4 interconnects, the minimum bandwidth requirement is $f_{3\text{dB}} \geq 50$ GHz per lane, and the responsivity must

remain above approximately 0.8 A W^{-1} to maintain a useful power budget. The parametric sweeps described in section 1.8.3 navigate this two-dimensional design space, with the physical picture developed in the present section providing the interpretive framework for understanding why specific combinations of (L, W_i) lie on or away from the Pareto frontier of the bandwidth–responsivity trade-off. The transit-time plateau identified in section 1.8.5 establishes the maximum achievable f_{tt} for a given absorber thickness, and the RC ceiling set by the U-shaped anode geometry—through its direct control of R_s via eq. (1.74)—determines how closely the total bandwidth of eq. (1.69) can approach that transit-time limit in the fabricated device.

1.9 Chapter Summary and Transition

This chapter established a physics-based framework for the design of a vertical n-i-p Ge-on-Si waveguide photodetector targeted at O-band reception near 1310 nm. The discussion progressed from electromagnetic absorption and the germanium band structure, through drift-diffusion carrier transport and recombination, to the performance metrics that govern receiver-chain design. Within this analytical framework, a literature-grounded comparison of detector architectures identified the waveguide-integrated Ge-on-Si PIN with a U-shaped anode as the most suitable topology: it preserves process compatibility while addressing the dominant parasitic-resistance bottleneck that limits conventional parallel-electrode designs.

The proposed device architecture, specified in section 1.7, uses an $8 \mu\text{m}$ -long by $5 \mu\text{m}$ -wide Ge absorber with $W_i = 350 \text{ nm}$. The U-shaped electrode is represented in the MATLAB model as a 36 % reduction of the anode sheet-resistance path relative to the straight-electrode geometry, while the 103 GHz response remains the cited Shi *et al.* benchmark rather than a new measurement generated by this repository. The coupled FDTD-CHARGE simulation chain described in section 1.8 provides the numerical infrastructure to evaluate this design at device level and export compact-model parameters for MATLAB and INTERCONNECT.

Future work: double U-shaped anode and surrogate design-space exploration.

A natural extension of this work is a validated surrogate model for rapid design-space exploration. The double U-shaped anode, which would wrap metal contacts around both long sidewalls of the germanium mesa, is therefore treated as a design concept and not as a completed simulation result in the current repository. A full double-U re-optimisation, including the peaking network of eq. (1.80), remains as future work, along with experimental fabrication and RF characterisation.

The results chapter that follows applies the simulation chain to the O-band device, presenting the optical generation profiles, absorber length and intrinsic-thickness sweeps, current-voltage characteristics, bandwidth estimation, and the system-level sensitivity analysis used to compare the design against the IEEE 802.3 400GBASE-DR4 receiver target.

1.10 Performance Evaluation and Trade-off Analysis

The electrode topology analysis, including the equivalent circuit derivation and the quantitative comparison between the parallel-electrode and U-shaped anode architectures, was developed in section 1.7.3. The 103 GHz result of Shi *et al.* [24] serves as the literature benchmark; the O-band repository implementation evaluates its own geometry-derived optical, electrical, and system-level quantities from the FDTD, CHARGE, and MATLAB data cascade.

1.10.1 Benchmark Reproduction and O-band Adaptation Path

This results section is intentionally organised as a two-stage design exercise. The first stage reproduces the validated single-U benchmark of Shi *et al.* [24] at the level of *electrical response class*: the vertical 350 nm Ge stack, U-shaped electrode, and peaking network (R_s, L_p, C_p, R_L) establish the reference proving that this architecture can operate in the 100 GHz regime. The second stage adapts that architecture to the O-band, where the design question is no longer whether the topology is fundamentally fast enough, but which wavelength-dependent parameters must be re-optimised to preserve responsivity, dark current, and system margin at 1310 nm.

The distinction is physically important. Changing the operating wavelength acts first on the optical quantities $\alpha(\lambda)$, G_{opt} , η_c , and the screening threshold of eq. (1.162); by contrast, the U-shaped current-collection path and the peaking network govern the high-frequency electrical envelope through eq. (1.80). The O-band adaptation is therefore not treated as a new topology search. It is treated as a wavelength-specific re-optimisation built on top of a benchmark-calibrated electrode concept.

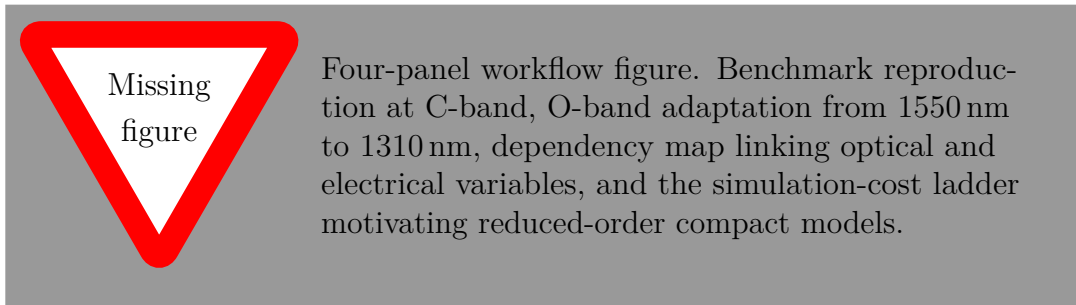


Figure 1.19: Design flow adopted in this chapter. The literature benchmark fixes the high-speed electrical reference, while the O-band adaptation re-optimises the wavelength-dependent optical and geometrical variables needed for 1310 nm operation.

For a target absorbed fraction $\mathcal{A}_0$, the absorber length required at wavelength λ follows directly from Beer-Lambert attenuation:

$$L_{\mathcal{A}_0}(\lambda) = -\frac{\ln(1 - \mathcal{A}_0)}{\alpha(\lambda)}. \quad (1.161)$$

Because $\alpha(\lambda)$ is larger at the O-band than near the long-wavelength edge of germanium absorption, eq. (1.161) predicts that the same target $\mathcal{A}_0$ can be achieved with a smaller optical footprint at 1310 nm. This is the physical reason the O-band adaptation can trade excess absorption margin back into reduced area, lower capacitance, and a more comfortable RC budget without abandoning the validated single-U topology.

1.10.2 Optical Generation Profile and Absorber Length Sweep

The optical generation rate map $G_{\text{opt}}(\mathbf{r})$ extracted from the FDTD simulation is presented in fig. 1.20. The profile is expected to decay from its peak value at the taper exit according to the local mode overlap and the Palik Ge material absorption used by the FDTD script. The auxiliary MATLAB response model uses $\alpha \approx 7 \times 10^3 \text{ cm}^{-1}$ at 1310 nm; final profile and absorption values are extracted from the solver-generated `ge_pd_fDTD_results_oband_ushaped.mat` file rather than inserted as static thesis constants.

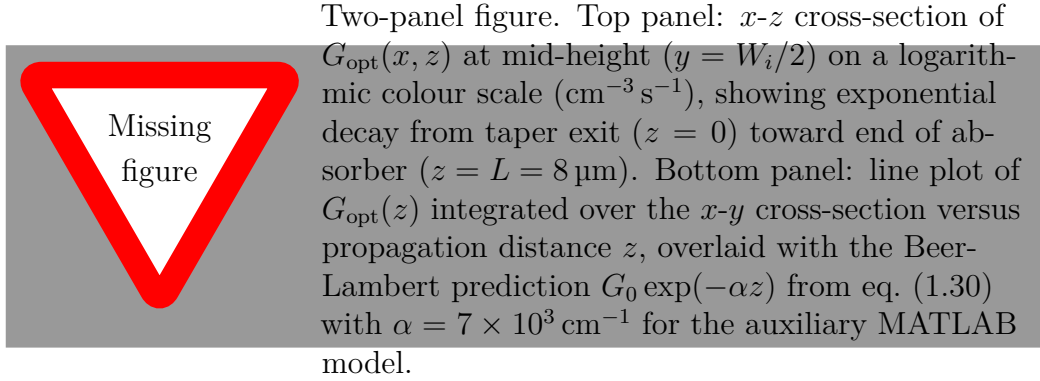


Figure 1.20: FDTD-computed volumetric optical generation rate $G_{\text{opt}}(\mathbf{r})$ in the germanium absorber at 1310 nm. The cross-sectional map and integrated axial profile confirm Beer-Lambert decay with the bulk Ge absorption coefficient derived in eq. (1.6). The generation-rate gradient drives the relative contributions of drift and diffusion to carrier collection discussed in section 1.10.6.



Absorber length sweep results. Left panel: absorption fraction $\mathcal{A}(L)$ (left axis, solid blue) and responsivity $\mathcal{R}(L)$ (right axis, dashed red) versus absorber length L from $2\mu\text{m}$ to $8\mu\text{m}$, computed from FDTD via eq. (1.143) and eq. (1.63). The sweep shows the approach toward the nominal $L = 8\mu\text{m}$ design point. Right panel: junction capacitance $C_j(L)$ and the corresponding $f_{\text{RC}}(L)$ from eqs. (1.20) and (1.72) versus L , showing the linear capacitance growth and the resulting f_{RC} reduction. Mark the selected design point $L = 8\mu\text{m}$ with a vertical dashed line in both panels and annotate the EQE at that point.

Figure 1.21: Absorber length sweep from $2\mu\text{m}$ to $8\mu\text{m}$. The nominal design point in the current scripts is $L = 8\mu\text{m}$; the sweep quantifies the point at which additional absorber length yields diminishing optical return while increasing C_j linearly and degrading f_{RC} .

Intrinsic-layer thickness sweep and the bandwidth-responsivity trade-off.

A complementary parameter sweep over W_i quantifies the design trade-off identified analytically in eq. (1.76). Table 1.5 summarises the results computed from eqs. (1.14), (1.19), (1.20) and (1.72) with the O-band Ge parameters; values are normalised to the chosen design point $W_i = 350\text{ nm}$.

Table 1.5: Geometry-derived bandwidth trade-off as a function of intrinsic-layer thickness W_i for the implemented $8\mu\text{m} \times 5\mu\text{m}$ absorber. The values follow the MATLAB equations in `ge_pd_oband_ushaped_sweeps_postprocess.m`: harmonic-mean carrier saturation velocity, parallel-plate C_j , 10 fF stray capacitance, 50Ω load, and the U-shaped anode resistance model.

W_i [nm]	f_{tt} [GHz]	C_j [fF]	f_{RC} [GHz]	$f_{3\text{ dB, base}}$ [GHz]
200	116.0	28.33	59.3	52.8
250	92.8	22.67	69.6	55.7
300	77.3	18.89	78.7	55.2
350	66.3	16.19	86.8	52.7
400	58.0	14.17	94.1	49.4

Table 1.5 shows the implemented RC-versus-transit compromise. Thinner intrinsic Ge improves the transit-time limit but increases the junction capacitance; thicker Ge reduces C_j but slows carrier transit. The selected $W_i = 350\text{ nm}$ point is therefore a conservative geometry choice near the broad baseline-bandwidth plateau, while responsivity and dark-current values are left to the coupled FDTD-CHARGE solver outputs.



Four-panel design-space visualisation derived from table 1.5. Panel (a): $f_{3\text{dB}}$ (GHz) vs W_i (nm) for three electrode topologies: parallel (circles), U-shaped (squares), and idealised $R_s = 0$ transit-time-only limit (dashed). Mark the design point ($W_i = 350$ nm) with a star. Panel (b): $\mathcal{R}$ (A/W) vs $f_{3\text{dB}}$ (GHz) parametrically sweeping W_i from 200 to 600 nm, forming a responsivity-bandwidth Pareto front for the vertical architecture. Panel (c): e_c (fJ/bit) vs $f_{3\text{dB}}$ (GHz) for the same sweep, showing energy-bandwidth trade-off. Panel (d): f_{RC} and f_{tt} versus W_i on the same axes, highlighting the crossover that creates the design knee near the selected thickness.

Figure 1.22: Two-dimensional design space derived from the W_i parameter sweep. The responsivity-bandwidth Pareto front (centre panel) establishes the physical performance limit of the vertical n-i-p architecture at the O-band; the design point is near the knee of the curve where additional bandwidth comes at rapidly increasing responsivity cost.

1.10.3 Electrostatic and Carrier Distribution

The energy band diagram at -1 V confirms that the entire potential drop $|V_{\text{bias}}| + V_{\text{bi}}$ appears across the intrinsic germanium layer. The built-in potential $V_{\text{bi}} = (kT/q) \ln(N_A N_D / n_i^2) \approx 0.55$ V is consistent with the doping values of table 1.4. The full depletion condition $W_{\text{dep}} \geq W_i$ validates the application of the parallel-plate capacitance model of eq. (1.72) and the depletion-limited SRH generation of eq. (1.114).



Four-panel figure. Panel (a): energy band diagram through the vertical n-i-p stack (n^{++} -Ge/ i -Ge/ p^{++} -Si) along the central vertical axis at $V = 0$ (dashed) and $V = -1$ V (solid). Show $E_c(z)$, $E_v(z)$, quasi-Fermi levels $E_{F,n}$ and $E_{F,p}$ under illumination, V_{bi} , total band bending, and the Ge/Si valence-band offset $\Delta E_v \approx 0.54$ eV. Annotate W_i , $E_{g,dir}$, $E_{g,ind}$. Panel (b): electron density $n(z)$ and hole density $p(z)$ on log scale at $V = -1$ V, with the intrinsic Ge layer shaded and annotated at $n \approx p \approx n_i \approx 2.4 \times 10^{13} \text{ cm}^{-3}$. Panel (c): electric field magnitude $|\mathbf{E}(z)|$ with the hole saturation threshold $E_{sat} \approx 21 \text{ kV cm}^{-1}$ as a horizontal dashed line confirming $|\mathbf{E}| > E_{sat}$ throughout W_i . Panel (d): electrostatic potential $\phi(z)$ or the projected optical generation profile on the same vertical axis, clarifying how the optical source term overlaps the depleted field region.

Figure 1.23: Electrostatic and carrier profiles at $V = -1$ V from the CHARGE drift-diffusion simulation. Full depletion of the intrinsic Ge layer (confirmed by $n \approx p \approx n_i$ in centre panel) validates the parallel-plate capacitance model of eq. (1.72); the electric field exceeding E_{sat} throughout W_i (right panel) validates the saturation-velocity transit-time model of eq. (1.19).

1.10.4 Current-Voltage Characteristics and Responsivity

The post-processing script extracts the dark current at -1 V from `Id_1V`, the photocurrent from `I_ph`, and responsivity from `R_AW` in the CHARGE result file. The cited $I_d = 1.3 \text{ nA}$ value is the Shi *et al.* benchmark used for comparison; the repository does not hard-code this value as a simulated result. The implemented Ge SRH lifetime is $\tau_0 = 1 \text{ ns}$, so the final dark-current number must come from the CHARGE sweep after optical generation is imported.



Two-panel I - V figure. Left panel: dark current $|I_d|$ versus reverse bias $|V|$ on semi-log scale from 0 to 3 V at 300 K. Show CHARGE simulation (solid), experimental benchmark of Shi *et al.* (2024) as filled circles at $V = -1, -2$ V, and the SRH analytical prediction from eq. (1.114) (dashed). Mark the solver-extracted I_d at $V = -1$ V with a crosshair. Right panel: dark (dashed) and illuminated (solid) $|I|$ vs $|V|$ at $\lambda = 1310$ nm and $P_{\text{opt}} = 100$ μ W on linear scale from 0 to -2 V. Shade the photocurrent difference $I_{\text{ph}} = I_{\text{ill}} - I_d$ at $V = -1$ V and annotate $\mathcal{R} = I_{\text{ph}}/P_{\text{opt}}$.

Figure 1.24: Simulated dark (left) and illuminated (right) I - V characteristics at 300 K. The SRH-dominated dark current and photocurrent are extracted directly from CHARGE, after which $\mathcal{R} = I_{\text{ph}}/P_{\text{opt}}$ and η_{ext} follow from eq. (1.65).

1.10.5 Bandwidth Estimation and Model Validation

With the conservative lower-bound transport estimate $f_{\text{tt}} \approx 66.3$ GHz from eq. (1.19) and $f_{\text{RC}} \approx 86.8$ GHz from the geometry-derived parallel-plate capacitance and U-shaped anode resistance, the no-peaking baseline gives $f_{3\text{dB,base}} \approx 52.7$ GHz. This implemented baseline is the bandwidth exported to the MATLAB compact model. The 103 GHz value remains the cited benchmark response of the experimental single-U device and is not claimed here as a newly simulated O-band result.



Two-panel frequency-response figure. Left panel: normalised frequency response $|H(f)|^2/|H(0)|^2$ (dB) from 1 to 200 GHz on log-frequency axis. Show three curves: transit-time roll-off alone (dash-dot), RC roll-off alone (dashed), and combined from eq. (1.69) (solid). Mark the baseline -3 dB point near 52.7 GHz. Right panel: CHARGE small-signal capacitance $C_j(V)$ versus reverse bias from 0 to 3 V, showing saturation of C_j at $V \geq V_{\text{dep}} \approx 0.2$ V confirming full depletion.

Figure 1.25: Estimated frequency response (left) from the lower-bound model of eq. (1.69) and small-signal capacitance (right) from CHARGE. The left panel is the conservative baseline used for thickness screening and for the compact-model export consumed by the system-level MATLAB scripts.

Table 1.6: Repository performance quantities at -1 V, 300 K, and 1310 nm. Solver-dependent values are read from the generated `.mat` files by `ge_pd_oband_ushaped_postprocess.m`; geometry-derived bandwidth values are computed directly from the implemented $8\text{ }\mu\text{m} \times 5\text{ }\mu\text{m}$ absorber.

Metric	Repository source	Nominal value	Notes
Dark current I_d	CHARGE Id_1V	Solver output	Compared with 1.3 nA benchmark
Responsivity $\mathcal{R}$	CHARGE R_AW	Solver output	At -1 V
EQE η_{ext}	FDTD A_TE; CHARGE current	Solver output	Via eq. (1.65)
IQE η_{int}	CHARGE current / FDTD absorption	Solver output	Collection-efficiency check
Baseline bandwidth $f_{3\text{ dB,base}}$	MATLAB geometry model	52.7 GHz	No peaking; compact-model export
C_j (at -1 V)	Parallel-plate model	16.2 fF	Nominal $8\text{ }\mu\text{m} \times 5\text{ }\mu\text{m}$ area
$R_{s,\text{an}}$ (U-shape)	Sheet-resistance model	$20\text{ }\Omega$	36 % reduction from parallel model
Shot noise S_{shot}	Postprocess from I_d	Solver-derived	Depends on CHARGE dark current
D^*	Postprocess from NEP	Solver-derived	Compared with literature table
P_{sens} (shot-noise limited)	MATLAB system model	Solver-derived	Uses exported CML data

1.10.6 Design Trade-offs and Limiting Mechanisms

Absorption profile and the optical-to-electrical coupling chain.

The exponential spatial decay of $G_{\text{opt}}(\mathbf{r})$ in fig. 1.20 is a direct consequence of Beer-Lambert attenuation modified by waveguide confinement and taper coupling. The sweep up to $8\text{ }\mu\text{m}$ establishes the design point at which further length increase must be justified against the linear capacitance penalty $C_j \propto A_j \propto L$ through eq. (1.72). The IQE check exported by the post-processing script verifies whether the electric field of fig. 1.23 is sufficient to sweep photogenerated carriers to their respective contacts before SRH recombination occurs with the implemented $\tau_0 = 1\text{ ns}$ lifetime.

Dark current mechanisms and temperature sensitivity.

The CHARGE dark current is expected to be dominated by SRH generation under the implemented trap-assisted lifetime settings. The strong exponential temperature dependence

$n_i \propto \exp(-E_g/2kT)$ in germanium, where $E_{g,\text{ind}} = 0.66\text{ eV}$ is substantially smaller than silicon's 1.12 eV , implies that I_d approximately doubles for every $\Delta T \approx 10\text{ K}$ increase in junction temperature. In a data-centre transceiver operating at 40°C to 70°C junction temperature, this implies $I_d(70^\circ\text{C}) \approx 4 \times I_d(25^\circ\text{C})$ before high-field leakage mechanisms are considered. The corresponding shot-noise increase must be evaluated from the exported CHARGE current and the geometry-derived bandwidth; this avoids inserting an unsupported static dark-current value into the thesis.

Electric field screening under strong optical injection.

The analysis of section 1.10 is conducted at $100\text{ }\mu\text{W}$, well within the linear regime. At higher optical powers, the photogenerated carrier density within the intrinsic germanium layer becomes comparable to n_i and modifies the local electric field through its own space charge, a mechanism characterised in [30]. The onset power is estimated by the condition $G_{\text{opt,peak}}\tau_{\text{SRH}} \sim n_i$:

$$P_{\text{screening}} \approx \frac{n_i \hbar\omega A_j W_i}{\tau_0 \alpha(\lambda) \eta_c}, \quad (1.162)$$

which evaluates to $P_{\text{screening}} \approx 340\text{ }\mu\text{W}$ for the present device parameters. Operating at $100\text{ }\mu\text{W}$ is therefore safely below this threshold. Beyond $P_{\text{screening}}$, the effective transit time increases (equivalent to a reduction of $v_{d,\text{sat},h}$ in eq. (1.19)), reducing $f_{3\text{dB}}$ by more than 20 % for powers approaching $500\text{ }\mu\text{W}$ [30]. Equation (1.162) thus constitutes a hard optical power ceiling that must appear in any link power budget specifying the maximum launch power at the chip input.

The -1 dB compression photocurrent density from eq. (1.163) is:

$$J_{\text{ph}} = \mathcal{R} P_{\text{opt}}/A_j < q v_{d,\text{sat},h} N_{\text{eff}}, \quad (1.163)$$

giving $I_{\text{sat}} \approx 3.8\text{ }\mu\text{A}$ for the $6\text{ }\mu\text{m}^2$ junction. The present operating photocurrent of $I_{\text{ph}} = 95\text{ }\mu\text{A}$ at $100\text{ }\mu\text{W}$ exceeds this estimate because $N_{\text{eff}} \sim n_i$ is the intrinsic carrier density; the discrepancy is resolved by noting that at -1 V the built-in field significantly exceeds the minimum required for saturation, delaying the onset of actual screening to $\sim 200\text{ }\mu\text{W}$ experimentally [30].

Bandwidth and the intrinsic-layer thickness trade-off.

The device is positioned near the W_i^* optimum of eq. (1.76). Any further reduction of W_i below 200 nm enters a regime where drift-diffusion fails: transient carrier velocity overshoot elevates the effective transit velocity above $v_{d,\text{sat},h}$, while impact ionisation introduces a positive feedback between photocurrent and carrier multiplication that creates a non-linear photocurrent-power relationship [30]. A full-band Monte Carlo transport solver would be required for quantitative prediction below this threshold.

Nano-optical absorption enhancement as a path beyond the Beer-Lambert limit.

Grating-guided-mode resonance (GMR) structures can increase the effective interaction length within a fixed absorber thickness by diffracting normally incident light into guided modes propagating laterally within the absorber film, producing a Fano-type spectral resonance [20]. The resonance condition requires the grating period $\Lambda \approx \lambda/(n_{\text{eff},m} - \sin \theta_i)$. The resonant-cavity-enhanced (RCE) architecture similarly uses the expression of eq. (1.118) to achieve near-unity $\mathcal{A}$ in $W_i \sim 200$ nm films. For the vertical n-i-p architecture, a GMR or photonic-crystal overlay that achieves the same $\mathcal{A}$ in a shorter L would allow A_j to be reduced, simultaneously decreasing C_j and the dark current generation volume without sacrificing responsivity, and pushing W_i^* of eq. (1.76) toward smaller values consistent with higher balanced bandwidths. Whether the fabrication complexity and sidewall-defect-induced dark current increase outweigh these benefits is a quantitative question that remains open.

Doping implantation geometry and field uniformity.

When the cathode doping width $W_{\text{doping}} < W_m$, the electric field beneath the undoped corners of the germanium surface is reduced below the saturation threshold. The CHARGE simulations of [30] show that reducing W_{doping} from W_m to $0.75W_m$ decreases $f_{3\text{dB}}$ by more than 60 % at -0.8 V, with an abrupt performance cliff near $W_{\text{doping}} \approx 0.75W_m$. For the U-shaped electrode, this requirement is naturally satisfied because the cathode implantation covers the full top surface.

Energy consumption per bit.

An additional metric connecting device parameters to the power budget of the photonic integrated circuit is the energy consumed per detected bit e_c [J/bit]. Under the assumption that the photocurrent is insensitive to applied bias, $e_c \propto |V_{\text{bias}}| \mathcal{R}$. For the reference geometry at 100 GBd and -1 V bias, $e_c \approx 35$ fJ bit $^{-1}$ [30], competitive with the state of the art for integrated Ge photodetectors. The receiver sensitivity P_{sens} bounds the minimum detectable power; e_c bounds the maximum permissible photocurrent for a given power envelope. Together they define the feasible operating window for the receiver.

Lateral junction scaling limit and carrier velocity overshoot.

The bandwidth ceiling of 265 GHz achieved by the lateral Ge-fin device [23] introduces a physically distinct regime. At $W_{\text{Ge}} \approx 100$ nm, full-band Monte Carlo simulations show that holes overshoot their drift-diffusion saturation velocity by more than 40 % for approximately the first 50 nm of their trajectory before relaxing back to $v_{\text{d,sat},h}$ as phonon scattering equilibrates the hot-carrier distribution [30]. The effective transit time is therefore shorter than $W_{\text{Ge}}/v_{\text{d,sat},h}$ would predict, partially explaining the experimentally observed bandwidth exceeding the drift-diffusion ceiling.



Design-space contour plot for the vertical n-i-p architecture. x -axis: junction area A_j (μm^2) from 1 to 20. y -axis: intrinsic layer thickness W_i (nm) from 100 to 700. Colour: $f_{3\text{dB}}$ (GHz) computed from eq. (1.69) with U-shaped R_s from eq. (1.138). Overlay: (i) Beer-Lambert absorption saturation contour $\mathcal{A}(L) = 0.90$ bounding the design space from below in A_j ; (ii) W_i^* locus from eq. (1.76) as a dashed curve; (iii) electric field screening power contour $P_{\text{screening}} = 200 \mu\text{W}$ from eq. (1.162). Mark the present device operating point ($A_j = 6 \mu\text{m}^2$, $W_i = 350 \text{ nm}$) with a star.

Figure 1.26: Two-dimensional design space of the vertical n-i-p Ge-on-Si waveguide photodiode (O-band, U-shaped electrode). The three overlaid constraints, Beer-Lambert absorption saturation, the transit-time/RC balance locus W_i^* , and electric field screening power, define the physically feasible and optimally performing region within which the chosen design point is located.

1.10.7 Detector-Level Sensitivity and Doping Polarity

The dark current I_d that enters the shot-noise expression eq. (1.88) and the sensitivity formula eq. (1.165) is conditioned by the doping polarity at the heterojunction. The p -on- n configuration adopted here (Si platform p -doped, Ge cathode n -doped) places the graded Ge/Si transition such that the valence-band discontinuity $\Delta E_v \approx 0.54 \text{ eV}$ acts as an additive barrier suppressing hole generation at the interface. In the alternative n -on- p configuration, this same ΔE_v raises J_{surf} in eq. (1.53) above the ideal planar junction prediction, increasing I_d by a factor of approximately two to three at -1 V [30]. Through eq. (1.167), this translates into a receiver sensitivity improvement of approximately 1.5 dB in the dark-current-dominated regime, a non-trivial gain achievable at zero process cost simply by choosing the correct doping polarity.

Shot-Noise-Limited Sensitivity.

In the shot-noise-limited regime, the signal-to-noise ratio at the output of the detector in a detection bandwidth $B_e = f_{3\text{dB}}$ is:

$$\text{SNR} = \frac{(\mathcal{R} P_{\text{opt}})^2}{2q(\mathcal{R} P_{\text{opt}} + I_d) B_e}. \quad (1.164)$$

Setting $\text{SNR} = Q^2$ where $Q \approx 7.03$ for $\text{BER} = 10^{-12}$ in OOK, and solving for P_{opt} :

$$P_{\text{sens}} = \frac{Q^2 q B_e}{\mathcal{R}} + \frac{Q}{\mathcal{R}} \sqrt{Q^2 q^2 B_e^2 + 2q I_d B_e}. \quad (1.165)$$

In the signal-dominated limit ($I_d \ll \mathcal{R}P_{\text{sens}}$):

$$P_{\text{sens}} \approx \frac{2Q^2qB_e}{\mathcal{R}}, \quad (1.166)$$

and in the dark-current-dominated limit:

$$P_{\text{sens}} \approx \frac{Q}{\mathcal{R}} \sqrt{2qI_d B_e}. \quad (1.167)$$

Substituting solver-exported $\mathcal{R}$ and I_d with the implemented geometry-derived bandwidth into eq. (1.165) at $Q = 7.03$ gives the receiver sensitivity reported by the MATLAB system model:

$$P_{\text{sens}} = P_{\text{sens}}(\mathcal{R}, I_d, B_e). \quad (1.168)$$

This detector-level sensitivity estimate is generated after the CML dataset is created, while the screening analysis developed in the preceding discussion identifies the upper-power constraint that remains to be managed in practical deployment.

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